
Looked Up or Copied? Later Mentions Decide Which Attention Channel Carries an In-Context Value

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Abstract

A token that states a value in context, such as *shelf* in “the candle is moved to the shelf”, can be read by later tokens through its attention key, which a later query matches, or its value, which is copied. We clamp the two separately at this writing token in 10 instruction-tuned models from four families. When the candidates are mentioned after the writing token, by an option list or, from 7B up, a neutral sentence, the value’s identity is looked up through the key; otherwise it is mainly copied, even when the same list precedes it. At Qwen2.5-7B and Mistral-7B, 5 % of attention heads, whose core is canonical duplicate-token heads, read the writing token’s key at the option words (0.967 and 0.942 of the effect; random heads 0.004 and 0.007), and the answer reads the option words directly. Such attention is also present at 1.5B and 3B under the sentence format, where their key identity is about zero. A released learned remap behaves nearly the same in every format, yet whether its keys or values appear to carry it follows the format’s readers, also when refit without later mentions. On Prakash et al.’s binding swap, replicated at Qwen2.5-14B, a preregistered law predicting its key share from the natural read at that depth, which carries almost no key identity, failed: the swap is key-dominant in every evaluable format (in two, a margin shift that flips no answer), and later mentions shift its key share less than predicted. We mark exploratory results and report every failed prediction.

1. Introduction

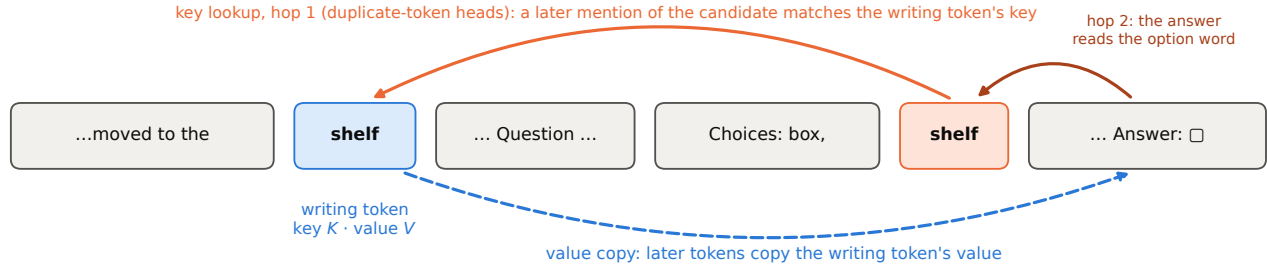
Mechanistic explanations are often stated at the level of components, such as attention heads or subspaces (Vig et al., 2020; Wang et al., 2022; Geiger et al., 2023), for instance that a value written in context is read through a particular attention channel, or that an intervention’s effect is carried by a particular component. A later token sees a context token only through its cached key, which a later query must match, and its value, which is copied when the token is attended to. A written value can thus be *looked up* (Figure 1a): a later token whose content matches the key attends to it, so the key itself carries which value was written. Or it can be *copied*: the reader attends for other reasons, and the identity travels in the value. In lookbacks, ordering IDs bind a state token and a later pointer retrieves it (Prakash et al., 2026). Without a later mention, we find their state-token patch (an address and payload swap, in their terms) reaches the answer mainly through the state tokens’ keys (value share 0.435), while an identity edit at the same tokens is copied through the value (Section 4.2). Binding information can sit in query/key subspaces or, in Llama models, mainly in keys (Oh & Demberg, 2026); what decides which read a prompt uses is unclear.

In the belief stories of Anonymous (2026), one token, the *writing token* (their “critical token”), states where an object was moved. We clamp its cached key and value separately to those of a run in which a different value was written; clamping both from the first layer reproduces that run exactly (Section 2). We vary whether and where the candidate values are mentioned (Figure 1b), in 10 instruction-tuned models (1.5B–72B, four families) on the belief task, and in five 7–14B models on two further templated tasks.

Later mentions decide (Section 3). Whenever the candidates are mentioned again after the writing token, by an option list, lettered options or, from 7B up, a neutral sentence, the value’s identity is looked up through the key;

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(a) Two ways later tokens can read the token that wrote a value



(b) Prompt formats (same story; candidates named after, before or not at all)

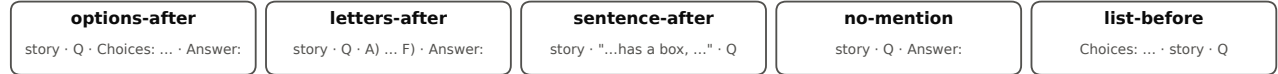


Figure 1. Two ways later tokens can read a value written in context. (a) The writing token *shelf* (blue) leaves a key K and a value V in the cache at every layer. A later mention of the same candidate (orange) can find it through its key, so the value’s identity is *looked up*: duplicate-token heads read the key there (hop 1), and the answer reads the option word (hop 2). Without such a mention, the identity reaches the answer through the writing token’s value, which is *copied*. We clamp K and V separately to those of another run. (b) The formats share the story and differ in whether and where the candidates are named, and in the answer instruction; Section 3.1 adds LIST-AFTER and SENTENCE-BEFORE.

without a later mention it is mainly copied through the value. The sentence effect was not predicted; a preregistered, instruction-matched 2×2 design in four 7–14B models then confirmed it and showed that position is causal: the same list or sentence placed before the writing token opens no key read (Table 1). The read has two hops (Section 3.4): at Qwen2.5-7B and Mistral-7B, 5 % of attention heads read the writing token’s key at the option words (alone they recover 0.967 and 0.942 of the key effect there), and the answer position then reads the option words directly (Figure 3). The re-mentioned words are the main readers in lists and in sentences naming at least three candidates (Section 3.3). Hop-1 attention is not new: canonical duplicate-token heads (Wang et al., 2022) form the readers’ core (Section 3.5), and with the causal mask this attention predicts the format dependence. What is new is that it decides which cache channel carries a value’s identity (by exact single-channel clamps), that it is not sufficient (it is present also at 1.5B and 3B, whose SENTENCE-AFTER key identity is about zero), and that its readers decide where a learned remap appears to act.

Where an intervention appears to act (Section 4). Anonymous (2026) fit a rank-16 distributed alignment search (DAS) remap of locations (Geiger et al., 2023) and, with options listed after the question, traced much of its difference from an unfitted PCA control to the writing token’s keys, their component-level explanation of the tested effects. Re-running their released interventions at 24B and 72B, we find the behavioural effect nearly the same in every format, while the channel that appears to carry it flips (Figure 2): keys with lettered options, values with no later

mention. Our reading, formed after these data, is that the edit is present in both channels and the format’s readers decide which one appears to carry it. Two preregistered tests then supported it: values carry the edit where keys do not read the identity, and a remap refit without later mentions shows the same crossover (Section 4.1). On the state-token binding swap of Prakash et al. (2026), which we replicate at Qwen2.5-14B, our preregistered extension failed (Section 4.2): the swap is key-dominant in each evaluable format (in two, a margin shift that flips no answer), while the natural read at that depth carries almost no key identity, so the law predicting an edit’s key share from that read fails (preregistration H, H7); later mentions shift its key share by +0.247, below the preregistered 0.4 (H9). A component-level explanation of an intervention is thus a property of the intervention, of what it changes, and of the prompt in which it is read. Table 33 (Appendix H) sets four accounts, ours included, against these observations; each is contradicted somewhere.

Contributions. (i) Exact key and value clamps, row-, head- and answer-row-restricted splices, and an exact key/value exchange on another lab’s patch. (ii) Later mentions decide look-up versus copy; their position is causal (A–E). (iii) The mechanism at 7B: sparse readers with a duplicate-token core, then the answer’s direct read of the option words, mainly through their keys (an unscored contrast); the duplicate-token match is not sufficient across scale (G, H1–H5). (iv) An identity edit’s attributed channel flips with the read format (C, E4, F; H11 at block 0); on the binding swap the preregistered law fails and later mentions shift the key share by only +0.247 (H7–H10; at block 28,

so edit kind and depth are confounded). (v) Eight preregistrations (A–H); every failed prediction is reported (Appendix A and Section 4.3).

2. Setup

Task and prompt formats. We generate belief stories with a port of the released generator of Anonymous (2026): “Everyone initially sees that the $\{o\}$ is in the $\{\text{init}\}$. [...] $\{A\}$ watches as the $\{o\}$ is moved to the $\{\text{LOC}\}$. $\{B\}$ does not see this happen.”, followed by “Where does $\{A\}$ believe the $\{o\}$ is?” (template in Appendix B). Locations are six single-token words. A base story B and a source story S differ only at the *writing token* $\{\text{LOC}\}$ at position p . Two further tasks write the value with one sentence: *paint* (“Later, $\{\text{agent}\}$ repainted the $\{\text{thing}\}$ $\{\text{colour}\}$.”, six colours) and *schedule* (“Later, $\{\text{agent}\}$ moved the $\{\text{thing}\}$ to $\{\text{day}\}$.”, seven days). Every prompt is wrapped as a system turn, a user turn, the generation prompt with thinking disabled and an “Answer:” prefill, and we read full-vocabulary log-probabilities at the answer position. The formats differ in where, if anywhere, the candidates are named and in the answer instruction (Appendix B): OPTIONS-AFTER (options after the question, “Answer with exactly one choice”, the format of Anonymous 2026); LETTERS-AFTER (options A–F after the question, “Answer with one letter”, scored on the letters); and, all with “Answer with one word”, SENTENCE-AFTER (the neutral sentence “The room has a box, a basket, ... and a closet.” after the story), NO-MENTION (no candidates) and LIST-BEFORE (the options list before the story). A *later mention* is any token after p that names a candidate. The instruction-matched 2×2 of Section 3.1 adds LIST-AFTER (OPTIONS-AFTER with the one-word instruction) and SENTENCE-BEFORE (the sentence before the story).

Clamping the key and value channels. A later token sees the writing token only through its cached key, which decides how much the token attends to p , and its value, which decides what is copied when it does. We call these the *key channel* and the *value channel*. The clamp $\mathcal{C}_K(Z)$ runs the base prompt but sets, at p and in every layer $\ell \geq \ell_0$, the pre-RoPE k_{proj} output (all KV groups) to the one cached from run Z ; $\mathcal{C}_V(Z)$ does the same for v_{proj} , and $\mathcal{C}_{KV}(Z)$ does both. Unless stated otherwise $\ell_0 = 0$, and then $\mathcal{C}_{KV}(S)$ reproduces the source run at every later position (Appendix C).

Measures. With $m_{\text{SB}} = \log p(S) - \log p(B)$ at the answer position, the effect of clamping channel C is measured against the *self-clamp row* $\mathcal{C}_{KV}(B)$, computed in the same batch:

$$d_C = m_{\text{SB}}(\mathcal{C}_C(S)) - m_{\text{SB}}(\mathcal{C}_{KV}(B)), \quad (1)$$

for $C \in \{K, V, KV\}$. The key share is $s_K = \bar{d}_K / (\bar{d}_K + \bar{d}_V)$, a ratio of means over stories. A foreign key can move m_{SB} without carrying the source value, for instance by weakening B. To isolate identity we also clamp from a story with a third value X that is absent from the story, and with $\Delta_Y(Z) = \log p(Y | \mathcal{C}_K(Z)) - \log p(Y | \mathcal{C}_{KV}(B))$ we define

$$\text{ID}_K = \frac{1}{2} [\Delta_S(S) - \Delta_S(X) + \Delta_X(X) - \Delta_X(S)]. \quad (2)$$

Any effect shared by all foreign keys cancels, so ID_K (in nats) is zero if keys carry no identity of the value; ID_V is defined in the same way with values. Because nats grow with the model’s confidence, we also report the scale-free *identity key share* $s_{\text{ID}} = \overline{\text{ID}}_K / (\overline{\text{ID}}_K + \overline{\text{ID}}_V)$, which, unlike s_K , excludes non-specific effects. The *row-restricted splice* runs attention twice per layer and splices output rows by a mask, so that only a chosen group of later tokens sees the swapped key; it is exact in FP32 to 10^{-4} (Appendix C).

Reader heads and the second hop. At Qwen2.5-7B and Mistral-7B, heads are ranked on 60 stories by a_3 , their attention change from the option rows to p under the key clamp, and tested on 60 others. *HeadSplice* lets only chosen heads, in the option rows, see $K_S: R(k)$ is the fraction of the option rows’ key effect recovered when only the top k heads see it, $\text{KO}(k)$ the fraction lost when only they do not, and k^* is 5% of heads. *HopSplice* keeps K_S at p but gives only the answer row the option rows’ base keys and values; r_{ans} is the fraction of the key-clamp effect this removes. Both are exact in FP32 (Appendix C).

The intervention of Anonymous (2026). Anonymous (2026) patch Mistral-Small-24B and Qwen2.5-72B at the output of (1-based) decoder block 4 over every token of the event span (from “ $\{A\}$ watches as” to the end of the story). For a rank-16 basis U with orthonormal rows, each span position becomes

$$h \leftarrow h_B + U^\top U (h_S - h_B). \quad (3)$$

In the learned remap M, U is a distributed alignment search basis (Geiger et al., 2023) fit with a six-choice cross-entropy objective so that the patched model ranks $T = \pi(S)$ first, where π is a fixed pair-swap of the six locations. In the control P, U is an unfitted PCA basis of source-minus-base differences, which transfers S. We reuse the released bases (three seeds each; the Qwen2.5-72B PCA bases are reconstructions, labelled so by Anonymous (2026)). Following their design, a *key-only exchange* replaces the writing token’s key in every block from (1-based) block 6 on by the key of the other patched run, while queries, values and residual states evolve freely; we add the symmetric *value-only exchange*. Unlike our clamp, these exchanges are not

exact, because the patch also changes the other span tokens. With $m_{TS} = \log p(T) - \log p(S)$, writing $P+K_M$ for P with M 's keys and $m(\cdot)$ for m_{TS} in a run,

$$\varphi = \frac{m(M) - m(P)}{m(T) - m(S)}, \quad (4)$$

$$\psi_K = \frac{m(P+K_M) - m(P)}{m(M) - m(P)}, \quad (5)$$

$$\psi_V = \frac{m(P+V_M) - m(P)}{m(M) - m(P)}. \quad (6)$$

φ is the shift of M beyond P towards T , as a fraction of the natural $S \rightarrow T$ shift; ψ_K and ψ_V are the fractions of the learned-minus-PCA difference that M 's keys or values add to P . All are ratios of means over the 96 of their 120 native story cores in which B , S and T are distinct, after averaging the three fits within each core.

The exchange on another lab's intervention. We also apply an exact exchange to the state-token patch of [Prakash et al. \(2026\)](#) at Qwen2.5-14B, on their template-2 stories, raw prompt and seed-10 pool (150 pairs). BIND, their patch, sets the residuals at the two state tokens and the punctuation token after each, at the output of (0-based) block ℓ^* , to those of a counterfactual in which the two binding sentences are exchanged. ID, an identity edit at the same positions, substitutes a drink S absent from the story for the queried state. With $m = \log p(\text{target}) - \log p(\text{queried state})$, Φ is the patch's effect on m , and ψ_K (ψ_V) the fraction of it that the clean run reproduces with the patched run's keys (values) at the patched positions from block ℓ^*+1 on; $\kappa = \psi_K / (\psi_K + \psi_V)$. ℓ^* is the earliest block of maximal interchange accuracy (IIA) under NO-MENTION. The formats are NO-MENTION, QNAMES (the question names four candidates) and OPTIONS-AFTER (details in Appendix G).

Models, statistics and preregistration. We study 10 instruction-tuned models from four families: Qwen2.5-Instruct ([Yang et al., 2024](#)) at 1.5B, 3B, 7B, 14B, 32B and 72B, Qwen3-8B, Mistral-7B-Instruct-v0.3, Mistral-Small-24B-Instruct-2501 ([Mistral AI, 2025](#)) and OLMo-2-1124-7B-Instruct ([Team OLMo et al., 2024](#)), in BF16 on GPU. We use $n = 150$ stories per format and model. The row-restricted splice uses $n = 60$ at 7–14B; at Qwen2.5-1.5B it ran, like preregistration A, on CPU in FP32 with $n = 40$. Intervals are 95% percentile intervals from 10,000 bootstrap resamples of stories (or pairs), with ratios recomputed within each resample. Eight preregistrations (A–H) were each committed to version control before their run, with the scoring script committed before its outputs were inspected; H was finalised before any output of G was inspected. Appendix A gives every prediction with its recorded outcome and classifies every main claim as confirmatory, exploratory or post hoc.

Table 1. Position decides the key read. The same list or neutral sentence, after or before the story, with “Answer with one word”; NO-MENTION for reference. ID_K (nats, 95% CIs), $n = 150$, keys clamped from layer 0. Preregistration E (E1).

Format	Qwen-7B	Qwen-14B	Mistral-7B	OLMo-2-7B
LIST-AFTER	+20.6 [20.2, 21.0]	+35.8 [34.9, 36.7]	+15.6 [15.1, 16.0]	+8.7 [8.4, 8.9]
SENTENCE-AFTER	+5.5 [5.2, 5.8]	+12.1 [11.5, 12.7]	+7.9 [7.3, 8.4]	+2.0 [1.8, 2.2]
LIST-BEFORE	−0.1 [−0.2, −0.1]	−0.1 [−0.2, 0.0]	0.0 [−0.1, 0.1]	−0.3 [−0.4, −0.2]
SENTENCE-BEFORE	0.0 [−0.1, 0.1]	−0.1 [−0.2, 0.0]	0.0 [−0.1, 0.1]	−0.2 [−0.3, −0.2]
NO-MENTION	+1.0 [0.9, 1.2]	+1.1 [0.9, 1.2]	+0.6 [0.5, 0.6]	+0.4 [0.3, 0.5]

3. How later tokens read a written value

Preregistered results name their prediction (B1 is prediction 1 of preregistration B; Appendix A); the others are marked exploratory or post hoc.

3.1. Later mentions decide which channel is read

Natural formats. Over all 10 models (Table 13), the formats fall into three regimes.

Candidates listed after the writing token. Under OPTIONS-AFTER and LETTERS-AFTER, ID_K is large and positive in every model, from 2.8 nats at Qwen2.5-1.5B to 39.0 at Qwen2.5-14B under OPTIONS-AFTER (A1 at 1.5B on fresh stories and B1 in five held-out models from 3B to 8B, both met; exploratory at 14B and 24–72B).

No later mention. Under NO-MENTION, ID_K is at most 1.25 nats, less than a tenth of its OPTIONS-AFTER value in every model (paired difference positive in 5/5 held-out models; B2, met), and the identity is mainly copied: ID_V exceeds ID_K more than tenfold (exploratory; Table 15). LIST-BEFORE names the same candidates before the writing token, which the causal mask keeps them from attending to; its ID_K is never reliably positive, and negative in nine of ten models (C4's 0.5-nat bound met; sign unexplained).

A neutral sentence after the story. SENTENCE-AFTER carries no positive key identity at Qwen2.5-1.5B and 3B, but an intermediate amount in every model from 7B up (2.0–12.1 nats), which we did not predict. The scale-free identity key share s_{ID} above 1.5B is 0.67–0.88 under OPTIONS-AFTER and at most 0.45 under SENTENCE-AFTER and 0.06 under NO-MENTION (exploratory; Table 14).

An instruction-matched 2×2 . Preregistration E crosses {list, sentence} with {before, after} under one instruction (Table 1); all parts of E1 were met. ID_K is 8.7–35.8 nats under LIST-AFTER and 2.0–12.1 under SENTENCE-AFTER,

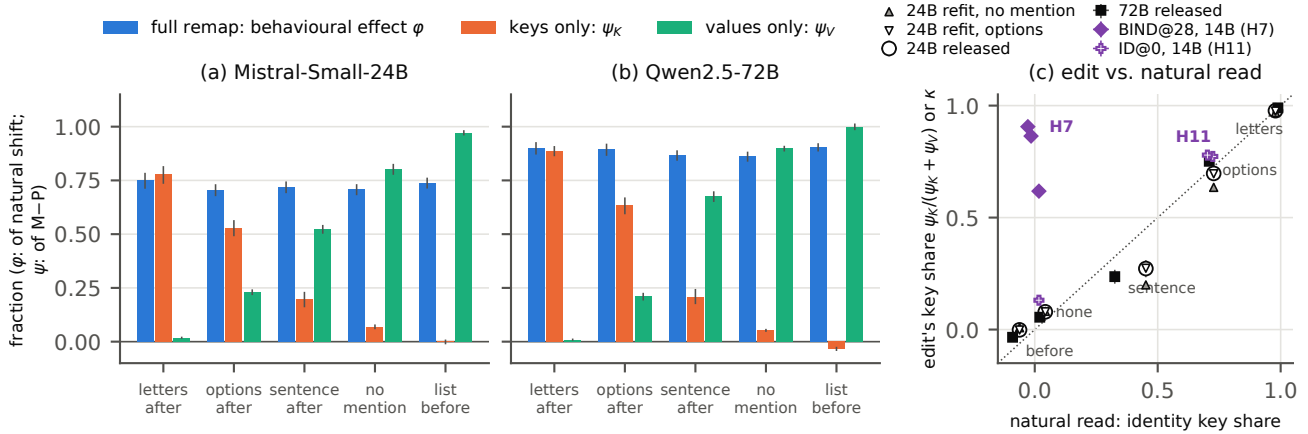


Figure 2. The same learned intervention has nearly the same behavioural effect in every format, but whether the writing token’s keys or values carry it depends on the later mentions. The released remap M of Anonymous (2026) against their PCA patch P at Mistral-Small-24B (a) and Qwen2.5-72B (b): φ , ψ_K and ψ_V (Equations (4) to (6)), exchanged from block 6 on; $n = 96$ cores $\times$ 3 fits, 95% bootstrap intervals. (c) The remap’s key share $\psi_K/(\psi_K + \psi_V)$ against the unpatched model’s s_{ID} in the same format: post hoc for the released remaps, preregistered (F4) for the refits (triangles; Section 4.1). Diamonds: the binding swap of Prakash et al. (2026) at Qwen2.5-14B, κ at block ℓ^* against s_{ID} from block $\ell^* + 1$ (H7, failed); crosses: the identity edit at block 0 against s_{ID} from block 1 (H11, met; nearly a restatement of the clamp). Values in Tables 12, 21 and 30.

every CI before the story contains zero or lies below it, and both paired differences exclude zero in 4/4 models. Position, not the instruction or the list format, decides whether the key channel opens.

3.2. Across tasks, families and scale

On the paint and schedule tasks, in five 7–14B models, the paired OPTIONS-AFTER–NO-MENTION difference in ID_K is positive in 5/5 models per task (D1, met), and LIST-BEFORE stays within the preregistered 0.5-nat bound in 10/10 cells (D2, met; Table 18). On IOI (Wang et al., 2022), plain sentences carry almost no key identity (G18, met), but the preregistered test that a later list opens a key read was not met (G19; Mistral-7B failed the competence gate; Appendix F). Our claims are limited to the three templated tasks. Within Qwen2.5, the OPTIONS-AFTER key share s_K rises from 0.37 at 1.5B to 0.91 at 14B and falls to 0.76 at 72B (B3, exploratory; Figure 4). The fall is not an artefact of the software or GPU setup (E3, met), but its cause is unknown. Other families give 0.73–0.92 (deeper clamp onsets: Table 16).

3.3. Which later tokens read the writing token’s key

Letting only the option words see the source key recovers 0.92–0.98 of the full key effect at 7–14B under OPTIONS-AFTER, and every other group at most 0.01; under SENTENCE-AFTER the sentence’s candidate words recover 0.67–0.73 (D3, met; Figure 3a). Conversely, forbidding the six candidate words to attend to the writing token removes the key identity (r_K , the fraction left, is -0.105

to 0.003 in 9/9 format–model cells; G5, met), and cutting a matched control column does not (G6, met). The words must name the candidates at stake: listing B with two other candidates instead of S and X gives no read (G9, met). In lists two named candidates suffice (G10, met for lists); in sentences the read grows with the number named (G10, not met for sentences), and with only S and X named the words carry 0.38–0.40 of the splice effect, most of it not identity-specific (G12, not met; Appendix F). The candidate words are thus the main readers in lists and in sentences naming at least three candidates.

What a re-mention must be. Varying the re-mentions of SENTENCE-AFTER in four 7–14B models shows a read of the word form, largely up to case (r_K : ID_K relative to an exact repeat). Capitalised and upper-case forms read nearly fully (0.83–1.14; G14, which predicted at most 0.75, not met), but a modified noun phrase such as “wooden shelf” reads only partly (0.32–0.64; G13, not met), as does a plural. Synonyms and mixed-language translations carry little on our English-scored measure (at most 0.15; G15, met). No preregistered token or concept outcome holds (G16, not met). Where the read is lost, the value identity rises (G17, met), but no more than after a sentence naming other words (exploratory; Appendix F.4).

3.4. Reader heads and the second hop

Preregistration H looked inside the option rows at Qwen2.5-7B and Mistral-7B under OPTIONS-AFTER (Table 27); H1–H5 were met in both. When only the top 5% of heads by a_3 (40 and 52) see the source key, they recover $R(k^*) = 0.967$

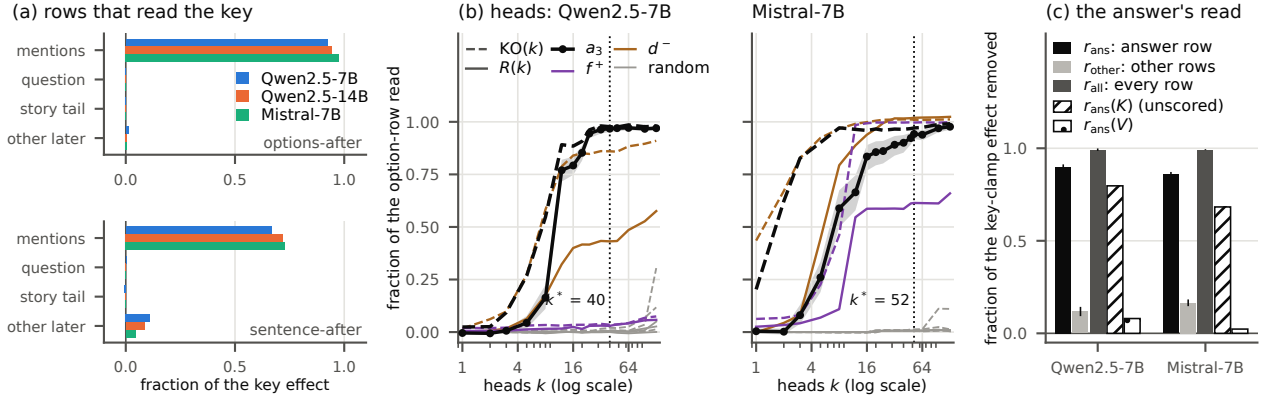


Figure 3. The later mentions read the writing token’s key through a small set of heads, and the answer reads the later mentions. (a) Row-restricted splice ($n = 60$; D3): the fraction of the full key effect recovered when only one group of later tokens sees the source key (top OPTIONS-AFTER, bottom SENTENCE-AFTER). (b) OPTIONS-AFTER, 60 held-out stories: $R(k)$ (solid) and $KO(k)$ (dashed) for heads ranked by a_3 (band: 95% CI), by single-head effect f^+ and leave-one-out effect d^- , and three random sets (grey); dotted: k^* (H1, H2). (c) The fraction of the key-clamp effect removed when only the answer row, all other later rows or all rows get the option rows’ base keys and values (H5); hatched and open bars: keys or values only (not scored). Values in Tables 20 and 27.

and 0.942 of the option rows’ read (H1), and blinding them removes 0.977 and 0.971 (H2); random sets of equal size recover 0.004 and 0.007 and remove 0.011 and 0.010 (three-draw means; Figure 3b). Ranking by single-head effect f^+ does not find the set ($R(128) = 0.06$ and 0.66); leave-one-out d^- does at Mistral-7B (80 % with 12 heads) but not at Qwen2.5-7B ($R(128) = 0.58$), and no layer alone recovers more than 0.05 (exploratory): the read is joint across layers.

Mean-ablating them at the option rows cuts ID_K to $\rho_K = 0.238$ and 0.059 of its value (control sets ≥ 0.953), raises ID_V by 6.907 and 10.861 nats, and leaves the base answer in 0.87 and 1.00 of stories (H3). The lookup thus suppresses a copy route that stays available. Cutting the candidate words’ edges instead (G5) also restores the value identity in 3/3 models but keeps the answer only at Qwen2.5-14B (G7, not met: a preregistered partial takeover); we do not reconcile the two. We do not claim the lookup replaces the copy: its preregistered tests failed (G11; on IOI, G20).

The second hop. The answer need not attend to the writing token (forbidding it leaves r_K at 1.010–1.043; G8b, met at 7–14B); it reads the option words. With the source key clamped, giving only the answer row the option rows’ base keys and values removes $r_{ans} = 0.899$ and 0.859 of the key-clamp effect, against 0.118 and 0.165 for all other later rows (H5, met; Figure 3c). This rejects a three-hop route via the template rows. Unscored (H5), the answer reads them by key: $r_{ans}(K) = 0.797$ and 0.683, against $r_{ans}(V) = 0.080$ and 0.023. The SENTENCE-AFTER line is not evaluable (Appendix G). Without a later mention, the same cut leaves 0.134–0.417 of ID_V (G8a, met).

3.5. What the duplicate-token account explains

The first hop, a re-mention attending to the writing token, is the pattern of duplicate-token heads (Wang et al., 2022). Under SENTENCE-AFTER, the re-mentioned word attends to the writing token far more than the other candidates do, and this attention follows the clamped key one for one, at 7B and 14B (E , the excess at the top three heads, = 0.869, 0.915; F/E , its share following the clamp, = 1.00, 0.98; G1) and equally at 1.5B and 3B ($E = 0.789, 0.879$; $F/E = 1.03, 0.99$; G2, G3), whose SENTENCE-AFTER key identity is about zero (within the preregistered ± 1 -nat margin). All three were met: the match is present without the read. The core of the causally selected readers is of this kind (H4, met): over the top 20 and 16 heads by a_3 , which carry 80% of the read, the median duplicate-token score D (on random repeated sequences) is 0.370 and 0.232 (whole-model medians 0.028 and 0.007, recomputed), the median induction score (Olsson et al., 2022) 0.011 and 0.022, and 6 and 5 of the top 10 by a_3 are in the top 10 by D . Only this core is canonical: over the full 5% set the median D is 0.179 and 0.065 (recomputed).

We therefore claim no new head type. What we add is that generic duplicate-token heads write an identity flag into the option words’ keys, which the answer then reads, mainly by key (H5, secondary); and that the scale dependence enters downstream of hop-1 attention (G2). The channels carry which value was written, not its belief role (E2, met; Table 19).

4. Where an intervention appears to act

[Anonymous \(2026\)](#) read their key exchange between M and P, under OPTIONS-AFTER, as causal evidence that keys induced by the learned patch contribute to the behavioural difference between the two; we reproduce their result (Appendix H). We then change only the format and add a value-only exchange. Two comparisons are confirmatory and met in both models: without a later mention M’s keys carry less than half of what they carry under OPTIONS-AFTER (C3), and M’s values carry most of the effect (E4); the rest is exploratory (Table 21).

Behaviour barely changes across formats; the carrying channel flips. φ lies between 0.70 and 0.75 at Mistral-Small-24B and between 0.86 and 0.91 at Qwen2.5-72B, and the PCA control transfers the source location in every format. Yet at Qwen2.5-72B, M’s keys reproduce 0.89 of the M–P difference under LETTERS-AFTER but 0.05 under NO-MENTION, while M’s values do the reverse (Figure 2b). Mistral-Small-24B and the removal fractions (Appendix C) show the same crossover. Across the ten format-model cells, the remap’s key share $\psi_K/(\psi_K + \psi_V)$ tracks the identity key share s_{ID} of the unpatched model in the same format (Figure 2c; $r = 0.98$). This is post hoc here, preregistered for the refits below, and failed out of sample for a binding edit (Section 4.2).

Interpretation. In LETTERS-AFTER, OPTIONS-AFTER, SENTENCE-AFTER and NO-MENTION the prompt is identical up to the end of the story, so under the causal mask M writes the same key and the same value at the writing token in all four; only their readers differ, and which channel appears to carry the edit follows them. This reading is post hoc (C); its value half (E4) and the refit test below (F) were then preregistered and confirmed. It concerns edits that change which value is written; on the material of [Prakash et al. \(2026\)](#) we show it only for an identity edit at block 0 (Section 4.2). That the readers are the same at 24B and 72B is an inference.

4.1. A remap fit without later mentions

The remap edits block 4, upstream of every exchanged key and value. Preregistration F asked whether the format it was fit in, rather than its readers, decides its channel: at Mistral-Small-24B it refit the remap with the released code under NO-MENTION (fit_{none}) and, as a same-code control, under OPTIONS-AFTER (fit_{opt}; Appendix D). All gates and all four predictions were met (Tables 5 and 12). fit_{none} is carried by keys under LETTERS-AFTER ($\psi_K = 0.77$) and by values under NO-MENTION ($\psi_V = 0.80$), transfers to lettered options ($\varphi = 0.67$), shows the control’s crossover (difference in differences $-0.008 [-0.027, +0.009]$), and its key share tracks the natural read ($r =$

0.96; Figure 2c). Post hoc, the fitting format still lowers fit_{none}’s ψ_K under OPTIONS-AFTER and SENTENCE-AFTER (by 0.09 and 0.07). The reading format decides which channel dominates.

4.2. A binding edit from another lab

Preregistration H applied the exchange to an edit we did not design: the state-token patch BIND of [Prakash et al. \(2026\)](#), at Qwen2.5-14B on their material, with an identity edit ID as a control (Table 30 and Appendix G.2). BIND replicates (IIA 0.993 at block 28 under NO-MENTION; Gate b1). It exchanges the two binding sentences, changing the state word’s ordering information but not its identity; in their account it swaps addresses (read through QK) and a payload (moved through OV).

The binding swap is key-dominant; four predictions failed. Without a later mention, keys carry most of BIND’s effect and values a substantial share ($\psi_K = 0.703$, $\psi_V = 0.435$, $\kappa = 0.618$). Under QNAMES and OPTIONS-AFTER, $\kappa = 0.906$ and 0.864 , but BIND flips 0/150 answers there, so these decompose a shift of the logit margin ($\Phi = 5.814$ and 6.523 nats); LETTERS-AFTER failed the effect-size gate. Not met: values carrying BIND without a later mention ($\psi_V - \psi_K = -0.268 [-0.317, -0.219]$; H8); a crossover of at least 0.4 (later mentions raised κ by 0.247 [0.211, 0.281], and QNAMES is not in between; H9); the rival $\kappa \leq 0.25$ (H10; nor is $\kappa \geq 0.75$ throughout); and the law $\kappa \approx s_{ID}$, which missed by 0.60–0.93 (H7; Figure 2c; against s_{ID} from block 0, secondary: 0.600, 0.181, 0.163).

The identity edit, and depth. ID at block 0 matched the law ($\kappa_{ID} = 0.131, 0.773, 0.779$ against $s_{ID} = 0.017, 0.724, 0.702$), and later mentions raised κ_{ID} by 0.648 [0.630, 0.666] (H11, met). This is consistency on their material, not out-of-sample support: at block 0 ID nearly equals the natural clamp, and H11’s block-28 part compares values near zero. Post hoc, at block 28 BIND is key-dominant while ID is value-carried ($\psi_V \geq 0.811$ in the evaluable formats), which is consistent with the H7 failure but does not explain it (Appendix G.2). Depth is an alternative explanation: under OPTIONS-AFTER, s_{ID} falls from 0.702 at block 3 to -0.014 at 27 and ID flips no answer from block 20, while BIND flips none below block 24 under NO-MENTION and at most 0.01 of answers under QNAMES or OPTIONS-AFTER (blocks 18–40). No depth has both the binding swap and a key-read identity route, so whether the kind of edit or its depth sets how far the format moves the attribution is untested.

4.3. Preregistered predictions that failed

A transfer law from clamping-estimated value completeness (C1, C2) failed at Mistral-Small-24B and held only

marginally at Qwen2.5-72B; $s_K \geq 0.80$ under OPTIONS-AFTER (C4) held only at Qwen2.5-32B of the three largest. Of the 34 predictions of G and H, 14 failed (H6 not evaluable, H12 not run; Tables 6 to 8): two kept the predicted direction but missed a threshold (G20, H9), six were met in part (G7, G10, G12–G14, G21), one failed on a competence gate (G19), one rejected the rival (H10), and four falsified our prediction (G11, G16, H7, H8).

5. Related Work

Options, binding and retrieval. Heads select among options (Lieberum et al., 2023; Wiegrefe et al., 2024; Tulchinskii et al., 2024; Sun et al., 2026), choosing a winner before binding it to a symbol (Wong et al., 2026). Binding IDs (Feng & Steinhardt, 2023) explain how bound attributes are stored and retrieved, without separating keys from values; key-side rebinding (Oh & Demberg, 2026), value-slot routers (Steele et al., 2026a;b) and cached-KV edits (Li, 2026) concern storage.

Closest work. Prakash et al. (2026) describe their state-token patch as swapping addresses, read through QK, and payloads, moved through OV; split between keys and values it is key-dominant, with a substantial value share without a later mention (Section 4.2). This fits both halves of their account without separating them; the swap changes ordering information, not identity, and an identity edit at the same tokens is copied through the value there (Appendix H). Wu & Shomali (2026) find an operation’s cache read mainly as an address, moving keys and values together in one format; Ma et al. (2025) find keys to be sparse routers and values dense payloads. With the channels separated and the format varied, whether keys route identity depends on the prompt: it is a payload without a later mention and a key read with one (at hop 2 by an unscored contrast). Zhou (2026) finds that questions address facts by order of mention, compatible with (not tested by) the key-dominant binding swap. Our readers’ core is duplicate-token heads, not induction heads (Olsson et al., 2022); whether the read is token- or concept-level (Feucht et al., 2025) is open (Appendix F.4). Options before the context cannot attend to it (Ok & Lee, 2026); a list moved before the writing token opens no key read (E1). Steering is brittle to prompt changes (Tan et al., 2024); here behaviour stays stable while the channel flips.

Validity, format dependence and patching. Interchange interventions (Geiger et al., 2021; 2023) may act through dormant or planted pathways (Makelov et al., 2023; Wu et al., 2024; Liu & Chen, 2026), need not reproduce natural activations (Shih et al., 2026) and are not identifiable from behaviour (Venkatesh & Kurupath, 2026); diagnostics include control tasks (Hewitt & Liang,

2019), benchmarks (Huang et al., 2024; Arora et al., 2024; Mueller et al., 2025) and input partitions (Li et al., 2026). Format changes alter performance, function vectors and what a steering gain identifies (Sclar et al., 2023; Opiełka et al., 2026; Gao et al., 2026); refusal steering acts mainly through OV (Cheng et al., 2026), introspection through injected key and value changes (Zou et al., 2026). Patching (Vig et al., 2020; Wang et al., 2022) depends on metric and corruption (Zhang & Nanda, 2023; Heimersheim & Nanda, 2024); precomputed KV caches substitute exactly for a joint pass (Pustovit, 2026), as our joint clamp does.

6. Discussion

A practical rule, and a hypothesis. Which channel carries a written value is set by the tokens after the writing token, so a component-level claim should state where the candidates are mentioned after it, and be tested with and without a later mention; moving them before the writing token is a cheap control that keeps the words but removes the readers (Table 1). A mechanism localised on multiple-choice benchmarks (Mueller et al., 2025) may use another channel than the free-form behaviour it should explain; attributions freezing attention patterns may, we expect (untested), miss the key channel (cf. Kamath et al. 2025).

Limitations. (i) Three synthetic, templated tasks with single-token values; on IOI a later list’s key read is not established. (ii) The sentence effect was found post hoc, then preregistered (E1c; its readers D3). (iii) A single-channel clamp may be off-distribution. (iv) The fall in key share beyond 14B and (v) the small key identity under NO-MENTION and its negative sign under LIST-BEFORE are unexplained. (vi) The revisit covers two edits: the remap of Anonymous (2026) and the binding swap. (vii) Heads and hop 2 were tested at two 7B models in BF16, in one format; the readers at 24B and 72B are inferred, and (viii) the same-heads claim across formats is not evaluable (H6). (ix) The binding swap was tested at one model and flips no answer when candidates are named; (x) H11’s identity edit at block 0 nearly restates the natural clamp; (xi) edit kind and depth are confounded (Gate b3).

Conclusion. Across four model families and three templated tasks, later mentions of the candidates decide whether a written value’s identity is looked up through its key or mainly copied from its value. At two 7B models the lookup has two hops: heads with a duplicate-token core read the key at the re-mentioned words, and the answer reads those words. The released remap of Anonymous (2026) and its refit appear to act in the channel the format’s readers use; the preregistered extension to another lab’s binding edit failed.

Impact Statement

This work aims to make explanations of language-model internals more reliable, by showing that the component that appears to carry a value or an intervention can depend on the prompt under which it is measured. More reliable component-level explanations could help researchers audit and debug models, and could reduce the risk that a mechanism found under one evaluation format is trusted to explain behaviour under another. Our results also caution against over-reading interpretability findings: an explanation validated under one prompt format should not, on its own, be used to certify a model’s behaviour under others. Methods for locating and editing what a model reads from its context could in principle also be used to steer models in unwanted ways, although our experiments use synthetic tasks, and the interventions we study are a released learned remap, refits of it made with its authors’ released code, and a released state-token patch from another lab.

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A. Preregistration log and claim status

We made eight preregistrations, A–H. Each was committed to version control before its run, and each scoring script was committed before its outputs were inspected. H was finalised before any output of G was inspected (item 10 below). Tables 2 to 7 give every prediction and gate with its outcome as recorded in the preregistration log of the repository (docs/PREREGISTRATION.md); a NOT MET is listed as plainly as a MET. The anonymised code release adds PREREGISTRATION_HISTORY.txt, which lists, without author information, every commit that changed the log or a scoring script, with its time and diff, and every commit that changed the results, experiment code or GPU scripts, so that the order of preregistration, run and scoring can be checked; each result file also records the commit of the code that produced it. We keep the wording of the log, with two changes: format names are replaced by ours, and long entries are shortened without changing their content. In the log, the “critical token” or “state token” is the writing token, identity(K) is ID_K , and l_0 is ℓ_0 ; in preregistration H, “state token” keeps the meaning of Prakash et al. (2026), the token that names a state (our writing token in their stories). The log rounds independently of the tables in Appendix E, so a few values and intervals differ in the last digit (for example, the interval of the Qwen2.5-32B key share). The tables of G and H are generated from the two score files; where a value was recomputed from the raw files for the log, we print the recomputed value and say so.

Table 2. Preregistrations A and B. A: Qwen2.5-1.5B-Instruct in FP32 on CPU, seed 1 (new story cores), $n = 40$ per format, all items, keys clamped from layer 0, 95% core-bootstrap CIs. Predictions committed at 13c1f1f, run at 71e8489. B: GPU stage 1 on held-out models (Qwen2.5-3B/7B, Qwen3-8B, Mistral-7B-v0.3, OLMo-2-7B), $n = 150$ per format, all items, 95% core-bootstrap CIs. Predictions committed at 13c1f1f, code at f706ef5, scoring script analysis/stage1_prereg.py committed before inspection at 004579c.

	Prediction	Observed	Verdict
A1	identity(K) > 1.0 nats with a CI excluding 0, in OPTIONS-AFTER and in LETTERS-AFTER	OPTIONS-AFTER +2.85 [2.38, 3.33]; LETTERS-AFTER +4.10 [3.45, 4.77]	met
A2	identity(K) CI within the equivalence margin ± 1.0 in NO-MENTION, LIST-BEFORE and SENTENCE-AFTER	NO-MENTION +0.10 [−0.01, 0.21]; LIST-BEFORE −0.20 [−0.28, −0.13]; SENTENCE-AFTER −0.41 [−0.56, −0.26]	met
A3	Key share ($l_0 = 0$) in OPTIONS-AFTER within [0.25, 0.50]	0.38 [0.35, 0.41]	met
B1	identity(K) in OPTIONS-AFTER and LETTERS-AFTER positive, CI excluding 0, in at least 4 of 5 models	5/5 models	met
B2	identity(K)(OPTIONS-AFTER) − identity(K)(NO-MENTION) > 0, CI excluding 0, in at least 4 of 5 models	5/5 models	met
B3	Exploratory, no directional claim: how the OPTIONS-AFTER key share ($l_0 = 0$ and $l_0 = 0.0625L$) varies with size within Qwen2.5	Rises steadily with size: 1.5B 0.37 [0.36, 0.39]; 3B 0.69 [0.68, 0.71]; 7B 0.82 [0.81, 0.83]; 14B 0.91 [0.90, 0.91]. The depth-matched l_0 gives identical values. Recorded later, with the outcome of C: non-monotonic beyond 14B.	–

Notes recorded with outcome A. LIST-BEFORE and SENTENCE-AFTER are slightly but reliably negative. They sit inside the margin but are not exactly zero. The seed-0 LETTERS-AFTER interaction (+2.0 [0.3, 3.7]) did not replicate: seed 1 gives +0.6 [−1.0, 2.2]. This confirms the post hoc option-listing claim in the same model on new data; generalisation across models is preregistration B.

Unpredicted observations recorded with outcome B (reported, not reinterpreted). SENTENCE-AFTER carries large key identity in the larger models (Qwen2.5-7B +5.5, Qwen2.5-14B +12.1, Qwen3-8B +8.8, Mistral-7B +7.9, OLMo-2-7B +2.0 nats) but about 0 at Qwen2.5-1.5B/3B (−0.47 and −0.04). So the post hoc “listed answer options only” refinement from 1.5B does not generalise. LIST-BEFORE is at or below 0 in every model (−0.45 to +0.01). NO-MENTION carries small positive identity(K) in larger models (+1.0 to +1.25 nats at 7–14B), outside the CPU-era ± 1 margin, though under 5% of OPTIONS-AFTER.

Preregistration G: status of the preregistered alternatives. (a) H_{track} , context-selective heads at small scale: not declared (H_{diss} holds and G4b is met). (b) H_{replaced} : not met; neither it nor $H_{\text{redundant}}$ holds, so the preregistered outcome is a partial takeover. (c) Graded list-likeness: not supported overall (the arms that name B with two other candidates fall below NO-MENTION in both families, and lists read fully with two named candidates), but two features match it in part: in sentences at Qwen2.5-7B, r_2 is close to 2/6, and in S2 the rows after the question carry 0.46 / 0.49 / 0.38 of the splice effect. (d) Concept-level lookup: not established; the concept pattern holds on a_v in two models and for one variant (FRMIX at the two Qwen models), not in the three the decision table needs. (e) An inhibitory key read or an additive mixture at 7B:

Table 3. Preregistration C: GPU stage 2, the released bases of Anonymous (2026) at Mistral-Small-24B and Qwen2.5-72B across formats, and the natural factorial at 24–72B. Predictions and the scoring script `analysis/stage2_score.py` committed before any stage-2 run (5dc82f5); code at 2b1eca7. C4 is scored by `analysis/stage1_prereg.py` (committed at 004579c). Primary population: native story cores with B, S and T distinct; fits averaged within core; 95% core bootstrap. κ_V was estimated before the run from the fixed-value data released with Anonymous (2026). Here ψ is our ψ_K .

Prediction	Mistral-Small-24B	Qwen2.5-72B
Gate: in OPTIONS-AFTER, the argmax of M and of P agrees with the outputs released with Anonymous (2026) on ≥ 0.95 of (core, fit) items	M 0.994, P 1.000 ($n = 360$ each): passed	M 1.000, P 1.000: passed
C1, transfer law: $\varphi_{\text{NO-MENTION}}(\text{M})$ within ± 0.15 of κ_V	0.708 [0.681, 0.734] vs 0.53 ± 0.15 : not met	0.861 [0.837, 0.884] vs 0.73 ± 0.15 : met (at the edge)
C2: $\varphi_{\text{OPTIONS-AFTER}}(\text{M}) - \varphi_{\text{NO-MENTION}}(\text{M}) > 0$, paired CI excluding 0	$-0.003 [-0.019, +0.013]$: not met	$+0.034 [+0.018, +0.049]$: met (small)
C3: $\psi_{\text{NO-MENTION}} < 0.5 \cdot \psi_{\text{OPTIONS-AFTER}}$	0.069 vs 0.528: met	0.052 vs 0.633: met
Prediction (natural factorial)	Outcome	
C4: OPTIONS-AFTER key share ≥ 0.80 at Mistral-Small-24B, Qwen2.5-32B and Qwen2.5-72B	not met overall: Mistral-24B 0.78 [0.77, 0.79] no; Qwen2.5-32B 0.86 [0.84, 0.87] yes; Qwen2.5-72B 0.76 [0.75, 0.77] no.	
C4: LIST-BEFORE identity(K) ≤ 0.5 nats in all three	met in all three: $-0.53, -0.42, -1.99$.	

neither at Qwen2.5-7B; Mistral-7B not evaluable. The GPT-2 small parts of G18, G21c and G22 replicate a disclosed CPU pilot on fresh cores and are not counted as confirmatory. For G16 the scorer prints the label “near-token-level”; the entry says only “graded”, the label does not describe FRMIX at the Qwen models, and we do not adopt it.

Disclosures for preregistration G.

- *Re-finalisation.* The first GPU attempt stopped in a unit test before any model was loaded: one check compared batch rows with equal inputs at a tolerance of 10^{-5} , which batched matrix products on a many-core CPU can exceed. The entry was re-finalised with test-only changes (that check uses the 10^{-4} of the other batched checks, and the tests run on at most 16 CPU threads). No experiment, stimulus, scorer or threshold changed.
- *Disclosed pilots.* Before the draft, two CPU pilots of part (a) at Qwen2.5-0.5B already showed the H_{diss} pattern, so G2 is a confirmatory replication in sibling models. After the draft, a full GPT-2 small IOI run on CPU was scored while the code was built; the IOI run of the stage therefore uses fresh cores, and its GPT-2 small parts are replications.
- *Points the entry left open (no verdict depends on them).* In part (a), the emitted casing is the one with the larger clean-B candidate mass (it matters only at Qwen2.5-1.5B under SENTENCE-AFTER, where both casings pass the gate), and F is computed on the evaluation half while the key effects use all 150 cores. In part (b), the behaviour clause of H_{replaced} is applied per model in both list formats, the q_V thresholds are applied to point estimates, and G6’s value criterion uses cell means, which equal the paired means.
- *Eager against sdpa attention.* The key effects of Gate a come from the eager run of part (a). At Qwen2.5-7B they are lower than the sdpa values on the same cores, both on the lowercase answer ids (ID_K 4.50 against 5.52 under SENTENCE-AFTER, 19.39 against 21.16 under OPTIONS-AFTER; under SENTENCE-AFTER the casing the model emits gives 4.73, as in Table 22); the other models agree closely. The main text quotes sdpa values; the gate is far from its bound.
- *IOI at the 7B pair rests on one model.* Mistral-7B failed the competence gate in AFTER, BEFORE and IN-LINE-BEFORE, and its not-evaluable cells count as not met, so the not-met verdicts of G19–G21 reflect Qwen2.5-7B-Instruct together with that gate failure.
- *GPT-2 attention probe.* The exploratory GPT-2 small name-mover and duplicate-head probe failed on the GPU (a device mismatch) and was re-run afterwards on CPU with the same arguments, model revision, cores and code, in FP32 with eager attention, under a different torch and Python version. No gate or prediction uses it.

Table 4. Preregistrations D and E. D: GPU stage 3; predictions and the scoring script `analysis/stage3_score.py` committed before any stage-3 run (0f5cbff); run at b8815cb. New tasks: $n = 150$ per format in Qwen2.5-7B, Qwen2.5-14B, Qwen3-8B, Mistral-7B-v0.3 and OLMo-2-7B; localisation: belief task, $n = 60$, in Qwen2.5-7B, Qwen2.5-14B and Mistral-7B, the order of the three D3 values. E: GPU stage 3b; predictions and the scoring script `analysis/stage3b_score.py` committed before any stage-3b run, which ran at that commit (3ca63f1). E1 and E2 use Qwen2.5-7B, Qwen2.5-14B, Mistral-7B and OLMo-2-7B; E4 uses Mistral-Small-24B and Qwen2.5-72B.

	Prediction	Result	Verdict
D1	Paint: per model, OPTIONS-AFTER $ID_K > 0$ and OPTIONS-AFTER — NO-MENTION > 0 (CIs excluding 0); met if ≥ 4 of 5 models count	5/5 models (OPTIONS-AFTER $ID_K +6.5$ to $+42.7$ nats)	met
D1	Schedule: same	5/5 models (OPTIONS-AFTER $ID_K +4.5$ to $+28.5$ nats)	met
D2	LIST-BEFORE $ID_K \leq 0.5$ nats in ≥ 9 of the 10 task $\times$ model cells	10/10 task $\times$ model cells (max $+0.32$)	met
D3	OPTIONS-AFTER: choice-word rows recover ≥ 0.70 of the full key effect; question rows and the remaining tail rows each ≤ 0.15 ; in all 3 models	3/3 models: choice words 0.92 / 0.94 / 0.98; question 0.00; tail ≤ 0.01	met
D3	SENTENCE-AFTER, where the full key effect is positive: re-mention-word rows recover ≥ 0.50 in ≥ 2 of 3 models	3/3 models: 0.67 / 0.72 / 0.73	met
E1a	Instruction-matched 2×2 , every arm “Answer with one word”: $ID_K(\text{LIST-AFTER}) > 0$, CI excluding 0, in 4/4 models	$+8.7$ to $+35.8$ nats, 4/4	met
E1b	$ID_K(\text{LIST-BEFORE}) \leq 0.5$ and $ID_K(\text{SENTENCE-BEFORE}) \leq 0.5$, each in $\geq 3/4$ models	4/4 and 4/4 (max $+0.01$)	met
E1c	Paired LIST-AFTER — LIST-BEFORE > 0 in 4/4 models; paired SENTENCE-AFTER — SENTENCE-BEFORE > 0 in $\geq 3/4$; each CI excluding 0	4/4; 4/4	met
E2	Role control: for the direct question in OPTIONS-AFTER and NO-MENTION, $ f_K \leq 0.15$ and $ f_V \leq 0.15$ in $\geq 3/4$ models (Qwen2.5-72B exploratory)	4/4 and 4/4 (all fractions ≤ 0.02 ; also ≤ 0.01 at Qwen2.5-72B)	met
E3	Environment: each re-run within 0.03 of the original OPTIONS-AFTER key share (Qwen2.5-14B under transformers 5.9.0 on 2 GPUs, original 0.91; Qwen2.5-32B under transformers 5.18.0 on 1 GPU, original 0.86)	Qwen2.5-14B: 0.91 (original 0.91); Qwen2.5-32B: 0.86 (original 0.86)	met
E4	Value-only exchange in the frames of Anonymous (2026) : $\psi_V(\text{NO-MENTION}) \geq 0.5$ and $\psi_V(\text{NO-MENTION}) > \psi_K(\text{NO-MENTION})$ in both models	Mistral-24B 0.80 vs 0.07; Qwen-72B 0.90 vs 0.05	met

Exploratory parts of D and E, with no prediction: key shares on the new tasks; LETTERS-AFTER and SENTENCE-AFTER identity on the new tasks; layer-window and KV-group localisation at 7–14B; ψ_V and ρ_V in the other formats; the role control in LETTERS-AFTER and at Qwen2.5-72B. Before preregistration E, an exploratory CPU pilot of the role control was run at Qwen2.5-1.5B ($n = 40$; recorded in the log with E).

- *Combined lines.* The scorer prints a prediction as met only when every sub-verdict is met; the counts above use these combined lines. The sub-verdicts are: G10 lists met, sentences not met; G13 THE met, MODIF not met; G14 PLURAL, SYN, FRMIX and DEMIX met, TITLE and UPPER not met.
- *Checks.* The reproduction gate checks determinism: the reproduced cells are bitwise identical to stage 3b (recomputed). Five recomputations from the raw files, none importing the scorer, reproduce every gate value, point estimate and verdict.

Disclosures for preregistration H.

- *Timeline.* H was finalised before any output of G was inspected (item 10 below). Between the finalising commit and the commit of the results, no scorer, experiment, test or library file changed, and the text of entry H is byte-identical.
- *Gate a2 and the BF16 batch-shape offset.* Gate a2 compares rows of a batched pass with single passes. Its failure at Qwen2.5-7B under SENTENCE-AFTER ($|\text{none—clean}| = 0.770$ against 0.500) is a batch-shape offset, not a splice error

Table 5. **Preregistration F.** GPU stage 4: the remap of [Anonymous \(2026\)](#) refit at Mistral-Small-24B with their released training code, under NO-MENTION (fit_none) and under their own format (fit_opt, same-code control). Predictions, gates and the scoring script analysis/stage4_score.py committed before the run, which ran at that commit (1ef8696). Thresholds refer to point estimates except F3; every statistic is per run, with the three seeds averaged within core; 96 cores, 95% core bootstrap.

	Prediction or gate	Result	Verdict
G1	Each refit P spans the released P of the same seed: mean squared principal cosine ≥ 0.99	1.000 for all six (one P for all seeds)	met
G2	$\varphi \geq 0.5$ under each run’s fitting prompt	fit_none 0.81; fit_opt 0.72	met
G3	fit_opt reproduces the released crossover: $\psi_K(\text{LETTERS-AFTER}) \geq 0.5$, $\psi_V(\text{NO-MENTION}) \geq 0.5$	0.78, 0.81	met
F1	fit_none: $\psi_K(\text{LETTERS-AFTER}) \geq 0.5$ and $\psi_V(\text{NO-MENTION}) \geq 0.5$	0.77, 0.80	met
F2	fit_none: $\varphi(\text{LETTERS-AFTER}) \geq 0.5$	0.67	met
F3	$D = [\psi_K(\text{LETTERS-AFTER}) - \psi_K(\text{NO-MENTION})]_{\text{fit_none}} - [\cdot]_{\text{fit_opt}}$, paired over cores: CI within $[-0.25, +0.25]$	$-0.008 [-0.027, +0.009]$	met
F4	For each run, $\psi_K/(\psi_K + \psi_V)$ over the five formats correlates with the natural s_{ID} of Mistral-Small-24B at $r \geq 0.9$	fit_none 0.96; fit_opt 0.98	met

Stated alternative. If the fitting format decided which channel carries the remap, F1 and F2 would fail with $\psi_V(\text{LETTERS-AFTER}) \geq 0.5$ for fit_none and $D \leq -0.4$. This all-or-none pattern, which needs an edit that values can read and keys cannot, was not observed; the graded post hoc shift in OPTIONS-AFTER and SENTENCE-AFTER (Appendix D) is the more sensitive comparison. *Exploratory parts of F*, with no prediction: evaluation without the “Answer:” prefix; ρ_K and ρ_V ; principal cosines between the remaps; loss curves; the identity-objective fits that the training code makes alongside the remap.

(recomputed): rows with equal inputs agree within each batch, the reference rows of batches of different sizes differ under SENTENCE-AFTER but coincide under OPTIONS-AFTER, and the hop exactness row is 0.0000 in every story. Every scored quantity is a difference against an in-batch reference row; at Mistral-7B the gate values stay at or below 0.224. The rule still ties H5’s SENTENCE-AFTER line and H6 to the gate, and both stay not evaluable.

- *Narrow pass.* Qwen2.5-7B passes Gate a2 under OPTIONS-AFTER only narrowly: $|\text{all}_T - \text{full}| = 0.669$ against a bound of 0.719, which exceeds the 0.5-nat floor only because it scales with d_{full} (35.969 nats). Its bootstrap interval, [0.528, 0.821] (recomputed with the scorer’s resampling; the log reports an interval from an independent resampling), crosses the bound. The verdict uses the point estimate, as the entry specifies.
- *Null probabilities.* The entry quotes hypergeometric probabilities at the thresholds (overlap ≥ 3 of the top 10 for H4: 1.7×10^{-4} and 7.8×10^{-5} ; overlap ≥ 10 of the top 20 for H6: 1.3×10^{-12} and 9.3×10^{-14}). The scorer prints the probability of the observed overlap or more: 1.4×10^{-10} and 6.7×10^{-9} (H4), 9.2×10^{-19} and 9.9×10^{-23} (H6). Both are correct; they differ in conditioning.
- *Floors and pooling.* The H3 (d) floors of 2.7 and 2.8 nats are rounded from the stage-1 summaries, far below the observed rises. H4 (ii) pools T_{dup} over (story, word) pairs as a ratio of means, within the entry’s wording; per-story averaging gives 0.468 and 0.430 against 0.459 and 0.433 (recomputed). H5’s paired condition is checked as the lower bound of the paired difference.
- *No BOS token at Qwen2.5-7B.* The duplicate-score sequences of Qwen2.5-7B have no BOS token, so position 0 is both the attention sink and the first repeated query’s earlier copy. This can raise a head’s D by at most $1/30$, which cannot move the median below 0.2. Mistral-7B’s sequences have a BOS token.
- *Evaluability from the NO-MENTION sweep.* As the entry specifies, a format is evaluable in part (b) if the edit reproduces on the NO-MENTION sweep. BIND@28 and ID@28 are therefore evaluable under QNAMES and OPTIONS-AFTER, although neither flips an answer there; those cells decompose a shift of the logit margin (Appendix G.2).
- *Statistics, onsets and pairs.* κ (one-sided) and s_{ID} (symmetric) are different statistics by design; $\ell_{\text{ID}}^* = 0$ comes from the tie rule on a flat sweep; $\ell^* = 28$ rests on one pair; and the overlap of 70 of 80 pairs with the validation split of [Prakash et al. \(2026\)](#) is approximate, because it is computed with our filter, not theirs. Appendix G.2 gives the details and the depth confound of the identity-against-binding contrast (Gate b3).

Table 6. Preregistration G. GPU stage 5, scored by `analysis/stage5_score.py`. Predictions, scoring script and code were finalised together before any stage-5 GPU run; a first attempt stopped in the unit tests before any model was loaded, and the entry was re-finalised with test-only changes at 51e105e, the commit the run used. Parts (a)–(d): the 150 seed-0 story cores of stages 1 and 3b (the row splice and the form probe on the first 60); part (e): 200 fresh IOI cores. Triples are Qwen2.5-7B / Qwen2.5-14B / Mistral-7B (Instruct) unless the row says otherwise; G2 and G3 concern Qwen2.5-1.5B and 3B; part (d) adds OLMo-2-7B and is met if at least three evaluable models meet it. A prediction with sub-verdicts is not met unless every sub-verdict is met (the scorer’s rule); the Observed column gives the sub-verdicts. 95% core-bootstrap CIs. In all, 12 met and 10 not met; none was not evaluable. Details in Appendix F.

	Prediction	Observed	Verdict
G1	Anchors attend: $E \geq 0.20$ (lower > 0.10) and $F/E \geq 0.5$ under SENTENCE-AFTER at Qwen2.5-7B and 14B	E 0.869 [0.852, 0.885], F/E 1.00 (7B); E 0.915 [0.908, 0.921], F/E 0.98 (14B)	met
G2	Dissociation at 1.5B and 3B (H_{diss}): the same attention while ID_K (SENTENCE-AFTER) stays within ± 1 nat	E 0.789 [0.778, 0.802], F/E 1.03, ID_K (lowercase ids) -0.53 (1.5B); E 0.879 [0.866, 0.892], F/E 0.99, ID_K (lowercase ids) -0.02 (3B; 0.04 in the emitted casing)	met
G3	Magnitude under H_{diss} : $R_A \geq 0.5$ and $E(m) \geq 0.5 \times$ the smaller anchor E	R_A 1.05 / 1.06; $E(m)$ /anchor 0.91 / 1.01 (1.5B / 3B)	met
G4	Hop 2 (attention): (a) $G \geq 0.10$ (lower > 0.05) at the anchors; (b) $Q \geq 0.5$ (upper < 1) at 1.5B and 3B	(a) G 0.255 / 0.329; (b) Q 0.34 [0.29, 0.39] / 0.44 [0.37, 0.52]	met
G5	Cutting the candidate rows’ attention to the writing token (M1): $r_K \leq 0.20$ (upper ≤ 0.25) in the lists, ≤ 0.40 in SENTENCE-AFTER	LIST-AFTER 0.001 / -0.026 / -0.007 ; OPTIONS-AFTER 0.003 / -0.028 / -0.009 ; SENTENCE-AFTER -0.105 / -0.036 / 0.001	met
G6	Matched control column (M2, LIST-AFTER): r_K within [0.8, 1.2], small $ \Delta ID_V $, $ID_K(M1)-(M2) < 0$	r_K (M2) 1.094 / 0.997 / 0.990	met
G7	$H_{\text{redundant}}$: (a) the copy takes over under M1 in $\geq 2/3$ models in both list formats; (b) the answer is kept (acc_B , on $B \geq 0.90$) in 3/3	(a) 3/3: $q_V(M1)$ 0.390 / 0.440 / 0.448 (LIST-AFTER) against 0.391 / 0.335 / 0.433 under NO-MENTION; (b) 1/3: acc_B 0.37 / 1.00 / 0.86 (LIST-AFTER), 0.30 / 0.99 / 0.73 (OPTIONS-AFTER); partial takeover	not met
G8	Routes at the answer: (a) cutting answer $\rightarrow$ writing token (M3) under NO-MENTION leaves ≤ 0.6 of ID_V ; (b) $r_K(M3) \geq 0.8$ under LIST-AFTER; (c) M4 against M1	(a) 0.134 / 0.370 / 0.417; (b) 1.043 / 1.027 / 1.010; (c) 0.134 / 0.465 / 0.570	met
G9	Membership: naming the swapped candidates (S3, L3) reads the key; naming B with two others (S3out, L3out) does not	S3—NO-MENTION +2.44 / +8.47 / +4.70; L3—NO-MENTION +20.66 / +34.78 / +15.11; S3out—NO-MENTION -0.32 / -0.30 / -0.21	met
G10	Proportionality refuted, per family: lower bound of $r_2 > 1/3$ and of $r_3 > 1/2$	sentences r_2 0.28 [0.23, 0.32] / 0.58 / 0.46; lists r_2 0.96 / 0.98 / 0.98; lists met; sentences 2/3	not met
G11	The copy returns when S and X leave the list: $ID_V(L3out)-ID_V(L3) > 0$ and $R_V \geq 0.5$	R_V 0.11 / 0.03 / -0.08 ; 0/3	not met
G12	Reader rows (splice): $f_{\text{words}} \geq 0.5$ (lower > 0.25) in S2, S3, L2, L3	S2 0.40 / 0.38 / 0.39; S3 0.68 / 0.72 / 0.72; L2 0.91 / 0.94 / 0.92; L3 0.94 / 0.95 / 0.95; 9/12	not met
G13	Wrappers keep the read: $r_K \geq 0.75$ (lower > 0.5) for THE and MODIF, ≥ 3 of 4 models	THE 0.96–1.09; MODIF 0.32–0.64; THE met; MODIF not met	not met
G14	An exact repeat reads most: $r_K \leq 0.75$ (upper < 1) for each of six variants	TITLE, UPPER 0.83–1.14 (0/4 each); PLURAL 0.11–0.49; SYN, FRMIX, DEMIX ≤ 0.15 (4/4 each); 4/6 sub-verdicts	not met
G15	Token-level on the English measure: $r_K \leq 1/3$ and $\rho_s \leq 0.5$ for SYN, FRMIX, DEMIX	r_K -0.13 to 0.15 ; ρ_s 0.09–0.33	met
G16	Token pattern on r_K^{any} and a_v in ≥ 3 models, per variant	SYN Q7B t/g, Q14B t/g, M7B t/g, O7B t/t; FRMIX Q7B c/c, Q14B c/c, M7B g/g, O7B g/g; DEMIX Q7B c/g, Q14B c/g, M7B t/t, O7B t/t (t token, c concept, g graded; r^{any}/a_v); graded	not met
G17	Value compensation: paired $ID_V(v) - ID_V(\text{SENTENCE-AFTER}) > 0$ where the read is lost	FRMIX +2.58 / +1.37 / +0.31; DEMIX +2.02 / +1.13 / +0.39 (Qwen2.5-7B / Mistral-7B / OLMo-2-7B)	met
G18	No key read in plain IOI: $f_K(\text{PLAIN})$ within ± 0.10 (CI within ± 0.20), 3/3	-0.02 / $+0.02$ / $+0.00$ (GPT-2 small / Qwen2.5-7B / Mistral-7B)	met
G19	A later list opens a key read on IOI: (a) existence, (b) positive sign, 2/2	Qwen2.5-7B $ID_K(\text{AFTER})$ +7.05 [6.26, 7.81]; Mistral-7B not evaluable (Gate e); 1/2	not met
G20	The lookup replaces the copy on IOI: $f_V(\text{AFTER}) \leq 0.5$, $f_V(\text{QUESTION}) \geq 0.75$	Qwen2.5-7B f_V 0.45 (AFTER), 0.69 (QUESTION); Mistral-7B not evaluable; 0/2	not met
G21	Controls: (a) BEFORE $ ID_K \leq 0.5$; (b) QUESTION; (c) GPT-2 small INLINE_BEFORE $ ID_K \leq 0.5$	(a) -2.02 (Qwen2.5-7B); (b) met at Qwen2.5-7B, Mistral-7B not evaluable; (c) -0.25 ; (c) met	not met
G22	GPT-2 small in-sentence re-mention: (a) a key read, (b) inhibitory (replication of a disclosed pilot)	$ID_K(\text{INLINE})$ -2.35 [-2.47 , -2.23]	met

Table 7. Preregistration H. GPU stage 6, scored by `analysis/stage6_score.py`. Predictions, gates, scoring script and code were finalised at `cc3a3e0`, before any stage-6 run and before any output of G was inspected; the run used that commit. Part (a), gates a1–a3 and H1–H6: Qwen2.5-7B / Mistral-7B (Instruct), OPTIONS-AFTER unless stated, BF16, eager attention, 60 ranking and 60 held-out evaluation stories, $k^* = 5\%$ of heads; a prediction over both models is met only if it is met in both. Part (b), gates b0–b3 and H7–H12: Qwen2.5-14B-Instruct on the material of Prakash et al. (2026), $n = 150$ pairs, BF16; triples are NO-MENTION / QNAMES / OPTIONS-AFTER; $\ell^* = 28$, $\ell_{ID}^* = 0$. met*: met except in the cell named in the Observed column. 95% bootstrap CIs (10,000 resamples). In all, 6 met (H1–H5, H11), 4 not met (H7–H10), 1 not evaluable (H6) and 1 not run (H12). Details in Appendix G.

	Gate or prediction	Observed	Verdict
a1	FP32 exactness of the head and hop splices (unit tests) before any model	7 passed, 0 failed	met
a2	BF16 floor per model and format: $ \text{none} - \text{clean} $ and $ \text{all}_T - \text{full} \leq \max(0.5, 0.02 d_{\text{full}})$; hop exactness ≤ 0.1	Qwen2.5-7B OPTIONS-AFTER 0.606, $0.669 \leq 0.719$; SENTENCE-AFTER 0.770 > 0.500 ; Mistral-7B ≤ 0.224 in both; hop 0.0000; not met: Qwen2.5-7B SENTENCE-AFTER	met*
a3	$\bar{d}_{\text{full}} \geq 10$ (OPTIONS-AFTER), > 0 (SENTENCE-AFTER); $d_G/d_{\text{full}} \geq 0.8$	35.969 / 26.469; 10.438 / 12.254; 0.926 / 0.982	met
H1	Sparsity: $R(k^*) \geq 0.8$, lower bound ≥ 0.7 (top 5% by a_3)	0.967 [0.952, 0.982] / 0.942 [0.919, 0.961]	met
H2	Necessity and specificity: $KO(k^*) \geq 0.8$; random sets R , $KO \leq 0.25$	KO 0.977 / 0.971; random R 0.004 / 0.007, KO 0.011 / 0.010	met
H3	Ablation: $\rho_K \leq 0.5$; candidate mass kept; control sets $\rho_K \geq 0.75$; ID_V rises above the floor; answer kept ≥ 0.8	ρ_K 0.238 / 0.059; $\Delta ID_V +6.907$ / $+10.861$; kept 0.87 / 1.00	met
H4	Canonical duplicate-token heads over C : median $D \geq 0.2$, median $I \leq 0.1$; median $T_{\text{dup}} \geq 0.2$	D 0.370 / 0.232; I 0.011 / 0.022; T_{dup} 0.459 / 0.433	met
H5	Second hop: $r_{\text{ans}}(KV) \geq 0.5$; $r_{\text{ans}} > r_{\text{other}}$; $r_{\text{all}} \geq 0.8$	0.899 / 0.859; r_{other} 0.118 / 0.165; secondary (not scored) $r_{\text{ans}}(K)$ 0.797 / 0.683, $r_{\text{ans}}(V)$ 0.080 / 0.023; SENTENCE-AFTER line not evaluable	met
H6	The same readers for list and sentence: top-20 overlap ≥ 10	13 (Gate a2 failed) / 14	not evaluable
b0	Exchange exactness per cell: $ m(r_4) - m(r_1) $, $ m(r_0) - m(B) \leq 0.3$ nats	0.099–0.194; 0.141–0.203 (13 cells)	met
b1	Reproduction (NO-MENTION sweep): $IIA \geq 0.7$ at ℓ^* (BIND) and ℓ_{ID}^* (ID)	BIND 0.993 at $\ell^* = 28$; ID 1.000 at $\ell_{ID}^* = 0$	met
b2	Effect size $\Phi \geq 3$ nats per cell	met in 11 of 13 cells; LETTERS-AFTER at depth 28: 0.225, 0.034; not met: LETTERS-AFTER at 28	met*
b3	For H10 as a dissociation: identity key-read route at $\ell^* + 1$ ($s_{ID} \geq 0.5$) and $\kappa_{ID}(\ell^*) \geq 0.5$ under OPTIONS-AFTER	s_{ID} -0.015 ; κ_{ID} 0.006	not met
H7	The law on the binding swap: $ \kappa(f) - s_{ID}(f, 29) \leq 0.25$ for each evaluable f and $r \geq 0.9$	gaps 0.601 / 0.934 / 0.879; $r = -0.986$	not met
H8	Under NO-MENTION: $\psi_V \geq 0.5$, $\psi_K \leq 0.25$	ψ_V 0.435, ψ_K 0.703; $\psi_V - \psi_K$ -0.268 [-0.317 , -0.219]	not met
H9	Crossover: $\kappa(\text{OPTIONS-AFTER}) - \kappa(\text{NO-MENTION}) \geq 0.4$; QNAMES between	$+0.247$ [0.211, 0.281]; $\kappa(\text{QNAMES})$ 0.906 not between	not met
H10	The rival: $\kappa(f) \leq 0.25$ in every evaluable format	κ 0.618 / 0.906 / 0.864	not met
H11	Identity edit: Part 1 gaps to $s_{ID}(f, 1) \leq 0.25$ and κ_{ID} crossover ≥ 0.4 ; Part 2 gaps at depth 28	Part 1 gaps 0.114 / 0.048 / 0.077, crossover $+0.648$ [0.630, 0.666]; Part 2 gaps 0.113 / 0.036 / 0.020	met
H12	Optional replication at Llama-3-70B	not run (Gate b1 passed at 14B)	not run

Status of the preregistered alternatives. Part (a): a read distributed beyond 5% of heads (H1, H2 fail), task-tuned lookup heads instead of generic duplicate-token heads (H4 (i) fails), a three-hop route through the instruction and template rows (H5 fails with $r_{\text{all}} \geq 0.8$) and no head-level trade-off between the key and value routes (H3 (d) fails, as in the 0.5B pilot): none is supported, although the H4 margin at Mistral-7B depends on the size of the head set (Appendix G.1). Part (b): H10, the rival, is not met, and Gate b3 failed, so the reading “keys carry identity, values carry binding” is not available in either direction (“not evaluable as a dissociation at this depth”). The flat-high pattern, $\kappa \geq 0.75$ in every evaluable format, reported beside H10 without a verdict, does not hold either (NO-MENTION: 0.618). No evaluable cell is interaction-carried. H5’s SENTENCE-AFTER line, reported against $r_{\text{ans}}(KV) \geq 0.35$ and $r_{\text{all}} \geq 0.5$ and not scored, holds at both models but is not evaluable at Qwen2.5-7B (Gate a2), and so not evaluable overall.

Table 8. The 14 failed predictions of preregistrations G and H, by kind (Section 4.3). Values as in Tables 6 and 7; Qwen2.5-7B unless stated.

	Predicted	Observed	Kind
G20	On IOI, the lookup replaces the copy (2/2 of the 7B pair)	every clause met except the QUESTION baseline, $f_V = 0.69$ against ≥ 0.75 ; Mistral-7B not evaluable	direction kept, threshold missed
H9	$\kappa(\text{OPTIONS-AFTER}) - \kappa(\text{NO-MENTION}) \geq 0.4$ on the binding swap, $\kappa(\text{QNames})$ between	$+0.247$ [0.211, 0.281]; $\kappa(\text{QNames}) = 0.906$ is not between	direction kept, threshold missed
G7	With the lookup cut, the copy takes over and the answer stays (3/3)	copy returns 3/3; answer kept 1/3 (Qwen2.5-7B $\text{acc}_B = 0.37$); preregistered “partial takeover”	met in part
G10	The dose effect is more than proportional, per family	lists 3/3; sentences 2/3 ($r_2 = 0.28$)	met in part
G12	The named words carry ≥ 0.5 of the read in S2, S3, L2, L3	9/12 cells; S2 fails (0.38–0.40)	met in part
G13	Wrappers keep the read (THE, MODIF)	THE 0.96–1.09; MODIF 0.32–0.64	met in part
G14	An exact repeat reads most (six variants)	four of six; case variants read nearly fully (r_K 0.83–1.14)	met in part
G21	IOI controls: list before, question wrapper, GPT-2 inline-before	question and GPT-2 met; list before gives $\text{ID}_K = -2.02$, not about 0; Mistral-7B not evaluable	met in part
G19	On IOI, a later list opens a positive key read (2/2)	met at Qwen2.5-7B (+7.05 nats); Mistral-7B failed the competence gate (two-way 0.59)	competence gate
H10	Rival: values carry the binding swap ($\kappa \leq 0.25$)	$\kappa = 0.618, 0.906, 0.864$	rival rejected
G11	Without the readers in the list, the copy carries the identity again ($R_V \geq 0.5$)	$R_V \leq 0.11$ in every model	falsified
G16	A token pattern on both measures for synonyms and translations	graded, none of the three preregistered outcomes; the French-mixed variant shows the concept pattern at Qwen2.5-7B and 14B	falsified
H7	$\kappa \approx s_{\text{ID}}$ on the binding swap, depth-matched (≤ 0.25)	gaps 0.60–0.93	falsified
H8	Without a later mention values carry the swap ($\psi_V \geq 0.5, \psi_K \leq 0.25$)	$\psi_V = 0.435, \psi_K = 0.703$	falsified

- *FP32 re-check.* The command of the exploratory FP32 re-check passes the dtype option twice (BF16, then FP32); the last value applies, and both output files record FP32. The scorer skips these files by design, so their values are recomputed.
- *Disclosed pilots.* At Qwen2.5-0.5B (FP32), the top heads by a_3 were duplicate-token heads, consistent with H4 at 7B. The same pilot showed no head-level trade-off and a weak answer-only hop on two stories; at 7B, H3 (d) and H5 are met, unlike the pilot. The release of [Prakash et al. \(2026\)](#) gives per-layer IIA at Qwen2.5-14B that predicted $\ell^* = 27-28$, and $\ell^* = 28$, so the replication table compares against material seen before the entry was finalised: it is a reproduction check, not an independent prediction.
- *H12 not run.* By a rule agreed before the run, for reasons of budget, the optional Llama-3-70B run was to be made only if Qwen2.5-14B failed to reproduce the edit (Gate b1). Gate b1 passed (IIA 0.993), and the run was not made. It can still be scored against the fixed 14B verdicts; with none of H7, H9 and H10 met, it would test only Gate b1, H8 and H11 Part 1.
- *Cosmetic.* For the one cell that fails the κ rule (ID@28 LETTERS-AFTER), the scorer does not print the fraction of dropped resamples, 55.0% (recomputed); the cell enters no verdict.
- *Checks.* Four recomputations from the raw files, none importing the scorer, reproduce every gate value, point estimate and verdict, and the committed scorer reproduces the score file byte for byte.

History of the claim, including post hoc steps. The account of when keys carry the value was revised twice after seeing data, and its intervention half was narrowed after a preregistered failure (item 10). We list every step.

1. *Starting hypothesis.* Before the first format factorial, our hypothesis was that any re-mention of the candidates after the writing token opens the key channel.
2. *Post hoc narrowing at 1.5B.* In the first run at Qwen2.5-1.5B (seed 0, $n = 40$), SENTENCE-AFTER showed no key effect. We therefore narrowed the claim to “listed answer options after the writing token open the key channel”. This narrowing was made after seeing the data, and an internal go/no-go gate written before that run was not met as written.
3. *Preregistration A* tested the narrowed claim on fresh story cores in the same model, and all three predictions were met. This confirms the narrowed claim in one model on new data, not across models.
4. *Preregistration B* tested held-out models. Both directional predictions were met in 5/5 models, but from 7B up SENTENCE-AFTER carries an intermediate amount of key identity, so the “listed options only” account does not generalise. The account we now give, that from 7B up a later mention of the candidates opens the key channel, whether a list or a neutral sentence (a list most strongly), while at 1.5B and 3B only a list does, was revised again, post hoc, after stage 1. It is closer to the starting hypothesis. The residual NO-MENTION identity in the Qwen models from 7B to 32B (up to 1.25 nats) extends just beyond the ± 1 -nat margin that A set at 1.5B; in the other models it stays inside. We report it as unpredicted, not as a confirmation.
5. *Preregistration C* added a transfer law derived from the fixed-value data of [Anonymous \(2026\)](#). It failed at Mistral-Small-24B (C1 and C2 not met) and held only marginally at Qwen2.5-72B. C3 was met in both models. The key-share threshold of C4 was met only at Qwen2.5-32B; its LIST-BEFORE control was met in all three models. The reading that the intervention is written into both channels, and that the format decides which one appears to carry it, was formed after seeing these data.
6. *Preregistration D* retested the options-list part of the account on two new tasks (D1, D2) and localised the readers at 7–14B (D3). It contains no test of whether SENTENCE-AFTER opens the key channel; SENTENCE-AFTER on the new tasks was exploratory.
7. *Preregistration E* was written in response to two internal reviews of a draft. It tested the revised account with an instruction-matched 2×2 (E1), including the first confirmatory test that a sentence after the story opens the key channel (E1c, SENTENCE-AFTER — SENTENCE-BEFORE). It also tested the value half of the post hoc reading of C (E4), the role control (E2) and the software environment (E3). All its predictions were met.
8. *Preregistration F* tested the other half of that reading: whether the format in which a remap is read, rather than the one in which it was fit, decides which channel appears to carry it, by refitting the remap without later mentions at Mistral-Small-24B. All gates and predictions were met; a smaller shift of the key share in the two mixed formats, in the direction the alternative predicts, was found post hoc.

9. *Preregistration G* was written in response to an external critique that expected the objection “duplicate-token attention in a new setting”. It conceded the first hop: the re-mentions attend to the writing token with a key-matched duplicate-token pattern at every scale, including 1.5B and 3B, whose key identity under SENTENCE-AFTER is about zero (G1–G3; with option lists it is positive), so the match is present without the read. The candidate words’ attention to the writing token proved necessary for the key read, and the answer’s own attention to it unnecessary (G5, G6, G8). A lead we had planned before the run, that the lookup largely replaces the value copy, was not supported (G7 a partial takeover; G11 and G20 not met) and was dropped. Three claims were narrowed: the readers to lists and to sentences naming at least three candidates (G12), the matched form to the word largely up to case (G13, G14), and the tasks to our three templated ones, since IOI did not confirm a later list’s key read (G19–G21).
10. *Preregistration H* identified the reader heads and the second hop at two 7B models (H1–H5 met) and applied the exchange to an intervention from another lab. On the binding swap of [Prakash et al. \(2026\)](#), the law that predicts an edit’s key share from the natural identity key share (H7), the shared value prediction (H8), the crossover (H9) and the rival (H10) all failed. The identity-edit control was met (H11), but at $\ell_{ID}^* = 0$ it nearly restates the natural clamp, so the law for identity edits out of sample is untested; the in-sample relation for the released remaps and the refits (F4) is unchanged. H was finalised (commit `cc3a3e0`, 01:16 UTC) before any output of G was inspected: the stage-5 run had started at 21:25 UTC the day before, its results were first committed at 08:41 UTC, after H, and the stage-6 run started at 10:57 UTC on `cc3a3e0`, unchanged. The reading that at block 28 the binding swap is key-dominant while the identity edit is value-carried (Section 4.2) was formed after these data and is post hoc.

Tables 9 to 11 list each main claim with its status. “Confirmatory” means that a preregistered prediction tested the claim in the stated models and was met. “Exploratory” means that the measurement was planned or run without a prediction. “Post hoc” means that the claim was formed after seeing the data it rests on. A claim whose preregistered test failed is listed with that failure.

Table 9. **Status of each main claim** (Section 3). Section of the main text, claim, and whether it is confirmatory (with the prediction that tests it), exploratory or post hoc. Continued in Tables 10 and 11.

Section	Claim	Status
Section 3.1	Candidates listed after the writing token (OPTIONS-AFTER, LETTERS-AFTER) give a large positive ID_K	Confirmatory at 1.5B (A1), in five held-out 3–8B models (B1) and, for LIST-AFTER, in four 7–14B models (E1a); exploratory at Qwen2.5-14B under OPTIONS-AFTER and LETTERS-AFTER and at 24–72B
Section 3.1	Without a later mention (NO-MENTION), keys carry little identity	Confirmatory as a paired contrast with OPTIONS-AFTER (B2) and at 1.5B (A2); the residual in the Qwen models from 7B to 32B (up to 1.25 nats), just beyond A’s ± 1 -nat margin, is unpredicted
Section 3.1	Without a later mention, the identity is mainly copied through the value channel ($ID_V \gg ID_K$), through the answer’s attention to the writing token	Exploratory (Table 15); that cutting the answer’s attention to the writing token removes most of ID_V is confirmatory (G8a, met in 3/3 at 7–14B), but the cut also removes most of the token’s whole effect
Section 3.1	Candidates before the writing token (LIST-BEFORE, SENTENCE-BEFORE) carry no positive key identity ($ID_K \leq 0.5$ nats in every model and task tested; ≤ 0 in most models on the belief task, down to -2.0 nats at Qwen2.5-72B)	Confirmatory: A2 at 1.5B (within ± 1 nat), and the ≤ 0.5 -nat bound in C4 second part (24–72B), D2 (new tasks) and E1b (7–14B); the negative sign was not predicted and is unexplained
Section 3.1	SENTENCE-AFTER carries intermediate key identity from 7B up, and about zero key identity at 1.5B and 3B	Post hoc (observed in B); the 1.5B null was predicted by A2 (equivalence margin, met), and the positive part from 7B up was then confirmed in four 7–14B models by E1c (SENTENCE-AFTER – SENTENCE-BEFORE > 0)
Section 3.1	With the instruction matched, moving a list or a sentence from before to after the writing token switches the key read on	Confirmatory in four 7–14B models (E1a–c); that a list reads more than a sentence, and that LIST-AFTER is close to OPTIONS-AFTER, are exploratory
Section 3.1	Key and value effects add up, with a small interaction	Exploratory (Table 15)
Section 3.2	The pattern holds on the paint and schedule tasks	Confirmatory for OPTIONS-AFTER, the OPTIONS-AFTER – NO-MENTION contrast and LIST-BEFORE (D1, D2); exploratory for SENTENCE-AFTER, LETTERS-AFTER and the key share
Section 3.2	Plain IOI carries almost no key identity; that a later list opens a key read on IOI is not established	G18 met (3/3); G19, G20 and G21 not met (Mistral-7B not evaluable by its competence gate; G20 fails on its baseline clause; the list before the sentence gives a negative read). A positive key read at Qwen2.5-7B-Instruct is a per-model observation under the failed G19
Section 3.2	Within Qwen2.5, the OPTIONS-AFTER key share rises with size to 14B, then falls	Exploratory (B3; Figure 4); the threshold $s_K \geq 0.80$ failed (C4)
Section 3.2	The fall beyond 14B is not a software-environment artefact	Confirmatory (E3)
Section 3.2	The key read is not only the input embedding	Exploratory (onset sweep, Table 16)

Table 10. Status of each main claim (Section 3), continued from Table 9; continued in Table 11.

Section	Claim	Status
Section 3.3	The later-mention words read the key at 7–14B, in lists and in sentences naming at least three candidates	Confirmatory for OPTIONS-AFTER and SENTENCE-AFTER in three 7–14B models (D3); exploratory for LETTERS-AFTER. In a sentence naming only S and X the named words carry only 0.38–0.40 of the splice effect, most of it not identity-specific (G12, not met, 9/12)
Section 3.3	Cutting the candidate words’ attention to the writing token removes the key read; a matched control column does not; the answer needs no direct attention to the writing token	Confirmatory: G5, G6 and G8b met (3/3; Qwen2.5-7B, Qwen2.5-14B, Mistral-7B)
Section 3.3	The read needs the swapped candidates themselves among the later mentions; in lists two suffice, in sentences it grows with the number named	Confirmatory: G9 met (3/3 in both families, replicated at seed 1); G10 met for lists, not met for sentences (2/3)
Section 3.3	At 1.5B, option words read the key in layers 4–15, one KV group dominant	Exploratory (no prediction)
Section 3.3	The read is bound to the word form on our English measure, largely up to case; a modified noun phrase or a plural reads only partly; token against concept is model-dependent	G15 met (4/4 per variant); G13 not met (THE met, MODIF not met); G14 not met (TITLE, UPPER); G16 not met, no preregistered outcome holds (per-model profile exploratory); G17 met
Section 3.4	About 5% of heads, ranked on held-out stories by attention change, are sufficient and necessary for the option rows’ key read; random sets of equal size are not	Confirmatory: H1, H2 met (Qwen2.5-7B, Mistral-7B, OPTIONS-AFTER)
Section 3.4	The read is a joint property of heads in many layers; ranking by single-head effect f^+ does not find the set, and leave-one-out d^- finds it only at Mistral-7B	Exploratory (Appendix G.1)
Section 3.4	Ablating the reader heads removes most of the key identity, the value identity rises, and the answer is kept	Confirmatory: H3 met (2/2). Edge-level counterpart: G7 not met (G7a 3/3, G7b 1/3; preregistered “partial takeover”); the two interventions differ and are not reconciled
Section 3.4	The answer then reads the later mentions	Confirmatory at Qwen2.5-7B and Mistral-7B under OPTIONS-AFTER (H5 met); that it reads them through their keys is H5’s secondary contrast, reported, not scored; the SENTENCE-AFTER line is not evaluable (Gate a2); the cross-scale difference at hop 2 is attention evidence only (G4)
Section 3.5	The re-mentions attend to the writing token with a duplicate-token pattern at every scale, including 1.5B and 3B, whose key identity under SENTENCE-AFTER is about zero	Confirmatory: G1, G2 (H_{diss}) and G3 met; G2 replicates disclosed 0.5B pilots
Section 3.5	The reader heads’ core is canonical duplicate-token heads	Confirmatory: H4 met (2/2); that the full 5% set is mixed is exploratory
Section 3.5	The same heads read a list and a sentence	Not evaluable (H6; Gate a2 failed at Qwen2.5-7B under SENTENCE-AFTER); met at Mistral-7B, exploratory otherwise
Section 3.5	The writing token’s channels carry identity, not belief role	Confirmatory for OPTIONS-AFTER and NO-MENTION in four models (E2); exploratory for LETTERS-AFTER and Qwen2.5-72B

Table 11. Status of each main claim (Section 4), continued from Table 10.

Section	Claim	Status
Section 4	Reproduction of Anonymous (2026) under OPTIONS-AFTER	Confirmatory gate (C), passed
Section 4	ψ_K is much smaller without a later mention	Confirmatory (C3)
Section 4	Without a later mention, values carry the intervention	Confirmatory (E4)
Section 4	φ varies little across formats (0.70–0.75 at Mistral-Small-24B, 0.86–0.91 at Qwen2.5-72B); the full crossover of ψ_K and ψ_V over five formats	Exploratory for the released remaps (the refit test is F); the one preregistered contrast (C2, OPTIONS-AFTER – NO-MENTION) was null at Mistral-Small-24B and small but positive at Qwen2.5-72B (+0.034 [+0.018, +0.049])
Section 4	An edit that changes which value is written is present in both channels; the format decides which appears to carry it	Post hoc reading of C; then confirmed for its value half (E4) and for a remap fit without later mentions (F1–F4); on the material of Prakash et al. (2026) shown only for an identity edit at block 0 (H11, met but near-tautological)
Section 4	The released remaps’ key share tracks the natural identity key share ($r = 0.98$)	Post hoc, exploratory; preregistered only for the refits (F4). Out of sample it fails for a binding edit (H7); the identity-edit control (H11) is met but nearly tautological at $\ell_{ID}^* = 0$, so the out-of-sample law for identity edits is untested
Section 4.1	A remap fit under NO-MENTION shows the same crossover as one fit under OPTIONS-AFTER: the format in which it is read decides which channel dominates	Confirmatory at Mistral-Small-24B (F1–F4)
Section 4.1	The fitting format shifts the key share in OPTIONS-AFTER and SENTENCE-AFTER, and transfer under NO-MENTION and LETTERS-AFTER	Post hoc (key share) and exploratory (φ)
Section 4.2	The state-token binding swap of Prakash et al. (2026) replicates at Qwen2.5-14B	Gate b1 met; the per-layer comparison with their release is exploratory (recomputed)
Section 4.2	Their binding swap is key-dominant in each of the three evaluable formats ($\psi_V = 0.435$ under NO-MENTION); a later mention shifts κ by +0.247 [+0.211, +0.281], below the preregistered 0.4	Exploratory description; the preregistered H8, H9 and H10 were not met. LETTERS-AFTER failed Gate b2; under QNAMES and OPTIONS-AFTER the swap flips 0/150 answers, so those cells decompose a margin shift
Section 4.2	The remap-share law ($\kappa \approx s_{ID}$) predicts a new intervention’s key share	Confirmatory test failed (H7). The identity-edit control H11 is met but near-tautological at $\ell_{ID}^* = 0$, so the law for identity edits out of sample is untested
Section 4.2	At block 28 the binding swap is key-dominant and the identity edit value-carried	Post hoc, exploratory (not preregistered; Gate b3, built for the reverse reading, not met). Both flip no answer under QNAMES and OPTIONS-AFTER. One reading consistent with H7’s failure, not an explanation of it; confounded with depth
Section 4.3	The transfer law from clamping-estimated κ_V fails	Confirmatory test, failed at Mistral-Small-24B (C1, C2) and marginal at Qwen2.5-72B

B. Prompts

Belief task. The story generator (`ckeys/story.py`) is ported from the released generator of [Anonymous \(2026\)](#). Each story core draws an agent and an other agent (Alice, Bob), an object and a distractor object from twenty objects, and four locations from the six candidates *box, basket, shelf, drawer, cabinet, closet*. The initial location, the base location and the source location must differ, and the distractor’s location must differ from the base and source locations. The two “initially sees” sentences are sorted by object name. The direct question is “Where does {*A*} believe the {*o*} is?”. Base B and source S differ only at the writing token, the word after “moved to the”. The third value X used for ID_K and ID_V is drawn, with a seed fixed by the core, from the candidates other than the base, source, initial and distractor locations, so it appears nowhere in the story. We keep a core only if the B, S and X prompts have the same token length and differ at exactly one token.

We show the first core generated with seed 0: agent Alice, other agent Bob, object *lamp*, distractor *coin*, initial location *basket* (the distractor is also in the basket), base location *cabinet*, source location *drawer*, third value *box*. Below are the raw user-turn prompts of the base run in all seven formats, as built by `ckeys/encoding.py`. Line breaks inside the story are added here for width; in the prompts the story is one line.

OPTIONS-AFTER (the format of the key-exchange experiments of [Anonymous 2026](#)).

```
Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone
initially sees that the lamp is in the basket. Alice watches as the lamp is
moved to the cabinet. Bob does not see this happen.
Question: Where does Alice believe the lamp is?
Choices: box, basket, shelf, drawer, cabinet, closet
Answer with exactly one choice.
Answer:
```

LIST-AFTER. As OPTIONS-AFTER, with the last instruction line replaced by “Answer with one word.”.

```
...
Question: Where does Alice believe the lamp is?
Choices: box, basket, shelf, drawer, cabinet, closet
Answer with one word.
Answer:
```

LETTERS-AFTER.

```
Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone
initially sees that the lamp is in the basket. Alice watches as the lamp is
moved to the cabinet. Bob does not see this happen.
Question: Where does Alice believe the lamp is?
Choices: A) box, B) basket, C) shelf, D) drawer, E) cabinet, F) closet
Answer with one letter.
Answer:
```

SENTENCE-AFTER.

```
Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone
initially sees that the lamp is in the basket. Alice watches as the lamp is
moved to the cabinet. Bob does not see this happen. The room has a box, a
basket, a shelf, a drawer, a cabinet and a closet.
Question: Where does Alice believe the lamp is?
Answer with one word.
Answer:
```

SENTENCE-BEFORE.

1265 Read the story and answer the question.

1266
1267 Story: The room has a box, a basket, a shelf, a drawer, a cabinet and a
1268 closet. Everyone initially sees that the coin is in the basket. Everyone
1269 initially sees that the lamp is in the basket. Alice watches as the lamp is
1270 moved to the cabinet. Bob does not see this happen.

1270 Question: Where does Alice believe the lamp is?

1271 Answer with one word.

1272 Answer:

1273
1274 **NO-MENTION.**

1275 Read the story and answer the question.

1276
1277 Story: Everyone initially sees that the coin is in the basket. Everyone
1278 initially sees that the lamp is in the basket. Alice watches as the lamp is
1279 moved to the cabinet. Bob does not see this happen.

1280 Question: Where does Alice believe the lamp is?

1281 Answer with one word.

1282 Answer:

1283
1284 **LIST-BEFORE.**

1285 Read the story and answer the question.

1286 Choices: box, basket, shelf, drawer, cabinet, closet

1287 Story: Everyone initially sees that the coin is in the basket. Everyone
1288 initially sees that the lamp is in the basket. Alice watches as the lamp is
1289 moved to the cabinet. Bob does not see this happen.

1290 Question: Where does Alice believe the lamp is?

1291 Answer with one word.

1292 Answer:

1293 **Chat wrapper and scoring.** Each raw prompt is wrapped as in the `Answer:-prefill` interface of [Anonymous \(2026\)](#)
1294 and its released encoder: a system turn “You are a helpful assistant.”, a user turn holding the raw prompt, the model’s
1295 generation prompt, and the assistant prefill “Answer:”. We also disable thinking (`enable_thinking=False`, which
1296 matters for Qwen3 and is ignored by other templates). The raw prompt itself ends in “Answer:”, so this string appears
1297 twice. For Qwen2.5 the full input is

1298 `<|im_start|>system`
1299 `You are a helpful assistant.<|im_end|>`
1300 `<|im_start|>user`
1301 `Read the story and answer the question.`
1302 `...`
1303 `Answer:<|im_end|>`
1304 `<|im_start|>assistant`
1305 `Answer:`

1306 The wrapper asserts that the raw prompt occurs exactly once in the templated text. Templates that reject a system role fall
1307 back to merging the system text into the user turn; every output file records whether this happened, and no model reported
1308 here needed it. We read full-vocabulary log-probabilities at the last position and score the candidates as single tokens with
1309 a leading space (“ box”, “ basket”, ..., or “ A” to “ F” under LETTERS-AFTER); the code asserts that each candidate is one
1310 token.

1311 **Role-swap control.** The role-swapped story R exchanges the two names in the event and keeps the location word. For
1312 the core above, the base event and the role-swapped event are

1313
1314 B: Alice watches as the lamp is moved to the cabinet. Bob does not see this happen.

1315 R: Bob watches as the lamp is moved to the cabinet. Alice does not see this happen.

1316 The question still asks for Alice’s belief, so the correct answer moves from the writing token’s location (*cabinet*) to the
1317 initial location (*basket*). The role effect is measured with $m = \log p(\text{initial}) - \log p(\text{LOC})$, and f_C is the effect of clamping
1318 channel C at the writing token to R’s, divided by the full R–B difference in m .
1319

Intervention frames. The stage-2, stage-3b and stage-4 runs with the bases of [Anonymous \(2026\)](#) use their released native story cores, rendered with the same generator, the same format builders and the same wrapper. The patched span is the move event, from the observer’s name to the end of the story; the writing token is the location word inside it.

Paint and schedule tasks. The two new tasks (`ckeys/tasks.py`) follow the same structure: two sentences give the starting values of two things, sorted by name, and a final sentence writes the new value at the writing token. They use the formats OPTIONS-AFTER, LETTERS-AFTER, SENTENCE-AFTER, NO-MENTION and LIST-BEFORE, built exactly as above with the task’s candidates; the neutral sentence of SENTENCE-AFTER is task-specific. The agent is drawn from six names (Alice, Bob, Carol, David, Emma, Frank) and the things from ten per task.

	paint	schedule
Candidates	red, blue, green, yellow, black, white	Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday
Things	fence, door, bench, shed, gate, chair, table, wall, boat, bike	meeting, dinner, appointment, party, class, interview, concert, match, lecture, trip
Starting value	“The {thing} was painted {value}.”	“The {thing} was scheduled for {value}.”
Writing sentence	“Later, {agent} repainted the {thing} {value}.”	“Later, {agent} moved the {thing} to {value}.”
Question	“What color is the {thing} now?”	“On which day is the {thing} now?”
SENTENCE-AFTER sentence	“The shop sells red, blue, green, yellow, black and white paint.”	“The calendar shows Monday, Tuesday, Wednesday, Thursday, Friday, Saturday and Sunday.”
LETTERS-AFTER options	A) red, ..., F) white	A) Monday, ..., G) Sunday

The first cores generated with seed 0 give, under OPTIONS-AFTER and SENTENCE-AFTER,

Read the story and answer the question.

Story: The boat was painted yellow. The shed was painted yellow. Later, Emma repainted the shed black.

Question: What color is the shed now?

Choices: red, blue, green, yellow, black, white

Answer with exactly one choice.

Answer:

Read the story and answer the question.

Story: The interview was scheduled for Friday. The match was scheduled for Monday. Later, Frank moved the interview to Wednesday. The calendar shows Monday, Tuesday, Wednesday, Thursday, Friday, Saturday and Sunday.

Question: On which day is the interview now?

Answer with one word.

Answer:

with source values *green* and *Thursday* and third values *blue* and *Tuesday*.

C. Implementation and checks

Clamp hooks. Each clamp is a forward hook on the `k_proj` and/or `v_proj` module of every decoder block $\ell \geq \ell_0$. The hook overwrites the output row at the writing position p with the vector cached from the donor run (B, S or X) at the same layer. These outputs come before RoPE and before key normalisation (Qwen3, OLMo-2), and they contain all KV groups, so under grouped-query attention every group is clamped. All clamp conditions for one story run as rows of a single batch of the base prompt: the self-clamp row, $\mathcal{C}_K$, $\mathcal{C}_V$ and $\mathcal{C}_{KV}$ to S at each onset ℓ_0 , and the same three to X at $\ell_0 = 0$. The onsets are $\ell_0 = 0$, $\ell_0 = \text{round}(0.0625 L)$, which matches block 6 of 80, where the exchange of Anonymous (2026) starts in Qwen2.5-72B, and $\ell_0 = \text{round}(0.3 L)$. Forward passes use `use_cache=False` and compute logits only at the last position. From preregistration A onward, every natural-factorial output file records the number of skipped cores, which is zero in every such run.

Self-clamp row. The self-clamp row $\mathcal{C}_{KV}(\text{B})$ sets the writing token’s keys and values to their own values, so mathematically it equals the clean base run. In BF16, however, a batched forward pass can differ slightly from an unbatched one. From preregistration A onward we therefore measure every effect against this row of the same batch (Equation (1)), not against a separately run clean base, so that batching noise cancels. Each result summary records the gap between the self-clamp row and the clean run as a noise floor. The first CPU factorials (FP32) and all row-restricted splices measure against the clean base run.

Exactness of the clamp. $\mathcal{C}_{KV}(\text{S})$ from $\ell_0 = 0$ reproduces the source run at every later position. Positions before p have identical inputs in B and S and hence identical keys and values; at p both are set to S’s in every layer; by induction over layers every later position receives the same inputs as in S. The clamp precedes key normalisation (Qwen3, OLMo-2) and RoPE, which both runs apply identically at p . In the FP32 CPU runs at Qwen2.5-1.5B, the recovery ratio $d_{KV}/[m_{\text{SB}}(\text{S}) - m_{\text{SB}}(\text{B})]$ equals one to the printed precision in every format (`results/format_factorial/`). The unit tests (`tests/test_interventions.py`, Qwen2.5-0.5B in FP32) check that an identity edit and a self key exchange leave the logits unchanged to within 10^{-4} , and that patching the residual stream at the writing token after one block transfers the source answer.

Row-restricted splice. The splice wraps the forward pass of each attention module. It runs the module once with the base key at p and once with the source key, which the `k_proj` hook inserts, and returns the source output in the rows of the chosen token group G and the base output elsewhere. In the layer-window and KV-group variants, the source key is inserted only in the chosen layers, or only in one group’s slice of the key. A unit test checks that $G = \text{all rows}$ reproduces the full key clamp and $G = \emptyset$ reproduces the clean run, each to within 10^{-4} in the logits. Every run also evaluates $G = \emptyset$ for each story and reports its mean effect, which is zero to the printed precision.

A KV-cache pitfall. Our first splice implementation was silently wrong. Transformers 5 builds a `DynamicCache` by default, even for a single forward pass. Running the attention module twice per layer therefore appended keys to the cache, and the second (source-key) pass silently read the base keys. The fix is to run every forward pass with `use_cache=False`. The exactness test above was added in the same change, and every splice result reported here was produced after it. Anyone who patches attention by calling a module more than once per forward pass should check for this.

Exchanges in the intervention frames. The patch of Anonymous (2026) replaces the output of decoder block 4 (1-based) over every token of the move-event span by Equation (3), materialised in BF16 as in their engine, with the released rank-16 bases for the learned remap M and the PCA control P, at seeds 101–103 (and, in stage 4, the refit bases of Appendix D, each with the shared P). The key-only exchange replaces the writing token’s `k_proj` output in blocks 6 to L (1-based) by the other patched run’s, giving $\text{P} + \text{K}_\text{M}$ (addition) and $\text{M} + \text{K}_\text{P}$ (removal); queries, values and residual states evolve freely. The value-only exchange does the same with `v_proj`. The removal fractions are

$$\rho_K = \frac{m(\text{M} + \text{K}_\text{P}) - m(\text{M})}{m(\text{P}) - m(\text{M})}, \quad \rho_V = \frac{m(\text{M} + \text{V}_\text{P}) - m(\text{M})}{m(\text{P}) - m(\text{M})},$$

with $m = m_{\text{TS}}$, averaged over the three fits within each core before the ratio of means is taken. The reproduction gate compares the argmax token of M and P under OPTIONS-AFTER with the outputs released with Anonymous (2026); they agree on 0.994–1.000 of items (preregistered, ≥ 0.95 ; Table 3). The remap was fit without the “Answer:” prefill of our formats, which Anonymous (2026) add only at evaluation.

Attention knockout (preregistration G). A knockout sets chosen (row, column) attention weights to exactly zero in every layer and head and renormalises the row, while the writing token’s key or value is clamped as in the factorial. The eight clamp rows of one story (the self-clamp row, a duplicate of it as the within-batch noise floor, and C_K, C_V, C_{KV} to S and to X) run in one batch per mask, each with its own four-dimensional attention mask, and every effect is measured against the self-clamp row under the same mask. The masks are listed in Appendix F.2. FP32 unit tests at Qwen2.5-0.5B check, to 10^{-4} in the logits, that the causal mask passed explicitly changes nothing, that the cut weights are exactly zero and the rest of each row is renormalised, that sdpa and eager attention agree, that per-row masks in one batch equal separate runs, and that with every edge to the writing token cut, each clamp row equals the self-clamp row. A sanity gate (Gate b) repeats the last check in every arm and model of the run: with all edges to the writing token cut, ID_K and ID_V are zero.

Head and hop splices (preregistration H). *HeadSplice* extends the row-restricted splice to heads: in the rows of the option words it returns, per head, the attention output computed with the source key at p for the chosen heads and with the base key for the others. Blinding a set of heads is the complement: all heads but these see the source key. Mean-ablation replaces a head’s output at the six option rows by its position-matched mean over the ranking stories. *HopSplice* clamps K_S at p and gives only the chosen later rows (the answer row, every other row, or all rows) the base run’s keys, values or both at the option rows, so that $r_{\text{ans}} = 1 - [m(\text{row}) - m(\text{ID})] / [m(K_S) - m(\text{ID})]$ measures how much of the key-clamp effect reaches the answer through the option rows. Gate a1 runs seven FP32 unit tests at Qwen2.5-0.5B before any model is loaded, each to 10^{-4} in the logits: HeadSplice with all heads equals the full key clamp and with none the clean run; all heads in the option rows equal the row splice, and KV-group slices the group splice; a batched grid equals single runs; self-ablation equals the clean run; HopSplice over all rows equals a plain clamp of the option rows’ keys and values, without overwrite it equals the clean and full runs, and the answer-only splice with the source run’s cache equals the full clamp. All seven passed.

The exchange on Prakash et al.’s patch (preregistration H). For each pair, format and depth, one batch holds seven rows: $r_0 = B + KV_B$ (self), $r_1 = M$ (the patched run), $r_2 = B + K_M$, $r_3 = B + V_M$, $r_4 = B + KV_M$, $r_5 = M + K_B$ and $r_6 = M + V_B$, where the keys and values of the patched positions are exchanged in every block after the patched one, and words-only variants of r_2 – r_4 exchange them at the state words alone. The residual patch is active only in r_1 , r_5 and r_6 . Then $\Phi = m(r_1) - m(r_0)$, $\psi_K = [m(r_2) - m(r_0)] / \Phi$, ψ_V likewise with r_3 , ρ_K and ρ_V from r_5 and r_6 , and $\kappa = \psi_K / (\psi_K + \psi_V)$, each as a ratio of means over pairs. Because the patch changes nothing before its own block and r_4 gives the clean run every patched key and value after it, r_4 equals r_1 exactly in exact arithmetic; Gate b0 requires the BF16 means of $|m(r_4) - m(r_1)|$ and $|m(r_0) - m(B)|$, with B run unbatched, to stay within 0.3 nats in every cell (observed 0.099–0.194 and 0.141–0.203). In FP32 at Qwen2.5-0.5B the same differences are asserted to be within 10^{-3} .

Code audit. Before the GPU runs, an independent review of the measurement code led to fixes made before any GPU result existed, among them the batched self-clamp baseline, depth-matched onsets, a seeded third value X, core-bootstrap CIs computed in the experiment scripts, the competence filter (now a secondary analysis) and provenance written into every output file.

Mistral-Small-24B tokenizer warning. When loading the Mistral-Small-24B tokenizer, transformers warns that its pre-tokenisation regex is incorrect and suggests `fix_mistral_regex=True`. We kept the default, under which the reproduction gate of preregistration C passed. In stage 4 we checked that the flag leaves the token ids of all 3,600 frames prompts (five formats, both prefills) unchanged; we have not checked the natural-factorial prompts.

Compute. All experiments except the stage-4 refits are forward passes; the refits train only rank-16 bases (Appendix D), and no model weights are ever trained. Pilots with models of at most 1.7B parameters, the first format factorials, preregistration A, the 1.5B row-restricted splices, the early attention probe and the FP32 unit tests ran on CPU in FP32; the preregistration-A output records transformers 5.18.0 and Python 3.11.15. All GPU runs used NVIDIA A100-SXM4-80GB GPUs in BF16, torch 2.11.0 with CUDA 12.8 and Python 3.12.14, as recorded in each stage’s `ENV.txt` and output files:

- stage 1 (preregistration B; Qwen2.5-1.5B to 14B, Qwen3-8B, Mistral-7B, OLMo-2-7B): one GPU, transformers 5.18.0, commit `f706ef5`;
- stage 2 (preregistration C; frames at 24B and 72B, natural factorial at 24–72B): two GPUs with `device_map="auto"`, transformers 5.9.0, pinned by the stage-2 script for the reproduction gate, commit `2bleca7`;

- stage 3 (preregistration D; new tasks and localisation at 7–14B): one GPU, transformers 5.18.0, commit b8815cb;
- stage 3b (preregistration E): two GPUs, commit 3ca63f1; the 2×2, the role control at 7–14B and the Qwen2.5-32B environment re-run under transformers 5.18.0 (the re-run on one GPU), and the Qwen2.5-14B environment re-run, the value-only frames and the role control at 72B under transformers 5.9.0;
- stage 4 (preregistration F): two GPUs, one per refit (55 minutes each), then the frames on both, transformers 5.9.0, commit 1ef8696;
- stage 5 (preregistration G): one GPU, transformers 5.18.0, commit 51e105e; eager attention for the re-mention attention and form probes, sdpa elsewhere; GPT-2 small and GPT-2 XL in FP32 with exactness asserted; the GPT-2 attention probe re-run on CPU (Appendix A);
- stage 6 (preregistration H): one GPU, transformers 5.18.0, commit cc3a3e0; eager attention for part (a), sdpa for part (b), and the exploratory re-check sweeps of part (b) in FP32. The release of [Prakash et al. \(2026\)](#) is fetched at run time at its commit 0579347 and checked against stored file hashes; none of it is copied into our repository.

Model revisions. For the two models of [Anonymous \(2026\)](#) we pin the revisions used for the reproduction: Mistral-Small-24B-Instruct-2501 at 9527884be6e5616bdd54de542f9ae13384489724 and Qwen2.5-72B-Instruct at 495f39366efef23836d0cfae4f6e635880d2be31. The pins apply to the frames and to the stage-2 natural factorial at those sizes. All other runs, including Qwen2.5-32B and the role control at Qwen2.5-72B, load the default revision available at run time.

Released code. The code release contains the story generator, encoder, tasks and hook primitives (ckeys/); one script per experiment (experiments/); the preregistered scoring scripts (analysis/stage1_prereg.py, stage2_score.py, stage3_score.py, stage3b_score.py, stage4_score.py, stage5_score.py, stage6_score.py); the refit wrapper (experiments/refit_remap.py); a port of the story generator and counterfactuals of [Prakash et al. \(2026\)](#) that reads their release at run time (ckeys/causalcom.py); one shell script per GPU stage, which pins the library version and records the commit and environment (scripts/); the exactness tests (tests/); and paper/make_figures.py, which generates every result figure, every table in Appendices D to G, the tables of preregistrations G and H, and every result macro in the text from the saved results and score files (the schematic in Figure 1 is drawn by paper/make_schematic.py).

D. Refitting the remap (preregistration F)

Recipe. We ran the released training code of Anonymous (2026) (gpu/train.py and gpu/behavior_engine.py, unmodified) through experiments/refit_remap.py. Its recipe: a rank-16 basis at the output of (1-based) block 4 over every event-span token; the pair-swap objective with a six-way conditional cross-entropy at the start of the reply (no prefill); AdamW, learning rate 10^{-3} , batch 1, one shuffled epoch of their 1000 training pairs (2000 stories); seeds 101–103, which change only the order of the training pairs: every fit starts from the same PCA basis of source-minus-base differences (the randomised sketch differs by seed but returns the same subspace, mean squared principal cosine 1.000 across seeds), and that basis is the control P of every fit. The code also fits an identity-objective basis next to each remap (exploratory): its shift of the answer towards T, relative to P and as a fraction of the natural $S \rightarrow T$ shift, is at most 0.18 for fit_none (under LETTERS-AFTER) and 0.04 for fit_opt. The training stories are disjoint from the 120 evaluation cores, but most evaluation event sentences also occur in some training story, as for the released remap.

What the wrapper replaces. At import time, and recorded with hashes in each run’s FRAME.json: the prompt function, for fit_none only, by our NO-MENTION prompt, which agrees token for token with the original prompt up to the end of the story in all 1000 training pairs; the release-integrity check, by one that tolerates only the release’s mismatched README.md; and the model loader, by one with the same load arguments (BF16, sdpa, eval mode, no gradients, no cache, pad = eos, fix_mistral_regex) at the pinned revision, waiving the release’s exact Python, package, device and memory pins and its model-asset receipt (the weights are identified instead by the pinned commit, which the loader checks). Under the causal mask the activations up to the end of the story are the same in both runs, so both start from the same P; the PCA bases came out bitwise identical across runs.

Results. Table 12 gives every frame quantity for the two refits and for the released bases re-run in the same job; Table 5 gives the gates and predictions. The rest of this paragraph is exploratory. The control matches the released remap to within 0.03 in every quantity and format. Every fitted remap keeps 0.34–0.36 of its subspace (mean squared principal cosine) in the shared PCA start, against 0.003 for random subspaces, so overlaps between fits lie above that floor: fit_opt overlaps the released remap at 0.71–0.79, while fit_none overlaps the released remap at 0.37–0.39 and fit_opt at 0.37–0.39, less than two seeds of one fit overlap each other (0.41–0.44). Without the “Answer:” prefill, which matches the interface the remaps were fit at, the difference in differences is $+0.004$ [-0.014 , $+0.021$]. The training loss of the remap fits fell from 2.9–4.8 nats over the first 100 updates to 0.11–0.48 over the last 100; it starts above chance ($\ln 6 \approx 1.79$) because the PCA start transfers the source, which the pair-swap objective scores as wrong.

Where the fitting format still matters (post hoc). The fitting format shifts the balance where both channels read, in the direction the stated alternative predicts. fit_none’s ψ_K is lower than the control’s by 0.09 [0.07, 0.12] under OPTIONS-AFTER and by 0.07 [0.05, 0.10] under SENTENCE-AFTER. It transfers better under NO-MENTION ($\varphi = 0.81$ against 0.72) and worse under LETTERS-AFTER (0.67 against 0.77), where its values carry almost nothing ($\psi_V = 0.02$). These comparisons were made after seeing the data; the preregistered all-or-none alternative (Table 5) was not observed.

E. Additional results

Unless stated otherwise, intervals are 95% percentile intervals from 10,000 bootstrap resamples of stories (cores), with ratios recomputed within each resample. Each estimate is printed with its interval below it, except in Tables 15 and 16, where the intervals of the individual effects are in the result summaries released with the code.

Natural factorial in all models. Table 13 gives the OPTIONS-AFTER key share and ID_K in every format for all 10 models, and Figure 5 shows ID_K relative to OPTIONS-AFTER. LIST-BEFORE is negative in every model except Mistral-7B, where it is indistinguishable from zero; at Qwen2.5-72B it reaches -2.0 nats.

Scale and families. Within Qwen2.5, the OPTIONS-AFTER key share rises from 0.37 at 1.5B through 0.69 (3B) and 0.82 (7B) to 0.91 at 14B, then falls to 0.86 at 32B and 0.76 at 72B (Figure 4); the 14B and 72B intervals do not overlap. The size trend was exploratory (B3), and the preregistered threshold $s_K \geq 0.80$ for the three largest models was met only at 32B (C4). Re-running Qwen2.5-14B and 32B each in the other’s software and GPU setup gave 0.91 and 0.86, the originals to two decimals (E3, met). The fall beyond 14B is therefore not an artefact of the environment, and it is also seen in the

Table 12. **Refit remaps in the frames of Anonymous (2026)** (Mistral-Small-24B, “Answer:” prefix). fit_none: refit under NO-MENTION; fit_opt: refit under OPTIONS-AFTER with the same code; released: the released bases, re-run in the same job. All fits share one PCA control P. $n = 96$ cores, three fits averaged within core; 95% core bootstrap.

Remap	Format	φ	ψ_K	ρ_K	ψ_V	ρ_V
fit_none	LETTERS-AFTER	0.67 [0.64, 0.70]	0.77 [0.73, 0.80]	0.98 [0.97, 0.98]	0.02 [0.02, 0.03]	0.15 [0.13, 0.19]
	OPTIONS-AFTER	0.72 [0.70, 0.74]	0.43 [0.39, 0.47]	0.74 [0.73, 0.76]	0.25 [0.23, 0.26]	0.54 [0.50, 0.58]
	SENTENCE-AFTER	0.81 [0.79, 0.83]	0.12 [0.10, 0.15]	0.47 [0.44, 0.49]	0.50 [0.47, 0.52]	0.85 [0.81, 0.88]
	NO-MENTION	0.81 [0.79, 0.84]	0.07 [0.06, 0.08]	0.18 [0.16, 0.20]	0.80 [0.77, 0.82]	0.93 [0.91, 0.94]
	LIST-BEFORE	0.76 [0.74, 0.78]	0.01 [-0.00, 0.02]	-0.00 [-0.01, 0.01]	0.97 [0.96, 0.99]	0.95 [0.94, 0.96]
fit_opt	LETTERS-AFTER	0.77 [0.74, 0.80]	0.78 [0.73, 0.82]	0.98 [0.98, 0.99]	0.02 [0.01, 0.02]	0.16 [0.13, 0.19]
	OPTIONS-AFTER	0.72 [0.69, 0.74]	0.52 [0.48, 0.56]	0.76 [0.74, 0.77]	0.23 [0.22, 0.24]	0.44 [0.40, 0.48]
	SENTENCE-AFTER	0.73 [0.71, 0.75]	0.20 [0.16, 0.23]	0.45 [0.43, 0.47]	0.52 [0.50, 0.54]	0.77 [0.73, 0.81]
	NO-MENTION	0.72 [0.69, 0.74]	0.07 [0.06, 0.08]	0.17 [0.15, 0.19]	0.81 [0.78, 0.83]	0.93 [0.92, 0.94]
	LIST-BEFORE	0.77 [0.74, 0.79]	-0.00 [-0.01, 0.01]	0.00 [-0.01, 0.02]	0.98 [0.96, 0.99]	0.96 [0.95, 0.98]
released	LETTERS-AFTER	0.75 [0.71, 0.79]	0.78 [0.74, 0.82]	0.99 [0.98, 0.99]	0.02 [0.01, 0.02]	0.16 [0.13, 0.19]
	OPTIONS-AFTER	0.70 [0.68, 0.73]	0.53 [0.49, 0.57]	0.76 [0.75, 0.78]	0.23 [0.22, 0.24]	0.44 [0.41, 0.48]
	SENTENCE-AFTER	0.72 [0.69, 0.75]	0.20 [0.16, 0.23]	0.45 [0.43, 0.47]	0.52 [0.50, 0.54]	0.77 [0.73, 0.81]
	NO-MENTION	0.71 [0.68, 0.73]	0.07 [0.06, 0.08]	0.18 [0.16, 0.20]	0.80 [0.78, 0.83]	0.93 [0.92, 0.94]
	LIST-BEFORE	0.74 [0.71, 0.76]	-0.00 [-0.01, 0.01]	0.01 [-0.00, 0.02]	0.97 [0.96, 0.98]	0.96 [0.95, 0.98]

Table 13. **Natural factorial, all models.** Key share s_K under OPTIONS-AFTER and value identity carried by keys, ID_K (nats), in each format, with keys and values clamped from layer 0. $n = 150$ stories per format and model; 95% bootstrap CIs over stories (ratio recomputed in each resample for s_K). Models ordered by size. Rows up to 14B are from GPU stage 1, whose predictions B1–B2 cover OPTIONS-AFTER, LETTERS-AFTER and NO-MENTION in Qwen2.5-3B/7B, Qwen3-8B, Mistral-7B and OLMo-2-7B; rows from 24B are from stage 2, whose prediction C4 covers s_K and LIST-BEFORE. The other cells, including the 1.5B and 14B rows, are exploratory; preregistration A tested 1.5B in a separate run (Table 2).

Model	s_K (OPTIONS-AFTER)	ID_K , identity carried by keys (nats)				
		LETTERS-AFTER	OPTIONS-AFTER	SENTENCE-AFTER	NO-MENTION	LIST-BEFORE
Qwen2.5-1.5B	0.37 [0.36, 0.39]	+3.6 [+3.3, +3.8]	+2.8 [+2.6, +3.1]	-0.5 [-0.6, -0.4]	+0.2 [+0.1, +0.2]	-0.2 [-0.3, -0.2]
Qwen2.5-3B	0.69 [0.68, 0.71]	+11.4 [+10.4, +12.4]	+10.3 [+9.6, +11.1]	-0.0 [-0.4, +0.3]	+0.1 [-0.0, +0.2]	-0.5 [-0.6, -0.3]
Mistral-7B	0.92 [0.90, 0.93]	+15.2 [+14.6, +15.7]	+14.9 [+14.5, +15.3]	+7.9 [+7.3, +8.4]	+0.6 [+0.5, +0.6]	+0.0 [-0.1, +0.1]
OLMo-2-7B	0.73 [0.71, 0.74]	+7.9 [+7.7, +8.0]	+8.2 [+8.0, +8.4]	+2.0 [+1.8, +2.2]	+0.4 [+0.3, +0.5]	-0.3 [-0.4, -0.2]
Qwen2.5-7B	0.82 [0.81, 0.83]	+20.4 [+20.0, +20.8]	+21.2 [+20.7, +21.6]	+5.5 [+5.2, +5.8]	+1.0 [+0.9, +1.2]	-0.1 [-0.2, -0.1]
Qwen3-8B	0.84 [0.83, 0.84]	+33.9 [+33.4, +34.3]	+35.4 [+34.6, +36.2]	+8.8 [+8.3, +9.3]	+1.2 [+1.0, +1.4]	-0.3 [-0.4, -0.3]
Qwen2.5-14B	0.91 [0.90, 0.91]	+34.8 [+34.5, +35.1]	+39.0 [+38.0, +40.0]	+12.1 [+11.5, +12.7]	+1.1 [+0.9, +1.2]	-0.1 [-0.2, -0.0]
Mistral-24B	0.78 [0.77, 0.79]	+9.2 [+9.1, +9.3]	+9.6 [+9.4, +9.8]	+5.6 [+5.4, +5.9]	+0.4 [+0.3, +0.5]	-0.5 [-0.6, -0.5]
Qwen2.5-32B	0.86 [0.85, 0.87]	+34.8 [+34.5, +35.1]	+32.4 [+31.6, +33.2]	+10.3 [+9.6, +10.9]	+1.1 [+0.9, +1.3]	-0.4 [-0.6, -0.3]
Qwen2.5-72B	0.76 [0.75, 0.77]	+26.4 [+26.2, +26.6]	+25.6 [+25.0, +26.2]	+8.4 [+8.0, +8.9]	+0.5 [+0.4, +0.6]	-2.0 [-2.2, -1.8]

identity key share (0.88 at 14B, 0.71 at 72B), although we do not know its cause. The other families give similar shares (0.73–0.92), and every model except Qwen2.5-1.5B carries more than half of the OPTIONS-AFTER effect in the key.

Identity key share. Table 14 normalises ID_K by the total identity effect, $s_{ID} = \overline{ID}_K / (\overline{ID}_K + \overline{ID}_V)$, so that formats and models with different confidence can be compared. It tracks s_K closely except under NO-MENTION and LIST-BEFORE, where s_K also counts the non-specific effect of a foreign key. Above 1.5B it is 0.94–1.00 under LETTERS-AFTER and at most 0.00 under LIST-BEFORE (exploratory); the main text gives the other formats.

Decomposition. Table 15 gives, for each model and format, the effects of clamping the key, the value or both, the interaction $d_{KV} - d_K - d_V$, and ID_V . The interaction is small relative to d_{KV} in every cell, so the two channels’ effects add up. Under LETTERS-AFTER the value channel carries almost no identity in any model, and under NO-MENTION and LIST-BEFORE it carries most of it: there the identity is copied. In the instruction-matched 2×2 (Table 1), ID_K is larger under LIST-AFTER than under SENTENCE-AFTER in all four models, and LIST-AFTER is close to OPTIONS-AFTER in the same model; neither comparison was preregistered.

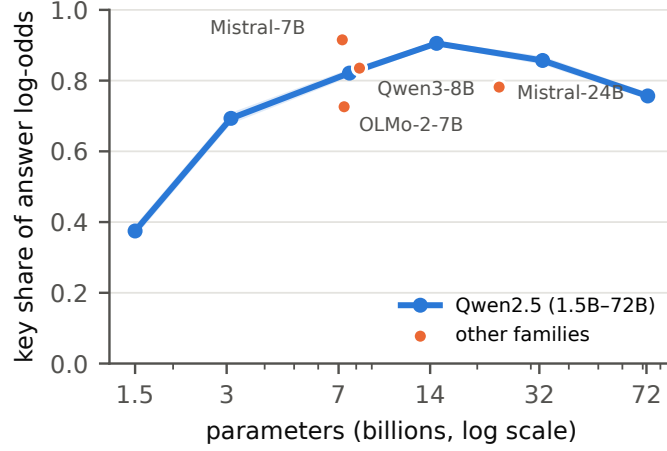


Figure 4. **Key share of the OPTIONS-AFTER answer across scale and families.** $s_K = \bar{d}_K / (\bar{d}_K + \bar{d}_V)$ with keys and values clamped from layer 0. Line and band: Qwen2.5-Instruct 1.5B–72B with 95% bootstrap CIs; points: other families. $n = 150$ stories per model; values with CIs in Table 13. s_K , the preregistered measure, includes non-specific effects; the identity key share s_{ID} shows the same rise to 14B and fall beyond (Table 14). Exploratory (B3), with the threshold of C4.

Table 14. **Identity key share s_{ID} , all models** (exploratory). Ratio of mean ID_K to mean $ID_K + ID_V$, keys and values clamped from layer 0; $n = 150$ stories per format and model; 95% bootstrap CIs, ratio recomputed in each resample.

Model	LETTERS-AFTER	OPTIONS-AFTER	SENTENCE-AFTER	NO-MENTION	LIST-BEFORE
Qwen2.5-1.5B	+0.83 [+0.81, +0.86]	+0.36 [+0.34, +0.38]	−0.08 [−0.10, −0.06]	+0.02 [+0.01, +0.02]	−0.04 [−0.05, −0.03]
Qwen2.5-3B	+0.98 [+0.97, +0.99]	+0.67 [+0.66, +0.69]	0.00 [−0.04, +0.03]	0.00 [0.00, +0.01]	−0.05 [−0.06, −0.04]
Mistral-7B	+0.94 [+0.92, +0.95]	+0.87 [+0.85, +0.88]	+0.41 [+0.39, +0.43]	+0.04 [+0.03, +0.04]	0.00 [0.00, 0.00]
OLMo-2-7B	+0.99 [+0.99, +0.99]	+0.69 [+0.68, +0.70]	+0.19 [+0.17, +0.20]	+0.03 [+0.03, +0.04]	−0.03 [−0.03, −0.02]
Qwen2.5-7B	+0.99 [+0.98, +0.99]	+0.79 [+0.77, +0.80]	+0.32 [+0.30, +0.33]	+0.06 [+0.05, +0.06]	−0.01 [−0.01, 0.00]
Qwen3-8B	+0.97 [+0.97, +0.98]	+0.81 [+0.80, +0.82]	+0.29 [+0.28, +0.30]	+0.05 [+0.04, +0.05]	−0.01 [−0.02, −0.01]
Qwen2.5-14B	+1.00 [+1.00, +1.00]	+0.88 [+0.87, +0.89]	+0.40 [+0.38, +0.41]	+0.05 [+0.05, +0.06]	0.00 [−0.01, 0.00]
Mistral-24B	+0.98 [+0.98, +0.98]	+0.73 [+0.72, +0.74]	+0.45 [+0.44, +0.46]	+0.04 [+0.03, +0.05]	−0.06 [−0.07, −0.05]
Qwen2.5-32B	+0.99 [+0.99, +0.99]	+0.82 [+0.81, +0.83]	+0.30 [+0.29, +0.32]	+0.04 [+0.04, +0.05]	−0.01 [−0.02, −0.01]
Qwen2.5-72B	+0.99 [+0.99, +0.99]	+0.71 [+0.70, +0.72]	+0.33 [+0.31, +0.34]	+0.02 [+0.02, +0.02]	−0.09 [−0.10, −0.08]

Clamp onset. Layer-0 keys are a function of the input embedding alone, so Table 16 starts both clamps deeper (exploratory). With $\ell_0 \approx 0.06L$ the shares are essentially unchanged in eight of ten models. With $\ell_0 \approx 0.3L$ the share stays at or above one half in seven models (up to 0.83 at Qwen2.5-14B), but falls to 0.19 at Qwen2.5-1.5B, 0.37 at Mistral-Small-24B and 0.03 at OLMo-2-7B, where the key effect depends on the lowest layers. Deeper keys therefore still carry the written word in most models. With $\ell_0 > 0$ the clamp is no longer exact, and at $\ell_0 \approx 0.3L$ the key and value effects no longer add up: the interaction is positive in every model (result summaries). The share at that onset is therefore descriptive only.

Layer windows and KV groups. Table 17 refines the row-restricted splice at Qwen2.5-1.5B (28 layers, two KV groups). Only the option-word rows see the source key, and the swap is further limited to a window of four layers (all KV groups) or to one KV group’s slice of the key (all layers). The read happens in layers 4–15, and KV group 1 carries more of it than KV group 0 in both formats. The seven windows together recover less than the option-word rows do with the swap

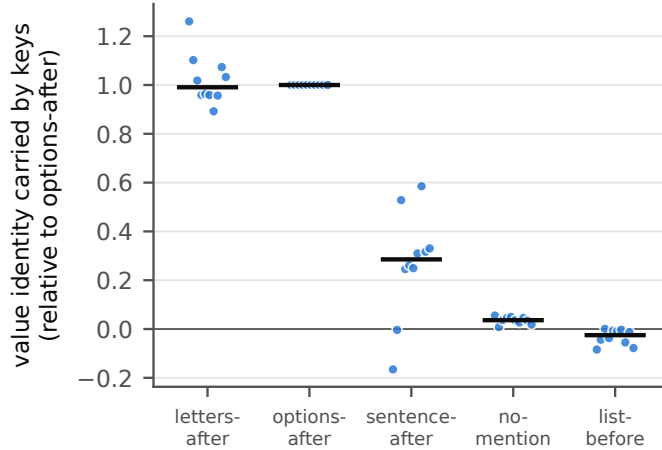


Figure 5. **Key identity by format, relative to OPTIONS-AFTER.** Each point is one of the 10 models: ID_K in that format divided by ID_K under OPTIONS-AFTER in the same model (so OPTIONS-AFTER is 1 by construction); points are jittered horizontally, and the horizontal line is the median over models. $n = 150$ stories per format and model; per-model values with 95% bootstrap CIs are in Table 13.

at all layers, and the two single-group swaps together recover less than both groups at once, so the read compounds across layers and groups. We ran the same window and KV-group splices, on the option-word rows under OPTIONS-AFTER and LETTERS-AFTER, at Qwen2.5-7B, Qwen2.5-14B and Mistral-7B (exploratory; result summaries). In these models no single four-layer window and no single KV group recovers most of the option-word effect, so the dominance of one KV group seen at 1.5B does not carry over.

New tasks. Table 18 gives the paint and schedule results. In every model and both tasks, ID_K is large under LETTERS-AFTER and OPTIONS-AFTER, intermediate under SENTENCE-AFTER, small under NO-MENTION and near zero under LIST-BEFORE. The OPTIONS-AFTER key share ranges from 0.47 to 0.92 (paint) and from 0.35 to 0.92 (schedule), lowest at OLMo-2-7B in both.

Role control. Could the writing token’s channels carry more than identity, such as whether the person asked about saw the move? The role-swap control exchanges the observer’s and the non-observer’s names but keeps the location word (Appendix B). Table 19 gives the role-swap effect and the fractions of it that the writing token’s key and value carry. The swap shifts the belief answer by 12.5 to 79.8 nats, depending on model and format, and clamping either channel alone to the role-swapped run carries at most 0.02 of that shift, under OPTIONS-AFTER, NO-MENTION and LETTERS-AFTER, in five models including Qwen2.5-72B. The prediction was preregistered for OPTIONS-AFTER and NO-MENTION in four models, with a bound of 0.15 (E2, met); LETTERS-AFTER and Qwen2.5-72B are exploratory. The writing token’s key and value thus carry *which* value was written, not the observer’s name; an ordering-ID address of the observer, the same in both runs, would not register (Appendix H).

Localisation. Table 20 gives the row-restricted splice for each token group. The later-mention words are the listed options under OPTIONS-AFTER and LETTERS-AFTER, and the candidate words of the neutral sentence under SENTENCE-AFTER. Under OPTIONS-AFTER, the question, the story after the writing token and the remaining tokens, including the answer position, each recover at most 0.01 at 7–14B. Under SENTENCE-AFTER the “other later tokens” recover 0.05–0.11, and 0.19–0.22 of the effect is recovered by no single group, so the groups do not add up. We did not localise the read at 24B or 72B. An earlier exploratory run at Qwen2.5-1.5B gives the same picture: the option words recover 0.87 of the key effect under OPTIONS-AFTER and 0.98 under LETTERS-AFTER. Restricting the splice further, the read is concentrated in layers 4–15 of 28, and one of the two KV groups carries most of it (exploratory; Table 17).

Intervention frames. Table 21 gives every frame quantity with its interval. At Qwen2.5-72B, M’s keys reproduce 0.89 of the M–P difference under LETTERS-AFTER and 0.63 under OPTIONS-AFTER, but only 0.21 under SENTENCE-AFTER, 0.05 under NO-MENTION and -0.03 under LIST-BEFORE, while M’s values do the reverse ($\psi_V = 0.01, 0.21, 0.67, 0.90$,

Looked Up or Copied? Later Mentions Decide Which Attention Channel Carries an In-Context Value

Table 15. Key/value decomposition. Mean effects (nats of m_{SB}) of clamping the writing token’s key (d_K), value (d_V) or both (d_{KV}) to the source run’s from layer 0, measured against the self-clamp row; the interaction $d_{KV} - d_K - d_V$ with its 95% bootstrap CI; and the value identity ID_V (nats). $n = 150$ stories per format and model.

Model	Format	d_K	d_V	d_{KV}	interaction	ID_V
Qwen2.5-1.5B	LETTERS-AFTER	+9.9	+4.2	+14.5	+0.4 [-0.5, +1.3]	+0.7
	OPTIONS-AFTER	+6.3	+10.4	+16.7	-0.0 [-0.5, +0.5]	+5.0
	SENTENCE-AFTER	+0.5	+14.4	+14.8	-0.0 [-0.4, +0.3]	+6.5
	NO-MENTION	+1.8	+21.8	+23.5	-0.1 [-0.7, +0.5]	+10.2
	LIST-BEFORE	+1.2	+13.6	+14.5	-0.3 [-0.6, +0.1]	+5.9
Qwen2.5-3B	LETTERS-AFTER	+22.3	+0.8	+22.8	-0.4 [-1.1, +0.3]	+0.2
	OPTIONS-AFTER	+20.4	+9.0	+29.1	-0.4 [-1.1, +0.3]	+5.0
	SENTENCE-AFTER	+2.9	+26.8	+29.9	+0.2 [-0.6, +1.0]	+11.8
	NO-MENTION	+4.3	+41.1	+45.9	+0.5 [-0.1, +1.1]	+18.1
	LIST-BEFORE	+2.8	+23.4	+26.4	+0.2 [-0.5, +0.8]	+9.6
Mistral-7B	LETTERS-AFTER	+31.3	+1.5	+32.2	-0.5 [-1.2, +0.1]	+1.0
	OPTIONS-AFTER	+28.0	+2.6	+30.8	+0.1 [-0.6, +0.8]	+2.3
	SENTENCE-AFTER	+12.9	+19.5	+31.8	-0.6 [-2.3, +1.1]	+11.5
	NO-MENTION	+2.7	+28.8	+31.5	-0.1 [-0.4, +0.3]	+13.6
	LIST-BEFORE	+1.2	+30.9	+32.1	+0.1 [-0.3, +0.5]	+14.6
OLMo-2-7B	LETTERS-AFTER	+15.6	+0.1	+15.7	+0.0 [-0.1, +0.1]	+0.1
	OPTIONS-AFTER	+15.5	+5.8	+21.0	-0.3 [-0.9, +0.2]	+3.7
	SENTENCE-AFTER	+3.8	+16.8	+20.7	+0.0 [-0.5, +0.6]	+8.6
	NO-MENTION	+2.7	+23.5	+26.1	-0.1 [-0.5, +0.3]	+11.0
	LIST-BEFORE	+0.2	+23.6	+23.9	+0.1 [-0.1, +0.4]	+11.2
Qwen2.5-7B	LETTERS-AFTER	+40.2	+1.1	+41.2	-0.1 [-0.7, +0.4]	+0.2
	OPTIONS-AFTER	+39.6	+8.6	+47.9	-0.4 [-1.1, +0.3]	+5.8
	SENTENCE-AFTER	+11.0	+23.1	+34.4	+0.4 [-0.4, +1.1]	+11.9
	NO-MENTION	+6.0	+36.8	+42.5	-0.3 [-0.7, +0.2]	+16.6
	LIST-BEFORE	+2.4	+34.8	+36.8	-0.4 [-0.8, -0.1]	+16.0
Qwen3-8B	LETTERS-AFTER	+66.8	+1.2	+67.9	-0.1 [-0.5, +0.3]	+1.0
	OPTIONS-AFTER	+67.1	+13.2	+80.5	+0.2 [-1.4, +1.7]	+8.2
	SENTENCE-AFTER	+14.3	+40.0	+53.3	-1.1 [-2.5, +0.3]	+21.8
	NO-MENTION	+5.7	+54.4	+59.4	-0.7 [-1.6, +0.2]	+25.3
	LIST-BEFORE	+2.9	+51.8	+54.4	-0.3 [-0.8, +0.2]	+24.0
Qwen2.5-14B	LETTERS-AFTER	+70.0	+0.2	+70.2	+0.0 [-0.2, +0.3]	+0.1
	OPTIONS-AFTER	+75.0	+7.8	+82.8	-0.1 [-0.7, +0.5]	+5.4
	SENTENCE-AFTER	+21.9	+33.8	+56.9	+1.1 [-1.4, +3.6]	+18.5
	NO-MENTION	+10.8	+46.5	+57.1	-0.2 [-0.9, +0.4]	+19.1
	LIST-BEFORE	+2.4	+65.9	+68.3	+0.0 [-0.5, +0.5]	+31.1
Mistral-24B	LETTERS-AFTER	+18.7	+0.4	+19.0	-0.1 [-0.2, -0.0]	+0.2
	OPTIONS-AFTER	+17.1	+4.8	+21.9	+0.1 [-0.2, +0.4]	+3.6
	SENTENCE-AFTER	+8.1	+10.4	+18.6	+0.1 [-0.7, +0.8]	+6.8
	NO-MENTION	+2.0	+20.8	+22.6	-0.2 [-0.4, +0.0]	+9.8
	LIST-BEFORE	-0.0	+19.4	+19.4	+0.1 [-0.1, +0.3]	+9.1
Qwen2.5-32B	LETTERS-AFTER	+69.5	+0.6	+70.3	+0.2 [-0.1, +0.5]	+0.3
	OPTIONS-AFTER	+61.4	+10.2	+71.6	-0.1 [-1.8, +1.7]	+7.1
	SENTENCE-AFTER	+18.7	+44.8	+63.1	-0.4 [-2.0, +1.3]	+23.4
	NO-MENTION	+9.8	+53.8	+63.4	-0.2 [-0.9, +0.5]	+23.4
	LIST-BEFORE	+2.8	+65.1	+68.1	+0.1 [-0.5, +0.7]	+30.6
Qwen2.5-72B	LETTERS-AFTER	+52.8	+0.2	+53.0	-0.0 [-0.2, +0.2]	+0.3
	OPTIONS-AFTER	+44.6	+14.3	+59.6	+0.7 [-0.3, +1.7]	+10.5
	SENTENCE-AFTER	+14.4	+30.9	+46.4	+1.1 [-0.3, +2.6]	+17.5
	NO-MENTION	+3.1	+47.7	+50.9	+0.1 [-0.4, +0.5]	+22.9
	LIST-BEFORE	-1.7	+51.2	+49.8	+0.3 [-0.2, +0.7]	+24.1

1.00; exploratory except C3 and E4). The removal fractions follow the addition fractions: ρ_K is large where ψ_K is, and ρ_V where ψ_V is. Among the six candidates, M’s top answer is the target T in 0.76–0.87 of (core, fit) items at Mistral-Small-24B

Table 16. **Key share by clamp onset.** $s_K = \bar{d}_K / (\bar{d}_K + \bar{d}_V)$ under OPTIONS-AFTER with both clamps applied from layer ℓ_0 on; the onset used in each model is given in brackets. $n = 150$ stories per model; point estimates (the $\ell_0 = 0$ intervals are in Table 13). Exploratory.

Model	layers L	s_K , onset $\ell_0=0$	s_K , $\ell_0 \approx 0.06L$	s_K , $\ell_0 \approx 0.3L$
Qwen2.5-1.5B	28	0.37 ($\ell_0=0$)	0.37 ($\ell_0=2$)	0.19 ($\ell_0=8$)
Qwen2.5-3B	36	0.69 ($\ell_0=0$)	0.69 ($\ell_0=2$)	0.56 ($\ell_0=11$)
Mistral-7B	32	0.92 ($\ell_0=0$)	0.91 ($\ell_0=2$)	0.53 ($\ell_0=10$)
OLMo-2-7B	32	0.73 ($\ell_0=0$)	0.72 ($\ell_0=2$)	0.03 ($\ell_0=10$)
Qwen2.5-7B	28	0.82 ($\ell_0=0$)	0.82 ($\ell_0=2$)	0.66 ($\ell_0=8$)
Qwen3-8B	36	0.84 ($\ell_0=0$)	0.84 ($\ell_0=2$)	0.73 ($\ell_0=11$)
Qwen2.5-14B	48	0.91 ($\ell_0=0$)	0.91 ($\ell_0=3$)	0.83 ($\ell_0=14$)
Mistral-24B	40	0.78 ($\ell_0=0$)	0.78 ($\ell_0=2$)	0.37 ($\ell_0=12$)
Qwen2.5-32B	64	0.86 ($\ell_0=0$)	0.86 ($\ell_0=4$)	0.82 ($\ell_0=19$)
Qwen2.5-72B	80	0.76 ($\ell_0=0$)	0.76 ($\ell_0=5$)	0.55 ($\ell_0=24$)

Table 17. **Depth and KV-group localisation of the option-word key read** (Qwen2.5-1.5B, FP32). Entries are fractions of the effect obtained when the option-word rows see the source key in every layer and both KV groups, with 95% bootstrap CIs over stories (ratio of means recomputed in each resample). $n = 40$ stories per format. Exploratory.

Rows/layers that see the swapped key	OPTIONS-AFTER	LETTERS-AFTER
layers 0–3	-0.02 [-0.02, -0.01]	-0.01 [-0.01, -0.00]
layers 4–7	+0.23 [+0.11, +0.35]	+0.36 [+0.26, +0.46]
layers 8–11	+0.14 [+0.10, +0.18]	+0.06 [+0.03, +0.09]
layers 12–15	+0.15 [+0.12, +0.19]	+0.04 [+0.02, +0.06]
layers 16–19	+0.02 [+0.01, +0.04]	+0.01 [+0.00, +0.02]
layers 20–23	+0.00 [-0.00, +0.00]	+0.00 [+0.00, +0.00]
layers 24–27	+0.00 [-0.00, +0.00]	+0.00 [+0.00, +0.00]
KV group 0 (all layers)	+0.22 [+0.17, +0.27]	+0.06 [+0.03, +0.09]
KV group 1 (all layers)	+0.53 [+0.40, +0.66]	+0.66 [+0.57, +0.75]

and 0.92–0.95 at Qwen2.5-72B, depending on the format.

Competent-only analysis. Our primary analyses pool all stories, whether or not the model answers the clean runs correctly. As a secondary analysis, every natural-factorial quantity is also computed on the stories whose base and source clean runs are both answered correctly (argmax over the candidates). From 7B up most stories are competent and the two analyses agree closely. The competent subset is smallest at Qwen2.5-1.5B and 3B, and the estimates move most at 3B. The sign of ID_K under OPTIONS-AFTER, LETTERS-AFTER, NO-MENTION and LIST-BEFORE is unchanged in every model. The near-zero SENTENCE-AFTER value at Qwen2.5-3B becomes slightly positive, with a CI that excludes zero, but stays far below the SENTENCE-AFTER values from 7B up. No conclusion in the main text changes.

Attention probe. An early exploratory probe at Qwen2.5-1.5B ($n = 25$ cores, no CIs, a different prompt) found that the option word naming the source location attends to the writing token as in the source run when the writing token’s key is swapped. Part (a) of preregistration G supersedes it (Appendix F.1).

Table 18. Paint and schedule tasks. Key share under OPTIONS-AFTER and ID_K (nats) in each format, clamped from layer 0. $n = 150$ stories per task, format and model; 95% bootstrap CIs. Preregistration D (D1, D2) for OPTIONS-AFTER, NO-MENTION and LIST-BEFORE; the rest exploratory.

Task	Model	s_K (OPTIONS-AFTER)	ID_K (nats)				
			LETTERS-AFTER	OPTIONS-AFTER	SENTENCE-AFTER	NO-MENTION	LIST-BEFORE
paint	Qwen2.5-7B	0.73 [0.72, 0.74]	+21.5 [+21.3, +21.7]	+21.4 [+21.2, +21.7]	+3.0 [+2.8, +3.1]	+1.2 [+1.1, +1.3]	+0.3 [+0.2, +0.4]
	Qwen2.5-14B	0.90 [0.90, 0.91]	+35.7 [+35.4, +36.0]	+40.3 [+39.9, +40.8]	+3.5 [+3.3, +3.7]	+0.8 [+0.7, +0.9]	+0.2 [+0.1, +0.2]
	Qwen3-8B	0.92 [0.92, 0.93]	+39.7 [+39.4, +40.1]	+42.7 [+42.3, +43.0]	+4.9 [+4.7, +5.2]	+1.1 [+1.0, +1.2]	+0.1 [+0.1, +0.2]
	Mistral-7B	0.86 [0.85, 0.87]	+16.0 [+15.7, +16.3]	+16.1 [+15.9, +16.4]	+2.2 [+1.9, +2.4]	+0.4 [+0.3, +0.5]	+0.1 [+0.1, +0.2]
	OLMo-2-7B	0.47 [0.46, 0.48]	+7.4 [+7.2, +7.5]	+6.5 [+6.3, +6.7]	+0.7 [+0.6, +0.8]	+0.1 [+0.1, +0.2]	−0.1 [−0.1, −0.1]
schedule	Qwen2.5-7B	0.58 [0.56, 0.59]	+19.6 [+19.3, +20.0]	+12.7 [+12.4, +13.0]	+4.1 [+3.8, +4.4]	+0.1 [+0.1, +0.2]	−0.1 [−0.2, −0.1]
	Qwen2.5-14B	0.87 [0.87, 0.88]	+35.6 [+35.3, +35.9]	+26.1 [+25.7, +26.5]	+10.9 [+10.3, +11.6]	+0.6 [+0.5, +0.6]	+0.2 [+0.1, +0.2]
	Qwen3-8B	0.92 [0.91, 0.92]	+39.6 [+39.2, +40.0]	+28.5 [+28.0, +29.0]	+7.3 [+6.9, +7.7]	+0.9 [+0.8, +1.0]	+0.2 [+0.1, +0.2]
	Mistral-7B	0.82 [0.81, 0.84]	+16.2 [+15.6, +16.7]	+13.2 [+12.9, +13.5]	+5.1 [+4.6, +5.6]	+0.1 [+0.1, +0.2]	0.0 [−0.1, 0.0]
	OLMo-2-7B	0.35 [0.34, 0.37]	+7.4 [+7.2, +7.6]	+4.5 [+4.3, +4.7]	+2.3 [+2.1, +2.4]	+0.1 [+0.1, +0.1]	0.0 [0.0, +0.1]

Table 19. Identity, not belief role. Role effect: mean change in $m = \log p(\text{initial}) - \log p(\text{LOC})$ when the observer and non-observer are swapped with the same location word (nats). f_K, f_V : fraction of that change reproduced by clamping only the writing token’s key or value (all layers) to the role-swapped run’s, as a ratio of means. Direct question; $n = 150$ stories per format and model; 95% bootstrap CIs. Preregistration E (E2) for OPTIONS-AFTER and NO-MENTION in the four 7–14B models; LETTERS-AFTER and Qwen2.5-72B exploratory.

Model	Format	role effect (nats)	f_K	f_V
Qwen2.5-7B	OPTIONS-AFTER	+34.4	+0.006 [+0.004, +0.008]	+0.010 [+0.007, +0.013]
	NO-MENTION	+28.7	+0.014 [+0.011, +0.018]	+0.001 [0.000, +0.003]
	LETTERS-AFTER	+29.8	+0.003 [0.000, +0.005]	+0.018 [+0.013, +0.025]
Qwen2.5-14B	OPTIONS-AFTER	+79.8	+0.001 [0.000, +0.001]	0.000 [−0.001, 0.000]
	NO-MENTION	+51.4	+0.001 [0.000, +0.002]	0.000 [−0.001, +0.001]
	LETTERS-AFTER	+66.0	0.000 [−0.001, 0.000]	0.000 [−0.001, 0.000]
Mistral-7B	OPTIONS-AFTER	+28.4	+0.003 [+0.002, +0.004]	+0.004 [+0.001, +0.008]
	NO-MENTION	+26.6	+0.019 [+0.014, +0.025]	+0.001 [0.000, +0.003]
	LETTERS-AFTER	+28.9	0.000 [−0.001, +0.002]	+0.011 [+0.006, +0.017]
OLMo-2-7B	OPTIONS-AFTER	+16.9	+0.004 [+0.003, +0.006]	+0.002 [0.000, +0.003]
	NO-MENTION	+17.7	+0.014 [+0.011, +0.017]	+0.002 [0.000, +0.004]
	LETTERS-AFTER	+12.5	0.000 [−0.001, +0.001]	+0.003 [+0.001, +0.004]
Qwen2.5-72B	OPTIONS-AFTER	+50.6	+0.001 [0.000, +0.002]	+0.002 [+0.001, +0.003]
	NO-MENTION	+38.4	+0.007 [+0.006, +0.009]	−0.001 [−0.002, +0.001]
	LETTERS-AFTER	+46.8	−0.001 [−0.002, 0.000]	+0.003 [+0.002, +0.004]

Table 20. Which later tokens read the writing token’s key. Fraction of the full key effect (all tokens see the source key, in every layer) recovered when only one group of later tokens sees it; ratio of mean log-odds changes. “Other later tokens” covers the instruction, chat template, prefill and answer position. Qwen2.5-7B, Qwen2.5-14B and Mistral-7B: BF16, $n = 60$ stories per format; preregistration D (D3) for OPTIONS-AFTER and SENTENCE-AFTER, LETTERS-AFTER exploratory. Qwen2.5-1.5B: FP32, $n = 40$, exploratory. 95% bootstrap CIs. The groups need not add up to one.

Model	Format	later-mention words	question	story after writing token	other later tokens
Qwen2.5-1.5B	OPTIONS-AFTER	+0.87 [+0.80, +0.95]	0.00 [−0.01, +0.01]	−0.01 [−0.01, 0.00]	0.00 [−0.04, +0.03]
Qwen2.5-1.5B	LETTERS-AFTER	+0.98 [+0.96, +1.00]	+0.01 [0.00, +0.01]	0.00 [−0.01, 0.00]	0.00 [0.00, 0.00]
Qwen2.5-7B	OPTIONS-AFTER	+0.92 [+0.90, +0.94]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	+0.01 [+0.01, +0.02]
Qwen2.5-7B	LETTERS-AFTER	+1.00 [+0.99, +1.01]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Qwen2.5-7B	SENTENCE-AFTER	+0.67 [+0.60, +0.73]	0.00 [0.00, +0.01]	−0.01 [−0.01, 0.00]	+0.11 [+0.09, +0.13]
Qwen2.5-14B	OPTIONS-AFTER	+0.94 [+0.93, +0.96]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	+0.01 [+0.01, +0.01]
Qwen2.5-14B	LETTERS-AFTER	+1.00 [+1.00, +1.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Qwen2.5-14B	SENTENCE-AFTER	+0.72 [+0.64, +0.80]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	+0.09 [+0.07, +0.10]
Mistral-7B	OPTIONS-AFTER	+0.98 [+0.96, +0.99]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	+0.01 [0.00, +0.01]
Mistral-7B	LETTERS-AFTER	+1.00 [+1.00, +1.01]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Mistral-7B	SENTENCE-AFTER	+0.73 [+0.67, +0.78]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	+0.05 [+0.03, +0.06]

Table 21. The intervention of Anonymous (2026) across formats. φ : behavioural shift of the learned remap M relative to the PCA patch P, as a fraction of the natural source-to-target shift (Equation (4)). ψ_K, ψ_V : fraction of the M–P difference that P reproduces when given M’s keys or values at the writing token (Equations (5) and (6)). ρ_K, ρ_V : fraction of that difference that M loses when given P’s keys or values (Appendix C). Exchanges from block 6 on. $n = 96$ story cores with B, S and T distinct, three fits averaged within each core; 95% bootstrap CIs over cores. Preregistrations C (C1–C3) and E (E4) for the cells they name; the rest exploratory.

Model	Format	φ	ψ_K	ρ_K	ψ_V	ρ_V
Mistral-Small-24B	LETTERS-AFTER	0.75 [0.71, 0.79]	0.78 [0.73, 0.82]	0.99 [0.98, 0.99]	0.02 [0.01, 0.02]	0.16 [0.13, 0.19]
	OPTIONS-AFTER	0.70 [0.68, 0.73]	0.53 [0.49, 0.57]	0.76 [0.75, 0.78]	0.23 [0.22, 0.24]	0.44 [0.41, 0.48]
	SENTENCE-AFTER	0.72 [0.69, 0.75]	0.20 [0.16, 0.23]	0.45 [0.43, 0.47]	0.52 [0.50, 0.54]	0.77 [0.73, 0.81]
	NO-MENTION	0.71 [0.68, 0.73]	0.07 [0.06, 0.08]	0.18 [0.16, 0.20]	0.80 [0.78, 0.83]	0.93 [0.92, 0.94]
	LIST-BEFORE	0.74 [0.71, 0.76]	−0.00 [−0.01, 0.01]	0.01 [−0.00, 0.02]	0.97 [0.96, 0.98]	0.96 [0.95, 0.98]
Qwen2.5-72B	LETTERS-AFTER	0.90 [0.87, 0.93]	0.89 [0.86, 0.91]	0.99 [0.99, 1.00]	0.01 [0.00, 0.01]	0.17 [0.13, 0.20]
	OPTIONS-AFTER	0.89 [0.86, 0.92]	0.63 [0.59, 0.67]	0.79 [0.78, 0.81]	0.21 [0.19, 0.23]	0.44 [0.39, 0.48]
	SENTENCE-AFTER	0.87 [0.84, 0.89]	0.21 [0.17, 0.25]	0.34 [0.31, 0.36]	0.67 [0.65, 0.70]	0.80 [0.76, 0.84]
	NO-MENTION	0.86 [0.84, 0.88]	0.05 [0.05, 0.06]	0.11 [0.09, 0.12]	0.90 [0.88, 0.91]	0.95 [0.94, 0.96]
	LIST-BEFORE	0.91 [0.88, 0.92]	−0.03 [−0.04, −0.02]	−0.02 [−0.04, −0.01]	1.00 [0.98, 1.01]	1.01 [0.99, 1.02]

F. Preregistration G: re-mention attention, knockout, membership, word forms and IOI

Preregistration G (GPU stage 5) asked what the duplicate-token account predicts and where it stops. Parts (a)–(d) use the 150 seed-0 story cores of stages 1 and 3b (the row splice and the form probe the first 60 of them), and part (e) 200 fresh IOI cores. Every contrast is paired by core, with 95% percentile intervals from 10,000 core-bootstrap resamples and ratios of means recomputed within each resample. Table 6 gives every verdict; here we define the measures, give the per-cell values and report the observations that the entry did not predict, without reinterpreting them. A shared gate required the stage-5 factorial to reproduce the stage-3b ID_K of SENTENCE-AFTER, NO-MENTION and LIST-AFTER in the four models of parts (c) and (d); every value agrees at the printed precision. Triples are Qwen2.5-7B / Qwen2.5-14B / Mistral-7B unless stated.

F.1. (a) Re-mention attention at every scale

Under SENTENCE-AFTER the neutral sentence mentions the written word B again. With eager attention, we take for each head the attention from that mention to the writing token, minus the mean attention to the writing token from the candidates not mentioned before it. The three heads with the largest mean excess on the odd-indexed cores form H^* . On the even-indexed cores, E is the mean excess at H^* , and F is the gain in the attention from the mention of S to the writing token under the key clamp $C_K(S)$, so $F/E = 1$ means that the attention follows the clamped key one for one. G is the same excess for the answer position’s attention to the mention of B (hop 2), at its own three heads, and Q is G at a small model divided by the smaller G of the two anchors, Qwen2.5-7B and 14B. Table 22 gives every cell.

Table 22. **Re-mention attention** (preregistration G, part (a)). E : excess attention of the later mention of B to the writing token at the three selected heads; F/E : the share of it that follows the key clamp; G : excess attention of the answer position to the mention of B (hop 2) at its own three heads; ID_K and the non-specific key effect d_K (nats) from the same eager run, scored in the casing the model emits. Heads selected on odd-indexed and evaluated on even-indexed cores of 150; 95% bootstrap CIs. Under SENTENCE-AFTER: G1 at 7B and 14B, G2 and G3 at 1.5B and 3B, G4 in all four; OPTIONS-AFTER is the gate and the reference of G3.

Model	Format	E	F/E	G (hop 2)	ID_K (nats)	d_K (nats)
Qwen2.5-1.5B	SENTENCE-AFTER	0.789 [0.778, 0.802]	1.03	0.086 [0.075, 0.098]	−0.53 [−0.64, −0.41]	+0.31
	OPTIONS-AFTER	0.750 [0.734, 0.767]	1.04	0.453 [0.423, 0.484]	+2.63 [2.41, 2.86]	+5.96
Qwen2.5-3B	SENTENCE-AFTER	0.879 [0.866, 0.892]	0.99	0.113 [0.095, 0.132]	+0.04 [−0.35, 0.44]	+3.03
	OPTIONS-AFTER	0.832 [0.821, 0.844]	1.03	0.395 [0.355, 0.435]	+10.37 [9.62, 11.13]	+20.57
Qwen2.5-7B	SENTENCE-AFTER	0.869 [0.852, 0.885]	1.00	0.255 [0.222, 0.288]	+4.73 [4.39, 5.07]	+9.21
	OPTIONS-AFTER	0.887 [0.874, 0.900]	1.00	0.770 [0.739, 0.798]	+19.39 [19.00, 19.78]	+36.90
Qwen2.5-14B	SENTENCE-AFTER	0.915 [0.908, 0.921]	0.98	0.329 [0.304, 0.353]	+13.48 [12.76, 14.20]	+24.09
	OPTIONS-AFTER	0.866 [0.856, 0.876]	0.99	0.865 [0.859, 0.872]	+38.95 [37.88, 39.97]	+74.79

At 1.5B and 3B the re-mention attends to the writing token about as strongly as at 7B and 14B, and the attention follows the clamped key, while ID_K under SENTENCE-AFTER stays within the preregistered ± 1 -nat margin (G2, H_{diss} ; G3). By the entry’s criterion the heads are not embedding-level, and the same heads serve both formats at 1.5B and 7B (overlap of the two H^* , as a Jaccard index, 1.00 and 1.00; 0.50 at 3B). Hop-2 attention is weaker at the small models ($Q = 0.34$ and 0.44; G4), but at 3B it reaches the presence bar that G4a sets for the anchors without any key read, so it does not alone account for the scale difference. G measures attention; it does not intervene on hop 2.

Unpredicted observations. (i) At 3B the key clamp has a non-specific effect without identity ($d_K = 3.03$ nats against $ID_K = 0.04$). (ii) Under SENTENCE-AFTER only 0.66 and 0.75 of the cores at the small models are answered correctly in both clean runs, and almost every wrong answer is the initial location; on the competent cores the 3B value is slightly positive, as in the stage-1 competent-only analysis (Appendix E; recomputed, not preregistered). (iii) Under the key clamp, the answer’s hop-2 attention does not follow the clamped key at 1.5B (change towards the mention of S 0.015, towards B 0.028), partly follows it at 3B (0.127 and −0.105) and follows it at 7B and 14B (0.343 and −0.243; 0.321 and −0.213). (iv) Under SENTENCE-AFTER at 7B and 14B, the answer row at the hop-2 heads attends mostly to the writing token itself

(0.534 and 0.468), not to the re-mentions; under OPTIONS-AFTER its five largest columns are all re-mentioned words. (v) The two formats' H^* overlap little at 14B (0.20), but each format's heads still work on the other (cross-format $E = 0.768$ and 0.734).

F.2. (b) Attention knockout

A knockout sets the attention from a set of rows to a set of columns to zero in every layer and head (Appendix C). The preregistered masks are M0, none; M1, the six candidate words after the writing token to the writing token; M2, the same rows to the earlier occurrences of the initial-location word, a matched control column; M3, the answer position to the writing token; M4, the union of M1 and M3; and M8, every later row to the writing token, a sanity gate. The exploratory masks are M2b, the candidate rows to the token before the writing token; M5, only the rows of the three clamped candidates B, S and X; M6, the other three candidate rows; M7, every later row that is not a candidate word; Mq, the question rows; and Mpost, the rows between the candidates and the answer position. r_K is ID_K under the mask divided by ID_K without it; $q_V = ID_V/d_{KV}$ is the value identity as a share of the joint clamp's effect; acc_B is the fraction of cores whose answer, restricted to the candidates, is still B. Table 23 gives every cell.

Cutting the candidate words' attention to the writing token removes the key read in every format and model, the control column does not, and cutting the answer's own attention leaves the read intact (G5, G6, G8b). The copy returns to its NO-MENTION share on the identity measure (G7a), but the answer is kept only at Qwen2.5-14B (G7b): the preregistered outcome is a partial takeover. Under NO-MENTION, M3 leaves 0.134 / 0.370 / 0.417 of ID_V (G8a), but it also removes most of the writing token's whole effect and of the models' ability to answer (Table 23).

Unpredicted observations. (i) At Qwen2.5-7B, the answer that replaces B under M1 is mostly the initial location (recomputed, descriptive); G7b's location-mass criterion cannot see these answers, because they are location words too. (ii) Under SENTENCE-AFTER, cutting only the answer's attention to the writing token (M3) raises the key read ($r_K = 1.660 / 1.800 / 1.394$), and so does M7. (iii) Cutting only the rows of the three clamped candidates (M5) removes the read as completely as M1 (under LIST-AFTER, $r_K = -0.008 / -0.025 / -0.007$), while cutting the other three (M6) leaves it intact (0.961 / 1.003 / 1.003). (iv) Under SENTENCE-AFTER, M1 raises q_V to 0.423 / 0.443 / 0.460. (v) At Qwen2.5-7B, cutting the control column (M2) slightly strengthens the key read ($r_K = 1.094$).

F.3. (c) Membership and dose

The arms replace the neutral sentence (family S) or the option list (family L) by one that names a subset of the candidates, in the canonical order and with "Answer with one word": S2 and L2 name S and X; S3 and L3 name B, S and X; S3out and L3out name B and two candidates other than S and X; S3half names B, S and one other; S4 and L4 name B, S, X and one other; S4out and L4out name S, X and two others, but not B; and S6 and L6 are SENTENCE-AFTER and LIST-AFTER. r_k is ID_K in the arm naming k candidates divided by ID_K in the arm naming all six. The row splice (first 60 cores) lets only the named words' rows see the source key, giving f_{words} , their share of the total splice effect d . Table 24 gives every arm.

The key read needs the swapped candidates themselves: naming B with two others gives none (G9, met in both families and replicated at seed 1). In lists two named candidates give the full read; in sentences the read grows with the number named, and at Qwen2.5-7B $r_2 = 0.28$ (0.32 at seed 1) is not above proportional (G10, met for lists, not for sentences). Removing S and X from a list does not bring the copy back ($R_V = 0.11 / 0.03 / -0.08$; G11, not met). In a sentence naming only S and X (S2), the named words carry 0.40 / 0.38 / 0.39 of the splice effect and the rows after the question 0.46 / 0.49 / 0.38 (G12, not met, 9/12). G12 splits the total key-swap effect d , not ID_K , and in S2 most of d is not identity-specific: ID_K/d_K is 0.14 / 0.29 / 0.38 in S2 against 0.44 / 0.56 / 0.61 in S3 (arithmetic on printed values). The main text's narrowing to "lists and sentences naming at least three candidates" rests on this.

Unpredicted observations. (i) Naming B with two other candidates pushes ID_K below NO-MENTION, although the written word is re-mentioned (S3out-NO-MENTION -0.32, L3out-NO-MENTION -0.35 at Qwen2.5-7B; Table 6). (ii) In sentences every added candidate raises ID_K : at Qwen2.5-7B the paired steps are S3-S2 = 1.96 (adds B), S4-S3 = 1.11 (adds one non-matching candidate) and S6-S4 = 0.93 (adds two), each with a CI excluding zero; a pure duplicate-token account does not predict the last two. Naming one of the two swapped candidates (S3half) gives about half of S3 (0.61 at Qwen2.5-7B). (iii) In lists that do not name the written word B, the key read is at full strength while the value identity stays near or above NO-MENTION: $ID_V(L4out)$ is 14.90 / 30.76 / 13.68 against 16.60 / 19.12 / 13.65 under NO-MENTION and 4.06 / 2.97 / 3.29 in L3. This suggests, post hoc, that the copy is suppressed when the written word itself is re-mentioned;

Table 23. **Attention knockout** (preregistration G, part (b); masks in the text). r_K : ID_K under the mask relative to no mask; q_V : ID_V as a share of the joint clamp’s effect d_{KV} ; acc_B : fraction of cores whose candidate-restricted answer is B. $n = 150$ cores per cell. M1, M2, M3, M4 and M8 enter G5–G8 and Gate b; the other masks, and every mask under SENTENCE-AFTER, are exploratory. Under NO-MENTION only the masks that do not need candidate rows are defined.

Format	Mask	Qwen2.5-7B			Qwen2.5-14B			Mistral-7B		
		r_K	q_V	acc_B	r_K	q_V	acc_B	r_K	q_V	acc_B
LIST-AFTER	M0	1.000	0.125	1.00	1.000	0.075	1.00	1.000	0.105	1.00
	M1	0.001	0.390	0.37	−0.026	0.440	1.00	−0.007	0.448	0.86
	M2	1.094	0.104	1.00	0.997	0.065	1.00	0.990	0.113	1.00
	M2b	1.001	0.125	1.00	1.000	0.075	1.00	1.000	0.107	1.00
	M3	1.043	0.022	1.00	1.027	0.017	1.00	1.010	0.053	1.00
	M4	0.001	0.512	0.00	0.001	0.470	0.66	−0.001	0.457	0.39
	M5	−0.008	0.418	0.73	−0.025	0.446	1.00	−0.007	0.457	0.89
	M6	0.961	0.117	1.00	1.003	0.071	1.00	1.003	0.106	1.00
	M7	0.947	0.003	0.97	1.022	0.004	1.00	0.948	0.020	0.97
	M8	0.000	–	0.00	0.000	–	0.13	0.000	–	0.17
	Mq	0.981	0.123	1.00	1.002	0.076	1.00	0.988	0.112	1.00
OPTIONS-AFTER	Mpost	1.001	0.119	1.00	1.002	0.068	1.00	0.999	0.096	1.00
	M0	1.000	0.120	1.00	1.000	0.066	1.00	1.000	0.074	1.00
	M1	0.003	0.377	0.30	−0.028	0.427	0.99	−0.009	0.422	0.73
	M2	1.071	0.099	1.00	0.998	0.057	1.00	0.972	0.083	1.00
	M2b	1.000	0.120	1.00	0.999	0.066	1.00	1.001	0.077	1.00
	M3	1.041	0.019	1.00	1.017	0.013	1.00	1.006	0.040	1.00
	M4	−0.000	0.523	0.00	0.002	0.464	0.58	0.001	0.446	0.29
	M5	−0.004	0.404	0.58	−0.027	0.435	1.00	−0.008	0.429	0.77
	M6	0.971	0.113	1.00	1.005	0.059	1.00	1.007	0.077	1.00
	M7	0.944	0.001	0.98	0.983	0.004	1.00	0.926	0.015	0.99
	M8	0.000	–	0.00	0.000	–	0.10	0.000	–	0.19
SENTENCE-AFTER	Mq	0.980	0.119	1.00	0.997	0.068	1.00	0.993	0.079	1.00
	Mpost	1.001	0.118	1.00	0.998	0.065	1.00	1.000	0.070	1.00
	M0	1.000	0.346	1.00	1.000	0.325	1.00	1.000	0.361	1.00
	M1	−0.105	0.423	1.00	−0.036	0.443	1.00	0.001	0.460	0.99
	M2	1.027	0.346	1.00	0.922	0.333	1.00	0.963	0.368	1.00
	M2b	1.007	0.346	1.00	0.994	0.327	1.00	0.991	0.362	1.00
	M3	1.660	0.099	0.77	1.800	0.117	1.00	1.394	0.188	0.99
	M4	0.007	0.496	0.07	0.014	0.480	0.91	0.002	0.446	0.29
	M5	−0.113	0.423	1.00	−0.035	0.443	1.00	−0.003	0.464	1.00
	M6	0.974	0.348	1.00	1.053	0.319	1.00	0.995	0.359	1.00
	M7	1.184	−0.036	0.35	1.939	−0.016	1.00	1.456	0.012	0.86
NO-MENTION	M8	0.000	–	0.00	0.000	–	0.14	0.000	–	0.05
	Mq	0.929	0.349	0.99	1.005	0.323	1.00	1.000	0.359	1.00
	Mpost	1.028	0.329	0.99	1.327	0.255	1.00	1.209	0.290	1.00
	M0	1.000	0.391	1.00	1.000	0.335	1.00	1.000	0.433	1.00
	M3	0.023	0.566	0.00	0.041	0.459	0.20	0.117	0.445	0.15
	M8	0.000	–	0.00	0.000	–	0.00	0.000	–	0.12
	Mq	0.967	0.388	1.00	1.005	0.331	1.00	0.988	0.432	1.00
	Mpost	1.040	0.378	1.00	1.004	0.318	1.00	0.995	0.402	1.00

it is a hypothesis for a future test, not a claim. (iv) Under NO-MENTION and every sentence arm, the clean answer puts almost no probability on the six lowercase candidates at the three primary models, so the sentence-family contrasts are scored on tokens the model is not about to emit, as in stage 3b; in lists the candidates carry most of it.

F.4. (d) Non-identical re-mentions

Each variant replaces the re-mentions of the SENTENCE-AFTER sentence: THE adds “the”; MODIF uses a modified noun phrase (“wooden shelf”); TITLE and UPPER change the case; PLURAL uses the plural; SYN a synonym; FRMIX and DEMIX put the French or German word in the English frame; FR and DE translate the whole sentence (exploratory); and three structure-matched floor sentences name other words. With the floor and the exact repeat (SENTENCE-AFTER) scored

Table 24. Membership and dose (preregistration G, part (c)). Top: ID_K and ID_V (nats) per arm, seed-0 cores, $n = 150$. Bottom: row splice ($n = 60$), the share of the total key-swap effect d read by the named words’ rows (f_{words}) and by the rows after the question. Qwen2.5-7B, Qwen2.5-14B and Mistral-7B enter G9–G12; OLMo-2-7B and the arms S3half, S4, S4out, L4, L4out, S6, L6 (as splice arms) and S3out, L3out (as splice arms) are exploratory.

Arm	Qwen2.5-7B		Qwen2.5-14B		Mistral-7B		OLMo-2-7B	
	ID_K	ID_V	ID_K	ID_V	ID_K	ID_V	ID_K	ID_V
NO-MENTION	+1.04	+16.60	+1.05	+19.12	+0.55	+13.65	+0.38	+10.97
S2	+1.52	+13.97	+6.99	+23.51	+3.66	+13.22	+0.86	+8.55
S3	+3.49	+11.94	+9.52	+19.67	+5.25	+11.83	+1.51	+8.75
S3half	+2.11	+12.13	+5.41	+16.16	+2.66	+11.79	+0.67	+8.37
S3out	+0.73	+12.61	+0.75	+13.40	+0.34	+12.18	+0.29	+8.84
S4	+4.59	+12.26	+10.88	+18.88	+6.15	+11.87	+1.64	+8.69
S4out	+4.18	+14.08	+10.82	+24.36	+6.17	+13.58	+1.59	+8.96
S6	+5.52	+11.91	+12.08	+18.46	+7.87	+11.46	+2.01	+8.63
L2	+19.76	+18.63	+35.19	+29.48	+15.27	+16.24	+5.87	+10.03
L3	+21.70	+4.06	+35.83	+2.97	+15.66	+3.29	+7.89	+5.14
L3out	+0.69	+5.49	+0.92	+3.49	+0.31	+2.47	+0.37	+4.30
L4	+21.50	+4.71	+35.87	+4.19	+15.38	+3.62	+8.35	+5.33
L4out	+20.29	+14.90	+35.45	+30.76	+15.45	+13.68	+8.00	+9.07
L6	+20.60	+5.93	+35.81	+5.69	+15.55	+3.44	+8.65	+5.93
<i>Reader rows (splice, $n = 60$): f_{words} / share read by the rows after the question</i>								
S2	0.40 / 0.46		0.38 / 0.49		0.39 / 0.38		0.26 / 0.62	
S3	0.68 / 0.13		0.72 / 0.11		0.72 / 0.10		0.90 / -0.01	
S6	0.67 / 0.11		0.72 / 0.09		0.73 / 0.05		0.76 / 0.06	
S3out	0.04 / 0.90		0.11 / 0.74		0.11 / 0.68		-0.16 / 1.40	
L2	0.91 / 0.14		0.94 / 0.17		0.92 / 0.08		0.72 / 0.13	
L3	0.94 / 0.01		0.95 / 0.00		0.95 / 0.01		0.91 / -0.01	
L6	0.91 / 0.01		0.93 / 0.01		0.95 / 0.01		0.88 / 0.00	
L3out	0.58 / 0.13		0.39 / 0.09		0.41 / 0.18		0.58 / 0.31	

on the English candidates, $r_K(v) = [ID_K(v) - ID_K(\text{floor})] / [ID_K(\text{SENTENCE-AFTER}) - ID_K(\text{floor})]$; r_K^{any} also scores the answer in the variant’s own form; ρ_s is the ratio of the identity key shares; and a_v is the variant’s attention to the writing token, summed over its tokens, relative to the exact repeat’s ($n = 60$). Gates required clean accuracy of at least 0.95 in the variant, its floor and its anchor, and that at least five of the six words map correctly in a six-way prompt (the passing words are the population of the cell). A prediction is met if at least three evaluable models meet it. Table 25 gives every cell.

On our English-scored measure the read is bound to the word form (G15): synonyms and mixed-language words carry at most 0.15 of an exact repeat’s key identity. It is largely insensitive to case (TITLE and UPPER read 0.83–1.14; G14 not met for them), sensitive to number (PLURAL 0.11–0.49), and a modified noun phrase reads only partly (MODIF 0.32–0.64; G13 not met for it, THE met). No preregistered token-or-concept outcome holds for any variant (G16, not met): the profile is graded and model-dependent. Per model, and exploratory: at the two Qwen models FRMIX shows the concept pattern on both measures ($r_K^{\text{any}} = 0.95$ and 1.06; $a_v = 0.86$ and 0.90), and scored in its own form it reads fully ($r_K = 0.99$ and 1.28); DEMIX is concept-like on r_K^{any} but graded on a_v there; at Mistral-7B and OLMo-2-7B the read is graded or token-level. Where the read is lost, the value identity rises (G17, met), but by no more than after the floor sentence that names other words (exploratory: the floor sentence raises ID_V over SENTENCE-AFTER by 2.71, 2.51 and 0.54 nats at Qwen2.5-7B, Mistral-7B and OLMo-2-7B, recomputed, against 2.58, 1.37 and 0.31 for FRMIX).

Unpredicted observations. (i) A capitalised or upper-case re-mention reads nearly as fully as the exact repeat, although its token ids differ; the CI lies below 1 in three of the eight cells. (ii) MODIF attends to the writing token about as much as the exact repeat ($a_v = 1.22, 1.24, 1.08$ and 0.94 in the four models) but reads the key only partly: attention without the read. Values of a_v above 1 come from summing attention over multi-token spans; with the last token only, the FRMIX attention leg falls below the concept bar (recomputed in the log). (iii) The raw rise of $\log p(S)$ under the key clamp does not separate the variants, so only the identity contrasts are reported.

Table 25. **Non-identical re-mentions** (preregistration G, part (d)). r_K : key identity relative to an exact repeat, above a structure-matched floor, on the English candidates; r_K^{any} : the same with the answer also scored in the variant’s own form; a_v : attention of the re-mention to the writing token relative to the exact repeat, summed over the re-mention’s tokens ($n = 60$). $n = 150$ cores per cell before the per-word competence gate; DEMIX and DE cover fewer cores. THE, MODIF, TITLE, UPPER, PLURAL, SYN, FRMIX and DEMIX enter G13–G16; FR, DE and the list variant are exploratory.

Variant	Qwen2.5-7B			Qwen2.5-14B			Mistral-7B			OLMo-2-7B		
	r_K	r_K^{any}	a_v	r_K	r_K^{any}	a_v	r_K	r_K^{any}	a_v	r_K	r_K^{any}	a_v
THE	1.09	1.09	1.09	0.99	0.99	1.02	1.06	1.06	1.11	0.96	0.96	1.02
MODIF	0.32	0.32	1.22	0.64	0.64	1.24	0.64	0.64	1.08	0.55	0.55	0.94
TITLE	0.94	1.19	0.92	1.09	1.37	0.97	1.07	1.53	1.79	0.87	0.89	0.90
UPPER	1.08	1.51	1.74	1.14	1.57	1.93	1.01	1.26	2.58	0.83	0.90	1.84
PLURAL	0.33	0.33	0.87	0.37	0.37	0.88	0.49	0.50	1.40	0.11	0.11	0.89
SYN	−0.03	0.09	0.41	0.00	0.09	0.41	0.06	0.26	0.53	−0.13	−0.13	0.21
FRMIX	0.15	0.95	0.86	0.10	1.06	0.90	0.07	0.56	0.55	0.09	0.44	0.39
DEMIX	−0.01	0.69	0.66	0.02	1.32	0.65	0.02	0.31	0.33	0.00	−0.13	0.09
FR	0.18	0.55	1.16	0.18	0.47	1.21	0.14	0.28	0.83	0.16	0.41	0.58
DE	0.00	0.59	0.92	0.06	1.06	1.01	0.05	0.55	0.58	−0.05	0.43	0.30
list SYN	0.05	0.33	−	0.05	0.23	−	0.07	0.47	−	−0.00	0.05	−

E.5. (e) IOI

The IOI cores follow the templates of Wang et al. (2022), alternating ABBA and BABA, with four names: the indirect object of the base and source runs and of the third-value run, and the subject. The clamps act at the indirect object’s mention. Arms: PLAIN (the sentence); AFTER (a “Choices:” list of the four names after the sentence, in the chat wrapper on instruct models); BEFORE (the list before it); QUESTION (“Which name completes the sentence?”, without candidates); and INLINE and INLINE_BEFORE (the list as a parenthetical right after the indirect object’s mention, or before the sentence; preregistered only at GPT-2 small). $f_K = \text{ID}_K / \text{ID}_{KV}$ and f_V likewise. A cell is evaluable only if the clean model passes Gate e: two-way accuracy at least 0.75, four-way at least 0.50, and a positive mean logit difference. GPT-2 small and GPT-2 XL ran in FP32 with exactness asserted, the 7B and 14B models in BF16. Table 26 gives every cell.

Plain IOI carries almost no key identity in the three preregistered models (f_K within -0.02 to 0.02 ; G18). At Qwen2.5-7B-Instruct a later list gives a large positive key read ($\text{ID}_K(\text{AFTER}) = 7.05$ [6.26, 7.81]), but G19 required it in both 7B models, and Mistral-7B failed the competence gate in AFTER. G20 fails at Qwen2.5-7B-Instruct on its baseline clause ($f_V(\text{QUESTION}) = 0.69$ against the predicted 0.75), and G21 because the list before the sentence gives a negative read (-2.02 nats).

Unpredicted observations. (i) BEFORE gives a negative key read in every model where it passes the gate (Qwen2.5-7B-Instruct -2.02 , Qwen2.5-14B-Instruct -3.83 , Qwen2.5-7B base -0.95 nats) and in the cells that fail it. (ii) The sign of the AFTER read varies by model: positive at Qwen2.5-7B- and 14B-Instruct (7.05 and 10.24; 14B exploratory, where G20 fails on its baseline clause as at 7B), absent at Qwen2.5-7B base (-0.03 , a weak null: the joint effect there is only 1.72 nats, against 11.96 at the instruct model) and negative in the gate-failed cells (Mistral-7B -1.98 , GPT-2 small -1.77 , GPT-2 XL -1.87). (iii) An in-sentence re-mention (INLINE) gives a negative key read in all six models (-3.29 to -2.33 nats), with the gate passed in every INLINE cell. (iv) In a row splice under AFTER ($n = 60$), the listed names carry the whole key effect (1.05 at Qwen2.5-7B-Instruct, 1.07 at Mistral-7B, whose AFTER cell failed the gate), the rows of the two clamped names alone 0.27 and 0.84. (v) In the GPT-2 small attention probe (re-run on CPU; Appendix A), the canonical name-mover heads attend from the end to the indirect object, and under INLINE the duplicate-token heads attend from the listed copy of the name to its earlier mention; under the key clamp this attention moves to the row of the listed swapped name.

Table 26. **IOI** (preregistration G, part (e); 200 fresh cores). Gate e: two-way / four-way accuracy of the clean run (failed cells are not evaluable). ID_K with 95% bootstrap CI, ID_V and ID_{KV} in nats; f_K , f_V : shares of ID_{KV} , unstable where ID_{KV} is near zero. GPT-2 small, Qwen2.5-7B-Instruct and Mistral-7B enter G18–G22 (the GPT-2 small parts replicate a disclosed pilot); Qwen2.5-7B base, Qwen2.5-14B-Instruct and GPT-2 XL are exploratory.

Model	Arm	Gate e (2-way / 4-way)	ID_K (nats)	ID_V	ID_{KV}	f_K	f_V
GPT-2	PLAIN	0.98 / 0.98	−0.15 [−0.18, −0.11]	+14.08	+8.61	−0.02	+1.64
	AFTER	0.48 / 0.36 (failed)	−1.77 [−1.84, −1.71]	+2.37	+0.93	−1.92	+2.57
	BEFORE	0.55 / 0.35 (failed)	−0.05 [−0.07, −0.04]	+1.00	+0.89	−0.06	+1.13
	QUESTION	0.49 / 0.49 (failed)	−0.04 [−0.06, −0.02]	+5.35	+4.37	−0.01	+1.23
	INLINE	0.94 / 0.91	−2.35 [−2.47, −2.23]	+6.71	+3.74	−0.63	+1.79
	INLINE_BEFORE	0.91 / 0.82	−0.25 [−0.29, −0.21]	+4.80	+3.27	−0.08	+1.47
Qwen2.5-7B	PLAIN	1.00 / 1.00	+0.20 [0.10, 0.31]	+6.72	+10.33	+0.02	+0.65
	AFTER	0.98 / 0.89	+7.05 [6.26, 7.81]	+5.34	+11.96	+0.59	+0.45
	BEFORE	0.97 / 0.82	−2.02 [−2.18, −1.86]	+6.37	+7.67	−0.26	+0.83
	QUESTION	1.00 / 1.00	+0.61 [0.43, 0.78]	+17.77	+25.90	+0.02	+0.69
	INLINE	1.00 / 0.67	−3.26 [−3.47, −3.05]	+2.63	+2.02	−1.61	+1.30
	INLINE_BEFORE	1.00 / 0.62	−0.74 [−0.80, −0.67]	+1.17	+1.43	−0.52	+0.82
Mistral-7B	PLAIN	0.99 / 0.99	+0.02 [−0.02, 0.07]	+6.06	+7.94	+0.00	+0.76
	AFTER	0.59 / 0.42 (failed)	−1.98 [−2.38, −1.58]	+3.35	+3.16	−0.63	+1.06
	BEFORE	0.81 / 0.27 (failed)	−0.96 [−1.12, −0.80]	−2.32	−1.01	+0.95	+2.30
	QUESTION	0.96 / 0.96	+0.11 [0.03, 0.18]	+11.33	+12.98	+0.01	+0.87
	INLINE	1.00 / 0.71	−2.33 [−2.46, −2.19]	+1.27	+1.01	−2.29	+1.25
	INLINE_BEFORE	0.99 / 0.32 (failed)	−0.42 [−0.48, −0.37]	−0.26	−0.01	+43.18	+26.44
Qwen2.5-7B base (expl.)	PLAIN	1.00 / 1.00	+0.18 [0.10, 0.26]	+5.45	+8.53	+0.02	+0.64
	AFTER	0.85 / 0.77	−0.03 [−0.15, 0.08]	+1.02	+1.72	−0.02	+0.59
	BEFORE	0.83 / 0.66	−0.95 [−1.01, −0.89]	+0.76	+0.92	−1.03	+0.82
	QUESTION	0.55 / 0.59 (failed)	+0.27 [0.19, 0.35]	+4.46	+8.25	+0.03	+0.54
	INLINE	1.00 / 0.74	−2.83 [−2.96, −2.70]	+1.72	+1.21	−2.34	+1.42
	INLINE_BEFORE	1.00 / 0.82	−0.76 [−0.82, −0.70]	+1.53	+1.68	−0.45	+0.91
Qwen2.5-14B (expl.)	PLAIN	1.00 / 0.99	+0.29 [0.18, 0.39]	+6.36	+9.66	+0.03	+0.66
	AFTER	0.99 / 0.92	+10.24 [8.60, 11.89]	+8.52	+21.47	+0.48	+0.40
	BEFORE	0.97 / 0.88	−3.83 [−4.07, −3.59]	+12.60	+14.31	−0.27	+0.88
	QUESTION	0.99 / 0.99	+0.88 [0.66, 1.10]	+20.86	+30.79	+0.03	+0.68
	INLINE	1.00 / 0.68	−2.85 [−3.01, −2.70]	+1.52	+1.14	−2.51	+1.33
	INLINE_BEFORE	1.00 / 0.46 (failed)	−1.06 [−1.16, −0.97]	−0.31	+0.33	−3.27	−0.94
GPT-2 XL (expl.)	PLAIN	0.99 / 0.99	−0.08 [−0.13, −0.03]	+8.82	+7.89	−0.01	+1.12
	AFTER	0.52 / 0.26 (failed)	−1.87 [−1.95, −1.79]	+0.72	+0.06	−30.82	+11.88
	BEFORE	0.52 / 0.32 (failed)	−0.57 [−0.62, −0.53]	+0.41	+0.31	−1.84	+1.30
	QUESTION	0.47 / 0.47 (failed)	−0.01 [−0.04, 0.03]	+4.00	+5.02	−0.00	+0.80
	INLINE	0.99 / 0.67	−3.29 [−3.41, −3.17]	+2.28	+1.09	−3.03	+2.09
	INLINE_BEFORE	0.99 / 0.55	−1.09 [−1.18, −1.01]	+1.41	+0.87	−1.25	+1.62

G. Preregistration H: reader heads, the second hop and another lab’s intervention

Preregistration H (GPU stage 6) has two parts: the heads that read the writing token’s key and the route from them to the answer, at two 7B models (Appendix G.1), and the key-only and value-only exchange applied to the binding intervention of Prakash et al. (2026) at Qwen2.5-14B (Appendix G.2). Table 7 gives every gate and verdict, and Appendix A the disclosures.

G.1. Reader heads and the second hop (part a)

Design. Qwen2.5-7B (784 heads) and Mistral-7B (1024 heads), BF16 with eager attention; OPTIONS-AFTER confirmatory, SENTENCE-AFTER reported. Heads are ranked on 60 ranking stories by a_3 , the attention change of the option rows of S and B towards p under the key clamp ($a_3 = \frac{1}{2}[(A^{K_S} - A^B)_{\text{row } S \rightarrow p} + (A^B - A^{K_S})_{\text{row } B \rightarrow p}]$), and every curve is evaluated on 60 other stories. $R(k)$ is the fraction of the option rows’ key effect d_G recovered when only the top k heads see the source key in the option rows (HeadSplice; Appendix C), and $KO(k)$ the fraction lost when only they are blinded; $k^* = \lceil 0.05 \times \text{heads} \rceil = 40$ and 52 , and k_{80} is the smallest k on the grid with $R(k) \geq 0.8$. The other rankings are the single-head effect f^+ (each head alone sees the source key) and the leave-one-out effect d^- (each head alone is blinded); the controls are three random sets (the first k^* heads of fixed random permutations) and the k^* heads most active at the option rows outside the top $2k^*$ by a_3 . The duplicate-token score D and the induction score I come from 100 random repeated-token sequences, and the in-task score T_{dup} is the attention from each option row to the earlier occurrences of the same word other than p . The causal set C is the top $\min(k_{80}, k^*)$ heads by a_3 . Table 27 gives H1–H5, Figure 6 the curves and Table 28 every ablation condition.

Table 27. **Reader heads and the second hop** (preregistration H, part (a); OPTIONS-AFTER, 60 held-out stories, BF16). $R(k^*)$, $KO(k^*)$: fraction of the option rows’ key effect recovered when only the top k^* heads by a_3 see the source key, or lost when only they are blinded. ρ_K : ID_K under mean-ablation of these heads at the option rows, relative to no ablation. C : the top heads by a_3 that carry 80% of the read. r_{ans} : fraction of the key-clamp effect removed when only the answer row gets the option rows’ base keys and values; r_{other} , r_{all} : the same for every other later row and for all rows. 95% bootstrap CIs over stories. The last row is H5’s secondary contrast, reported, not scored.

	Quantity (OPTIONS-AFTER)	Qwen2.5-7B	Mistral-7B
	heads k^* (of all heads)	40 (784)	52 (1024)
H1	$R(k^*)$, top- k^* by a_3 see K_S	0.967 [0.952, 0.982]	0.942 [0.919, 0.961]
	k_{80} by a_3 / by f^+ / by d^-	20 / none / none	16 / none / 12
H2	$KO(k^*)$, top- k^* blinded	0.977 [0.966, 0.987]	0.971 [0.954, 0.984]
	random sets: R / KO (mean of 3)	0.004 / 0.011	0.007 / 0.010
H3	ρ_K under mean-ablation of the top- k^*	0.238 [0.205, 0.270]	0.059 [0.040, 0.079]
	ρ_K , control sets (min–max)	0.953–1.018	1.002–1.009
	rise of ID_V (nats)	+6.907 [6.264, 7.561]	+10.861 [9.995, 11.677]
	answer kept (base argmax)	0.87	1.00
H4	$ C $ (top heads by a_3 carrying 80%)	20	16
	median duplicate score D over C	0.370 [0.361, 0.378]	0.232 [0.227, 0.238]
	median induction score I over C	0.011	0.022
	median in-task T_{dup} over C	0.459	0.433
	top-10 by $a_3 \cap$ top-10 by D	6	5
H5	r_{ans} : only the answer row restored	0.899 [0.886, 0.913]	0.859 [0.845, 0.872]
	$r_{\text{other}} / r_{\text{all}}$	0.118 / 0.992	0.165 / 0.992
	$r_{\text{ans}}(K) / r_{\text{ans}}(V)$ (secondary, not scored)	0.797 / 0.080	0.683 / 0.023

Sufficiency and necessity (H1, H2). The top 5% of heads recover almost all of the option rows’ read and blinding them removes almost all of it, while random sets of equal size do neither; no single random draw exceeds $R = 0.006 / 0.007$ or $KO = 0.020 / 0.013$ (recomputed). The ranking is stable: the split-half Spearman correlation of a_3 between ranking and

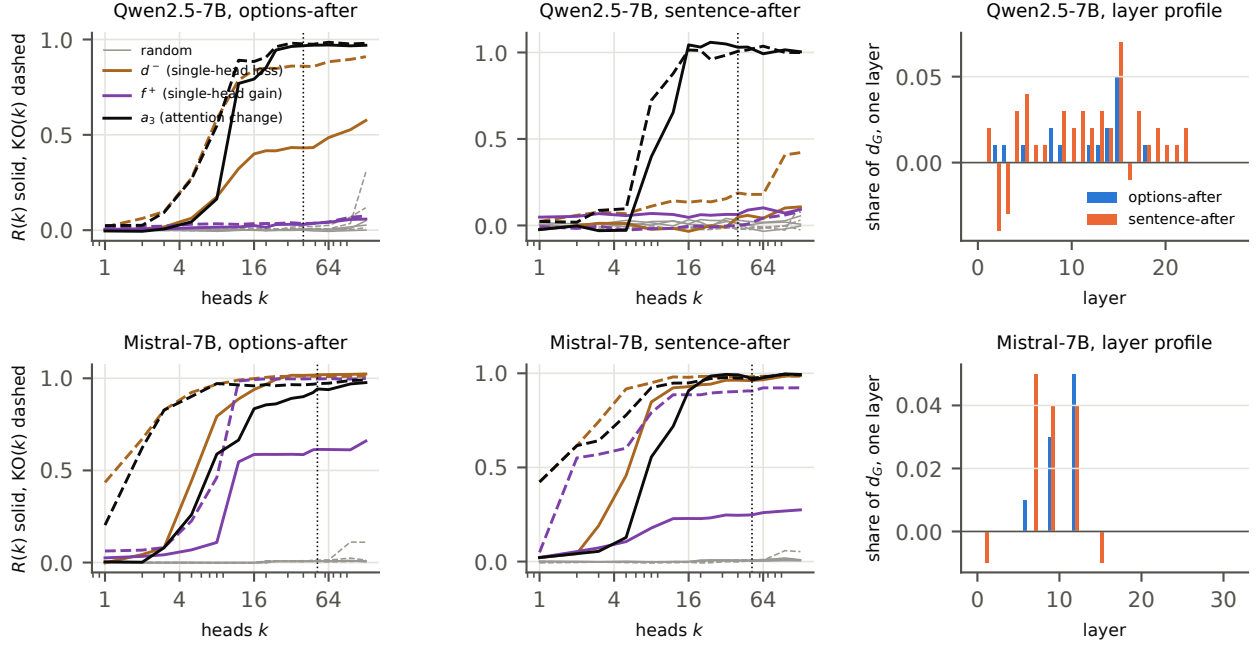


Figure 6. **Head curves and layer profiles** (preregistration H, part (a); exploratory except $R(k^*)$ and $KO(k^*)$ of the a_3 ranking under OPTIONS-AFTER). Left and middle: $R(k)$ (solid) and $KO(k)$ (dashed) against the number of heads k for the rankings by a_3 , single-head effect f^+ and leave-one-out effect d^- , and for three random sets (grey), under OPTIONS-AFTER and SENTENCE-AFTER; dotted: k^* . Right: the share of the option rows’ key effect recovered when all heads of one layer see the source key. 60 held-out stories per model and format.

evaluation stories is 0.973 / 0.956 under OPTIONS-AFTER and 0.985 / 0.981 under SENTENCE-AFTER.

The read is a joint property of many heads (exploratory). Ranking by single-head effect does not find the set, and ranking by leave-one-out effect finds it at one of the two models: at Qwen2.5-7B, f^+ reaches only $R(128) = 0.06$ and d^- 0.58; at Mistral-7B f^+ reaches 0.66, while d^- reaches $k_{80} = 12$, fewer heads than a_3 needs (16). The a_3 curve rises steeply (Qwen2.5-7B: $R(8) = 0.16$, $R(12) = 0.77$), and at Mistral-7B blinding the top three heads removes 0.83 of the read while the same three alone recover 0.08. Letting all heads of a single layer read the source key recovers at most 0.05 (layer 15) and 0.05 (layer 12) of d_G , and the single-layer shares sum to only 0.163 and 0.094 (recomputed); the heads of C span 12 and 10 layers (recomputed). No single head or layer is necessary or sufficient; the top 20 and 16 heads by a_3 , over many layers, are, jointly.

Table 28. **Every ablation condition** (preregistration H, part (a); 60 held-out stories). ρ_K : ID_K under ablation at the six option rows, relative to none; ΔID_V : rise of the value identity (nats); kept: fraction of stories whose candidate-restricted clean answer is still B. “mean”: mean-ablation with the head’s position-matched mean over the ranking stories; “zero”: zero-ablation. H3 uses the top- k^* mean row, the three random sets and the most-active set under OPTIONS-AFTER; every other cell is exploratory.

Condition	Qwen2.5-7B, OPTIONS-AFTER			Qwen2.5-7B, SENTENCE-AFTER			Mistral-7B, OPTIONS-AFTER			Mistral-7B, SENTENCE-AFTER		
	ρ_K	ΔID_V	kept	ρ_K	ΔID_V	kept	ρ_K	ΔID_V	kept	ρ_K	ΔID_V	kept
none	1.000	+0.000	0.97	1.000	+0.000	1.00	1.000	+0.000	1.00	1.000	+0.000	1.00
top- k^* (mean)	0.238	+6.907	0.87	-0.080	+2.295	1.00	0.059	+10.861	1.00	0.012	+2.640	1.00
top-10 (mean)	0.859	+1.978	0.98	0.616	+1.436	1.00	0.139	+12.326	1.00	0.233	+2.302	0.98
top-20 (mean)	0.654	+4.285	0.95	0.126	+2.282	1.00	0.104	+12.728	1.00	0.056	+2.648	1.00
top- k^* (zero)	0.081	+11.315	1.00	-0.060	+2.756	1.00	0.057	+9.030	0.95	0.006	+2.609	1.00
random set 1	1.002	+0.091	0.97	0.969	+0.081	1.00	1.002	-0.096	1.00	1.003	+0.048	1.00
random set 2	0.978	+0.851	1.00	0.655	+0.820	1.00	1.009	-0.292	1.00	1.289	-1.563	1.00
random set 3	0.953	+0.241	0.98	0.975	+0.114	1.00	1.006	-0.326	1.00	1.100	-0.336	1.00
most active at G	1.018	-0.177	0.97	0.934	+0.012	1.00	1.009	+0.049	1.00	0.949	+0.521	1.00
next k^* by a_3	1.019	-0.062	1.00	1.140	-0.028	1.00	0.996	+0.173	1.00	0.823	+0.682	1.00

Ablation (H3). Mean-ablating the top- k^* heads at the option rows removes most of the key identity, raises the value identity and keeps the answer; the random and most-active sets of equal size leave ρ_K near 1 (Table 28). Zero-ablation (exploratory) goes the same way ($\rho_K = 0.081$ and 0.057). Without ablation the answer is B in 0.97 of Qwen2.5-7B stories, against 0.87 under ablation. The rise of the value identity is partial at Qwen2.5-7B (ID_V under ablation 11.55 nats against 16.60 under NO-MENTION in stage 1), and at Mistral-7B it reaches about the NO-MENTION level (13.45 against 13.65); both comparisons are across stages and stories (arithmetic on printed values). The stage-5 edge knockout keeps the answer only at Qwen2.5-14B (G7b); the two interventions differ (mean-ablation at six rows leaves $\rho_K = 0.238$ at Qwen2.5-7B, the knockout leaves $r_K \approx 0$), and we do not reconcile them.

Duplicate-token heads (H4) and the size of C . Over C the median previous-token score is $0.017 / 0.004$ (largest $0.083 / 0.049$, recomputed), and the in-task score of control words that are not candidates is $0.027 / 0.065$, against $T_{\text{dup}} = 0.459 / 0.433$. T_{dup} is measured in the task the heads were selected on, so we treat it as supporting rather than independent evidence; D and I , on random sequences, are the test. H4 depends on the size of the head set (recomputed): only 13 of the 20 heads of C at Qwen2.5-7B and 9 of the 16 at Mistral-7B have $D \geq 0.2$, against 26 and 11 in the whole model, whose median D is $0.028 / 0.007$. Over the full top- k^* set the median D is $0.179 / 0.065$; at Mistral-7B it is 0.156 over the top 20 and 0.294 over the top 12. At Qwen2.5-7B, $R(16) = 0.792$, just under 0.8 , so C has 20 heads; over the top 16 the median D would also be 0.370 . Over all heads, a_3 correlates weakly with D (Spearman $0.185 / 0.230$) and with T_{dup} ($0.146 / 0.251$). We therefore say that the core of the readers, the top heads by a_3 that carry 80% of the read, is canonical duplicate-token heads, not that every reader is one.

The second hop in detail (H5). The routes add up: $r_{\text{ans}} + r_{\text{other}} = 1.017 / 1.024$ against $r_{\text{all}} = 0.992 / 0.992$. Restricted to single option rows, the source word’s row gives $r_{\text{ans}} = 0.605 / 0.557$, the base word’s row $0.414 / 0.391$ and the separators $0.049 / 0.016$ (exploratory). As H5’s secondary contrast, reported and not scored, the answer reads the option rows mainly through their keys ($r_{\text{ans}}(K) - r_{\text{ans}}(V) = 0.717 / 0.660$, CIs excluding zero), and the strong version $r_{\text{ans}}(KV) \geq 0.7$ holds in both models.

SENTENCE-AFTER (reported, not scored). Gate a2 failed at Qwen2.5-7B under SENTENCE-AFTER (Appendix A), so H6 and H5’s SENTENCE-AFTER line are not evaluable; the values below are exploratory. The option-row ceiling covers less of the full effect than under OPTIONS-AFTER ($d_G/d_{\text{full}} = 0.572 / 0.728$, recomputed). At Qwen2.5-7B, $R(k)$ and $KO(k)$ exceed 1 (Figure 6), and mean-ablation of the top k^* drives ρ_K to -0.080 ; one random set changes ρ_K substantially in each model (Table 28). The answer row’s read gives $r_{\text{ans}} = 0.788 / 0.780$, mainly through keys ($r_{\text{ans}}(K) = 0.875 / 0.732$, $r_{\text{ans}}(V) = -0.046 / 0.052$). The top-20 heads by a_3 of the two formats share 13 and 14 heads (H6’s threshold is 10; met at Mistral-7B, not evaluable at Qwen2.5-7B), the top 10 share 7 and 9, and the full a_3 grids correlate at Spearman $0.520 / 0.573$ (recomputed). In this free-form format the candidates carry little of the clean answer’s probability, so H3’s candidate-mass criterion fails in every condition, as expected.

G.2. The exchange on Prakash et al.’s intervention (part b)

Material and population. We use the template-2 stories of Prakash et al. (2026), their raw prompt (an instruction, the story, “Question:” and “Answer:”, without a chat template) and their pool of 320 reversed-sentence pairs drawn with their seed. In each story two characters each fill an opaque bottle with a drink, in two sentences, and the question asks what one character believes their bottle contains. The release is fetched at run time and checked by hash, and our port reproduces their pool byte for byte. Their language-model filter under NO-MENTION passed 315 pairs at Qwen2.5-14B-Instruct; the population is the first 150 passing pairs in pool order, of which about 70 are in their 80-pair validation split. The formats differ only after the story: NO-MENTION is their question; QNAMES names four candidate drinks in the question (the two story drinks and two drinks S and X absent from the story); OPTIONS-AFTER lists the same four as choices with “Answer with exactly one choice”; LETTERS-AFTER letters them (exploratory); and QNAMES2 (exploratory) names only the two story drinks, as the questions of their preliminary second dataset do.

The two edits. BIND is their state-token patch: at the output of block ℓ^* , the residuals at the two drink words and the punctuation after each are set to those of the counterfactual in which the two sentences are exchanged, each position taking the other sentence’s. The drink words’ identity does not change; their ordering information does. In their account the patch swaps the addresses (the character and object ordering IDs, read through QK) and the payload (the state ordering ID, moved through OV), so it is not a pure address edit. ID substitutes, at block ℓ_{ID}^* , the queried drink’s span from a donor

story in which that drink is replaced by S. With $m = \log p(\text{target}) - \log p(\text{queried drink})$, where the target is the other story drink for BIND and S for ID, IIA is the fraction of pairs whose patched answer is the target, and ℓ^* and ℓ_{ID}^* are the earliest blocks whose IIA under NO-MENTION is within 0.01 of the maximum.

Table 29. Replication of the binding swap (Qwen2.5-14B-Instruct, NO-MENTION, IIA of BIND by patched block). Released: the per-layer values published with Prakash et al. (2026) on their 80 validation pairs, transcribed from their release (not part of our results). Ours: 150 pairs, and the 70 pairs shared with their validation split. Recomputed from our sweep files; a reproduction check, since the released values were seen before preregistration H was finalised.

Block	Prakash et al. (80 pairs)	ours (150 pairs)	ours, 70 shared pairs
24	0.69	0.720	0.700
26	0.72	0.747	0.729
28	1.00	0.993	1.000
30–34	1.00	1.000	1.000
36	0.06	0.080	0.057

Replication and sweeps. The swap reproduces their released per-layer IIA (Table 29). Its IIA is 1.00 at blocks 29–34 and 0.993 at 28, where one pair fails; the tie rule selects $\ell^* = 28$, and one more failure would have selected 29. Below block 24 it flips no answer under NO-MENTION (IIA 0.00 at block 23, 0.72 at 24). Figure 7 shows every sweep. The swap reproduces only under NO-MENTION: under QNAMES, OPTIONS-AFTER and LETTERS-AFTER its IIA is at most 0.01 at every swept block from 18 to 40, and its effect on the margin peaks at 6.5 nats (OPTIONS-AFTER, block 28), 6.1 (QNAMES, block 31) and 2.8 (LETTERS-AFTER, block 23). The identity edit flips every answer under NO-MENTION at every block through 35, so its curve is flat and the tie rule picks $\ell_{\text{ID}}^* = 0$ by default rather than by localisation. When the candidates are named after the question it stops flipping answers from about block 20: under OPTIONS-AFTER its IIA is 1.00 through block 11, 0.89 at 14, 0.66 at 16, 0.17 at 19 and 0.00 from 20; under QNAMES it is 0.90 at block 0 and 0.00 from 20.

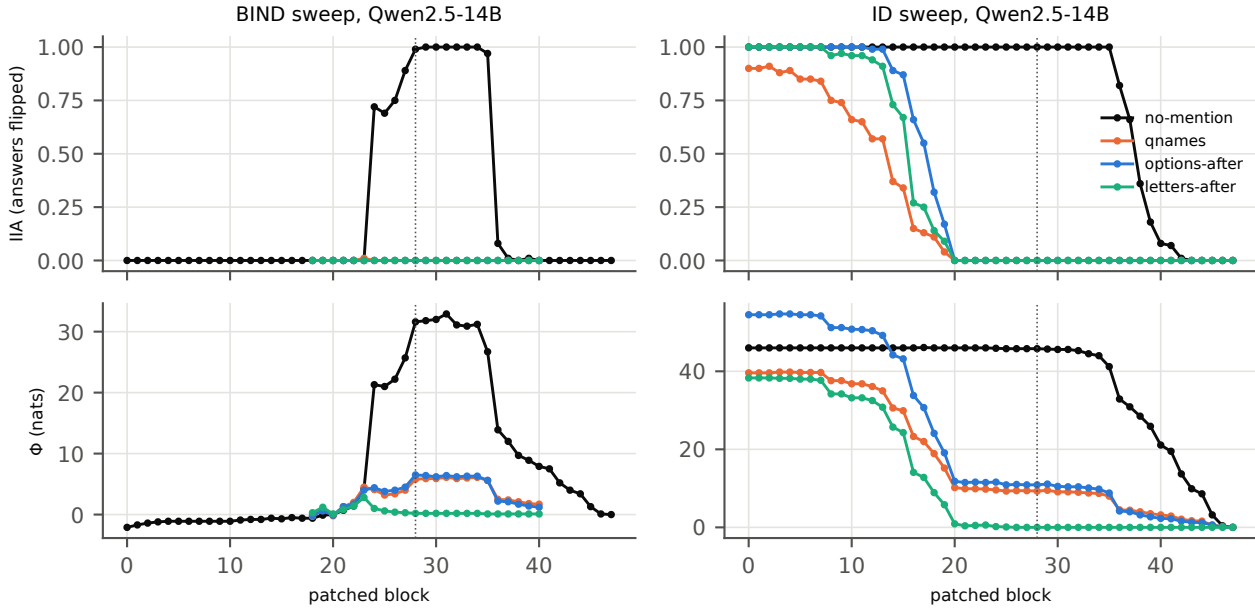


Figure 7. Sweeps of the two edits over the patched block (Qwen2.5-14B-Instruct, 150 pairs, BF16). Top: IIA, the fraction of answers flipped to the target; bottom: Φ , the mean shift of $m = \log p(\text{target}) - \log p(\text{queried drink})$. Left: the binding swap BIND (swept over blocks 18–40 in the formats that name candidates); right: the identity edit ID. Dotted: block $\ell^* = 28$. ℓ^* and ℓ_{ID}^* are chosen on the NO-MENTION sweeps (preregistered); the other formats are reported.

The exchange. The exchange (Appendix C) exchanges the patched positions’ keys or values in every block after the patched one. A cell is evaluable if Gates b0 and b2 pass in it ($\Phi \geq 3$ nats), $\psi_K + \psi_V \geq 0.5$ with both at least -0.1 , and the edit reproduces at that depth on the NO-MENTION sweep. Table 30 summarises the cells and Table 31 gives all thirteen.

LETTERS-AFTER fails Gate b2 at block 28 for both edits ($\Phi = 0.225$ and 0.034 nats), and ID@28 under LETTERS-AFTER also fails the κ rule; LETTERS-AFTER is exploratory throughout. BIND reproduces fully under NO-MENTION (it flips 149 of 150 answers), and that cell carries H8 by itself. Under QNAMES and OPTIONS-AFTER it flips 0 and 0 answers, so those cells decompose a shift of the logit margin ($\Phi = 5.814$ and 6.523 nats against base margins of -20.06 and -27.69 , recomputed), not a reproduced swap.

Table 30. The exchange on the interventions of Prakash et al. (2026) (Qwen2.5-14B-Instruct, 150 pairs). Each cell: $\kappa = \psi_K / (\psi_K + \psi_V)$ with (ψ_K, ψ_V) , the number of answers the full patch flips, and its effect Φ on the margin (nats). BIND@28: the binding swap at block $\ell^* = 28$; ID@0, ID@28: the identity edit at blocks 0 and 28. $s_{ID}(f, \ell_0)$: identity key share of the natural clamp at the queried drink from block ℓ_0 , with 95% CI. [†]Exploratory formats. [‡]Gate b2 failed ($\Phi < 3$ nats). n/e: the κ rule fails. H7 compares BIND@28 with $s_{ID}(f, 29)$, H11 ID@0 with $s_{ID}(f, 1)$ (Part 1) and ID@28 with $s_{ID}(f, 29)$ (Part 2).

	NO-MENTION	QNAMES	OPTIONS-AFTER	LETTERS-AFTER [†]	QNAMES2 [†]
BIND@28	κ 0.618 (0.703, 0.435) flips 149/150, Φ 31.616	κ 0.906 (1.020, 0.106) flips 0/150, Φ 5.814	κ 0.864 (1.025, 0.161) flips 0/150, Φ 6.523	κ 0.658 (1.652, 0.859) flips 0/150, Φ 0.225 [‡]	κ 1.014 (1.016, -0.014) flips 0/150, Φ 7.285
ID@0	κ 0.131 (0.131, 0.871) flips 150/150, Φ 46.005	κ 0.773 (0.777, 0.229) flips 135/150, Φ 39.606	κ 0.779 (0.789, 0.224) flips 150/150, Φ 54.550	κ 0.969 (0.977, 0.031) flips 150/150, Φ 38.255	—
ID@28	κ 0.129 (0.129, 0.872) flips 150/150, Φ 45.797	κ 0.007 (0.006, 0.827) flips 0/150, Φ 9.321	κ 0.006 (0.005, 0.811) flips 0/150, Φ 10.885	n/e (-0.171 , 3.024) flips 0/150, Φ 0.034 [‡]	—
$s_{ID}(f, 29)$	0.017 [0.007, 0.026]	-0.028 [-0.039 , -0.019]	-0.015 [-0.020 , -0.010]	0.098 [-2.155 , 1.750]	—
$s_{ID}(f, 1)$	0.017 [0.007, 0.026]	0.724 [0.710, 0.739]	0.702 [0.689, 0.716]	0.984 [0.976, 0.992]	—

Table 31. All thirteen exchange cells (Qwen2.5-14B-Instruct, 150 pairs). Φ : effect of the full patch on m (nats) with 95% CI; flips: answers moved to the target; ψ_K, ψ_V : fractions of Φ reproduced in the clean run by the patched run’s keys or values at the patched positions; interaction $1 - \psi_K - \psi_V$; κ with 95% CI; κ_w : κ with the drink words alone exchanged (punctuation positions excluded); ρ_K, ρ_V : fractions of Φ lost when the patched run gets the clean keys or values. LETTERS-AFTER and QNAMES2 are exploratory; κ_w, ρ_K and ρ_V are exploratory everywhere.

Edit	Format	Φ (nats)	flips	ψ_K	ψ_V	interaction	κ	κ_w	ρ_K	ρ_V
BIND@28	NO-MENTION	31.616 [30.193, 32.985]	149/150	0.703	0.435	-0.138	0.618 [0.595, 0.640]	0.779	0.565	0.297
	QNAMES	5.814 [5.187, 6.456]	0/150	1.020	0.106	-0.127	0.906 [0.879, 0.931]	0.919	0.894	-0.020
	OPTIONS-AFTER	6.523 [5.905, 7.171]	0/150	1.025	0.161	-0.186	0.864 [0.839, 0.889]	0.875	0.839	-0.025
	LETTERS-AFTER	0.225 [0.170, 0.282]	0/150	1.652	0.859	-1.511	0.658 [0.559, 0.854]	0.278	0.141	-0.652
	QNAMES2	7.285 [6.734, 7.861]	0/150	1.016	-0.014	-0.002	1.014 [0.992, 1.034]	1.019	1.014	-0.016
ID@0	NO-MENTION	46.005 [45.276, 46.725]	150/150	0.131	0.871	-0.002	0.131 [0.119, 0.142]	0.132	0.129	0.869
	QNAMES	39.606 [38.747, 40.488]	135/150	0.777	0.229	-0.006	0.773 [0.759, 0.786]	0.749	0.771	0.223
	OPTIONS-AFTER	54.550 [53.652, 55.462]	150/150	0.789	0.224	-0.012	0.779 [0.764, 0.793]	0.737	0.776	0.211
	LETTERS-AFTER	38.255 [37.690, 38.788]	150/150	0.977	0.031	-0.008	0.969 [0.960, 0.977]	0.957	0.969	0.023
ID@28	NO-MENTION	45.797 [45.069, 46.514]	150/150	0.129	0.872	-0.001	0.129 [0.118, 0.140]	0.131	0.128	0.871
	QNAMES	9.321 [8.653, 9.995]	0/150	0.006	0.827	0.166	0.007 [-0.004 , 0.018]	0.009	0.173	0.994
	OPTIONS-AFTER	10.885 [10.314, 11.513]	0/150	0.005	0.811	0.185	0.006 [-0.003 , 0.013]	0.003	0.189	0.995
	LETTERS-AFTER	0.034 [-0.012 , 0.078]	0/150	-0.171	3.024	-1.854	n/e	-0.263	-2.024	1.171

Further exchange results (exploratory). Part of BIND’s value share under NO-MENTION sits at the punctuation positions: with the drink words alone exchanged, ψ_V falls from 0.435 to 0.238 and $\kappa_w = 0.779$ (recomputed); under QNAMES and OPTIONS-AFTER, κ_w is close to κ (Table 31). Split by the queried character, BIND under NO-MENTION gives $\psi_K = 0.648$ and $\psi_V = 0.464$ for the first ($n = 71$) and 0.772 and 0.400 for the second ($n = 79$). Under QNAMES2, which names only the two story drinks, the margin shift is carried by keys alone ($\kappa = 1.014$, $\psi_V = -0.014$), but no answer flips. At block 28 the identity edit is value-carried in every evaluable format ($\psi_V = 0.872 / 0.827 / 0.811$), while BIND is key-dominant (Table 30); this is the post hoc reading of Section 4.2, with its caveats there: under QNAMES and OPTIONS-AFTER neither edit flips an answer, BIND also moves a payload, and Gate b3, built for the reverse dissociation, failed.

Depth. The identity key route at the drink word closes with depth when candidates are named after the question (Table 32): under OPTIONS-AFTER, s_{ID} is 0.702 for $\ell_0 \leq 3$, 0.587 at 14, 0.218 at 18, 0.041 at 24 and -0.014 at 27, while under NO-MENTION it stays near zero at every onset. Gate b3, which H10’s reading as a dissociation needed, failed: at $\ell_0 = 29$ under OPTIONS-AFTER, $ID_K = -0.092$ nats and $s_{ID} = -0.015$, and κ_{ID} at block 28 is 0.006. The binding swap

Table 32. Identity key share of the natural clamp on the stories of Prakash et al. (2026) (Qwen2.5-14B-Instruct, 150 pairs). $s_{ID} = \overline{ID}_K / (\overline{ID}_K + \overline{ID}_V)$ for keys and values clamped at the queried drink word from block ℓ_0 on. The onsets 0, 1, 3, 14 and 29 are preregistered in the four formats; the others form an exploratory sweep in NO-MENTION and OPTIONS-AFTER. Under LETTERS-AFTER at $\ell_0 = 29$ the denominator is near zero, so s_{ID} is undetermined (CI in Table 30).

Format	$\ell_0=0$	$\ell_0=1$	$\ell_0=3$	$\ell_0=6$	$\ell_0=12$	$\ell_0=14$	$\ell_0=18$	$\ell_0=24$	$\ell_0=27$	$\ell_0=29$	$\ell_0=30$	$\ell_0=33$	$\ell_0=36$	$\ell_0=42$
NO-MENTION	0.018	0.017	0.017	0.017	0.018	0.017	0.017	0.017	0.017	0.017	0.017	0.018	0.021	0.036
QNames	0.724	0.724	0.726	–	–	0.488	–	–	–	–0.028	–	–	–	–
Options-AFTER	0.702	0.702	0.702	0.688	0.634	0.587	0.218	0.041	–0.014	–0.015	–0.016	–0.015	–0.018	–0.070
Letters-AFTER	0.984	0.984	0.984	–	–	0.975	–	–	–	0.098	–	–	–	–

flips no answer below block 24 under NO-MENTION and at most 0.01 of answers at any swept block (18–40) when candidates are named, while the identity edit stops flipping answers from about block 20 when they are named, and the key route of identity has closed by block 24. No depth has both the binding swap and a key-read identity route, so the contrast between the identity edit at block 0 (a format shift of +0.648) and the binding swap at block 28 (+0.247) is confounded with depth: whether the kind of edit or its depth decides how far the format moves the attribution is untested.

κ and s_{ID} are different statistics. κ is one-sided (m compares the target with the queried drink) and exchanges the drink word and the punctuation after it; s_{ID} is the symmetric S-against-X identity contrast at the word alone. With $\ell_{ID}^* = 0$, the ID@0 exchange and the natural clamp from block 1 are nearly the same intervention: applied to both, the one-sided κ agree within 0.001 (NO-MENTION), 0.028 (QNames), 0.044 (Options-AFTER) and 0.012 (Letters-AFTER) (recomputed). The H11 Part 1 gaps therefore split as follows: under NO-MENTION almost the whole gap is the difference between the statistics (0.115); under QNames, 0.028 is exchange against clamp and 0.020 the difference between the statistics; under Options-AFTER, 0.044 and 0.032 (recomputed; the log’s split, from three-decimal printed values, differs in the last digit). H11 Part 2, at block 28, compares values within a few hundredths of zero under QNames and Options-AFTER ($\kappa_{ID} = 0.007$ and 0.006 against $s_{ID} = -0.028$ and -0.015) and repeats Part 1’s NO-MENTION cell (0.129 against 0.017). H11 is thus consistency of the exchange with the clamp on their material, not an independent test of the law. For H7, the one-sided clamp analogue from block 29 is 0.131 / 0.012 / 0.005 (recomputed), which would leave every gap far above 0.25; no verdict depends on the difference between the statistics. H7’s correlation, $r = -0.986$, is computed over three values of s_{ID} within a few hundredths of zero and carries no information.

FP32 re-check (exploratory, recomputed). Re-running the NO-MENTION sweeps in FP32 at blocks 26–30 gives BIND IIA of 0.747, 0.880, 0.993, 1.000 and 1.000, against 0.747, 0.887, 0.993, 1.000 and 1.000 in BF16, so the tie rule again selects block 28. At block 28 the flips agree in 0.987 of pairs (a different single pair fails in FP32), and Φ differs by 0.12 nats. The identity edit’s IIA is 1.000 at blocks 0–2 in both precisions.

H. Further discussion

Reproduction and scope of the intervention result. In the format of [Anonymous \(2026\)](#), our runs of their released bases reproduce their saved answers (a preregistered gate; Table 3), and we reproduce their key result: M’s keys alone reproduce a substantial part of the M–P difference in both models ($\psi_K = 0.53$ and 0.63 under OPTIONS-AFTER) and, in a comparison they did not make, more of it than M’s values do. They scoped their claim to their tested setting, which must include the prompt. We did not re-test their colour-lookup questions, and we score log-odds where they scored argmax agreement. The addition fractions sum to less than one and the removal fractions mostly to more than one (Table 21), so the channels do not combine additively, and part of the edit may travel through the other patched span tokens.

What four accounts predict. Table 33 sets four accounts against the observations of the main text. Each, ours included, is contradicted somewhere. The last row matters most: our account’s preregistered extension to a binding edit failed (H7, H9), and the lookback account is partly compatible with it, post hoc (one edit, one model).

Table 33. What four accounts predict. DT: duplicate-token attention alone; LB: lookback addresses and payloads ([Prakash et al., 2026](#); [Wu & Shomali, 2026](#)); FF: an edit’s channel is set by the format it was fit in; LM: later-mention readers (ours). ✓: predicted and observed; ×: predicted otherwise (refuted); –: silent; (ph): compatible, post hoc; ✓/×: the attention is predicted, its presence without a read is not; ✓/(ph): the attention is predicted, its presence without a read is explained post hoc (the read is decided downstream). Last row: Section 4.2.

Observation (evidence)	DT	LB	FF	LM
Hop 1: a sparse set of heads, whose core is duplicate-token heads, attends from the re-mentions to the writing token, key-matched, at every scale, also at 1.5B and 3B where $ID_K \approx 0$ (G1–G3, H1–H4)	✓/×	–	–	✓/(ph)
The same words placed before the writing token give no key read (E1)	✓	×	–	✓
The answer reads the option words, not the writing token (H5, G8b)	–	×	–	✓
Without a later mention the identity is copied through the answer’s edge to the writing token (G8a, E4)	–	✓	–	✓
Removing the readers raises the value copy (G7a, H3)	–	(ph)	–	✓
A remap refit without later mentions shows the same crossover (F1–F4)	–	–	×	✓
Prakash et al.’s binding swap is key-dominant in every evaluable format (κ 0.618, ψ_V 0.435 without a later mention); later mentions shift κ by +0.247 (H7–H10)	–	(ph)	–	×

Relation to lookback, address and payload accounts. In the lookback of [Prakash et al. \(2026, Sec. 2\)](#), an ordering-ID pointer at a later token is matched through the query–key circuit to an address at the recalled token, here the state token (our writing token), and the payload stored there is copied forward through the OV circuit; [Wu & Shomali \(2026\)](#) likewise find that an operation’s cache is read mainly as an address rather than a payload, moving keys and values together. In our terms their answer lookback reads identity by copy: attention is selected by ordering IDs, and the state’s identity travels in the value. The exchange of Appendix G.2 is consistent with this, and it shows that their patch is not carried mainly by the payload: their state-token patch, which in their account swaps the addresses and the payload together, reaches the answer mainly through the state tokens’ keys, that is, through where attention goes, with a substantial value share without a later mention ($\psi_K = 0.703$, $\psi_V = 0.435$ under NO-MENTION). Key dominance fits the address half of their account and the value share its payload half; because one patch moves both, the decomposition does not separate them, and it is one edit at one model. What is copied through the value without a later mention is the state’s identity: an identity edit at the same positions is value-carried under NO-MENTION ($\psi_V = 0.871$), as the remap of [Anonymous \(2026\)](#) is (E4). Later mentions add a second route for identity, a lookup in which a re-mentioned word finds the writing token through its key, as duplicate-token heads do ([Wang et al., 2022](#)); at two 7B models it runs through a sparse set of heads whose core is canonical duplicate-token heads, and the answer then reads the re-mentioned words (Section 3.4). On their material this route is visible for an identity edit at block 0 (κ_{ID} rises from 0.131 under NO-MENTION to 0.779 under OPTIONS-AFTER) but closes with depth, and no depth has both it and the binding swap (Appendix G.2), so whether it runs alongside an address-based lookback at matched depth is untested. In our QNAMES2 format, which names both candidate states as the questions of their preliminary second dataset do, the binding swap shifts the margin through keys alone ($\kappa = 1.014$) but flips no answer (exploratory; their second dataset itself was not run). As for the role-swap control: the observer is the first character named in both runs, so the control cannot detect an ordering ID of the observer at the writing token.